

Abstract

The general claim. Reference points and management rules are usually defended by *simulation*: a rule is run, the trajectory is inspected, and the rule is recommended. This paper defends them by *certification* instead, on two real resource systems, and argues that the difference is not stylistic. A simulation shows what happened on the runs that were tried; a certification states *why*, locates the verdict in a constant, and — most importantly — makes explicit *the horizon over which the verdict holds*. That last point is the paper’s central finding, and it recurs across two systems and independent pipelines: *certification has a horizon, and the horizon is set by the fitted map, not by the method*. On the Northern cod stock the certified layer has a finite horizon — six years under the two harsher floors, seven under the informative one — because the map is expansive at the reference point for every admissible carrying capacity. On the Edwards Aquifer, every positive-pumping certified kernel is empty beyond three years, an optimistic bound on a defect the out-of-sample audit exceeds. Two systems, different hydrologies, different institutions, the same shape of limitation. A certified verdict is therefore not a stronger version of a simulated one; it is a different kind of object, whose validity is *local in time* and whose locality can be quantified.

Why three parts, two systems. Certification is not one operation, and the three parts do three different ones on two real assessment records. Part I *characterises*: the robustness of a single-threshold reference point to persistent productivity shocks reduces to three constants, and the kernel computation collapses to algebra. Part II *certifies a historical event*: it certifies, in exact rational arithmetic, the obstruction structure that the Northern cod collapse exhibited. Part III *scores a policy family*: it computes robust viability kernels for competing governance rules on the Edwards Aquifer and retains a rule only if it meets a fixed criterion. Parts I and II are on Northern cod; Part III is on the Edwards Aquifer. The two systems have no hydrology, no biology and no institutional history in common — which is precisely why the recurrence of the horizon limitation is a finding rather than a property of one fit.

What is found. Part I: on the increasing branch of the surplus curve, the largest catch holding the reference point from itself is $C^* = g(\text{LRP}) - |e| = 91.59$ kt, and any rule’s protection margin is exactly C^* minus its catch there — at that point harvest and protection are one budget. A second constant, $g_{\max} - |e| = 215.2$ kt, separates viable-but-raised kernels from empty ones, and the two reproduce the kernel table in closed form. That identity makes the no-dominance verdict *structural, not empirical*: a reactive rule cannot out-supply a flat cap it matches in protection. Part II: the certified object is a harvest-free multiplier bracket, $[\rho, \max\{\rho + r, \rho/(1 - r)\}]$; both collapse steps certify as harvest-free contractions (upper bounds 0.754 and 0.372) under every timing convention, while the reference window’s worst step and the two prelude steps do not certify — a typology, not a blanket verdict. Part III: at the 618-ft threshold under the drought-of-record floor, training-mean pumping fails within roughly thirteen years while an interpolated 7.2% mean cut secures the threshold; the reactive rules are retained only *nominally*, on a hybrid criterion whose wet-year margin is exactly what the robust class excludes; and nothing is retained at 660 ft, where the rules are invisible to the kernel of their own trigger.

Structure. Part I is the harvest–protection budget, Part II the regime-viability certification, and Part III the Edwards policy-family scoring. Each is self-contained and reproduced in full from its companion paper without condensation, opening with that paper’s own abstract. A cross-system conclusion follows, stating what certification buys, what it costs, and where its horizon binds.

Certification, not simulation: the viability of harvest rules on two real resource systems, and the horizon of what can be certified

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Part I — The harvest–protection budget at the limit reference point

Abstract

Reference points are usually defended by simulation. Here the defence is a characterisation: on the increasing branch of the surplus curve, the robustness of a single-threshold reference point to persistent productivity shocks reduces to three constants, and the kernel computation collapses to algebra. Applied to Northern cod (NAFO 2J3KL; Schaefer fit 1983–2007; LRP 884.6 kt) under a protocol frozen before any score was computed: (1) the largest catch holding the reference point from itself is $C^* = g(\text{LRP}) - |e| = 91.59$ kt, and any rule’s protection margin is exactly C^* minus its catch there — at that point harvest and protection are one budget; a second constant, $g_{\max} - |e| = 215.2$ kt, separates viable-but-raised kernels from empty ones, and the two reproduce the kernel table in closed form. (2) That identity makes the no-dominance verdict structural, not empirical: a reactive rule cannot out-supply a flat cap it matches in protection. (3) The map is expansive at the reference point for every admissible $K \geq 2K^*$, so the certified layer has a finite horizon — six years under the two harsher floors, seven under the informative one — which is the crossing at which a geometrically growing erosion margin overtakes the worst-case trajectory, and which no catch reduction extends: it is the same for every admissible catch, including none at all. (4) Carrying capacity is not identified from above, but the functionals the results depend on are: across the profile set C^* lies in $[67.9, 95.2]$ kt — pinned to within 27 kt while K itself is not identified from above — whereas resampling the 24 transitions and refitting both parameters freely disperses the estimate over a far wider range (median 73.7 kt, 90% interval $[-89.4, 125.7]$, or 88.1 kt $([-5.6, 130.6])$ restricted to the expansive regime on which the results are conditional, 73% of replicates). The two answer different questions and the tight one is an identification result, not a precision one; 20-year survival from the reference point falls from 0.91 to 0.65 between moratorium-level and 120 kt removals. (5) Self-viability is regime-dependent: 171 kt on the post-moratorium window, 0 ± 8 kt on the modern series. On this map the reference point is protected by good years, and reactive design cannot buy protection with supply.

Keywords: northern cod; NAFO 2J3KL; surplus production; viability; robust viability kernel; harvest control rules; limit reference point

1. Introduction

After the 1992 moratorium, Northern cod (NAFO divisions 2J3KL) posed a clean governance question. Can any catch policy hold a collapsed stock’s spawning biomass above its limit reference point when productivity is depressed? Or is the reference point instead protected by good years rather than by demand management? Viability analysis supplies the natural instrument for such questions. It asks which states admit controls that keep every constraint satisfied over time, rather than which policy maximizes an objective (Aubin, 1991). It has been applied to fisheries in exactly this register: bioeconomic viability constraints (Béné, Doyen, and Gabay, 2001); viable recovery paths after collapse (Martinet, Thébaud, and Doyen, 2007); stochastic co-viability for ecosystem-based management (Doyen et al., 2012); and the numerical computation of viability kernels for fishery case studies (Krawczyk, Fox, W.W., 1970. An exponential surplus-yield model for optimizing exploited fish populations. *Transactions of the American Fisheries Society*, 99(1), 80–88. doi:10.1577/1548-8659(1970)99<80:aesmfo>2.0.co;2.

Pharo, Serea, and Sinclair, 2013; Krawczyk and Pharo, 2013). The reference point is a floor on the productive stock itself, namely the spawning biomass whose regeneration underwrites future yield. The question is therefore whether adjustments to the draw on that stock can hold the base that regenerates the yield, or whether the reference point is protected only by favourable productivity. This study answers the question by scoring rather than assertion.

Four questions organize the results. (Q1) Under which productivity regimes does any catch policy — zero catch included — hold the LRP, and where is the question vacuous because the floor exceeds the map’s maximum surplus? (Q2) Under the informative regime, what is the largest constant catch whose worst-case equilibrium remains at or above the LRP, and how do reactive rules fare against it under the declared dominance rule? (Q3) How fast does model error of the declared magnitude erode the certified safety margin when the governed map is expansive at the reference point? (Q4) Which answers are geometric properties of the rule and the threshold, which are identified properties of the fitted map, and which are sensitive to the model form or the carrying capacity?

The object is the governed surplus-production model fitted in a companion forecast-evaluation study (under separate review), which ended in a negative certificate with last-value persistence unbeaten on out-of-sample RMSE. Here the same fitted model, on the primary assessment specification (the 1983–2015 NCAM M-shift SSB series of DFO, 2016, Table A2, with LRP 884.6 kt), is scored as a closed-loop governance object. The deliverables are robust viability kernels of the LRP for a declared catch-policy family under persistent productivity-shock floors, plus a certified layer that converts the declared model defect into a safety-margin erosion. The intervention protocol — object, defect declaration, disturbance classes, policy family, and the selection rule restated in Definition 2.6 as a dominance partial order — was frozen (dated 2026-08-26) before any kernel, boundary, replay, or selection score was computed. The frozen protocol document is archived with the analysis code. Section 2 states the methods. Section 3 reports the kernels, the no-dominance verdict and the vacuous-class identity, the constructive boundary, the certified layer, and the stress replay, followed by the form comparison (Section 3.6), the carrying-capacity, stochastic, finite-duration, and uncertainty layers (Sections 3.7-3.10), and the second-specification row (Section 3.11). Section 4 discusses limitations.

Related work

Three literatures meet here: the viability analysis of managed renewable resources, the robust-viability treatment of uncertainty, and the evaluation of harvest control rules.

Viability in fisheries. That the governing question for a managed renewable resource is not which policy optimises but which states admit controls that keep every constraint satisfied over time is the founding move of viability theory (Aubin, 1991). It has been taken up in fisheries repeatedly: bio-economic viability, with the time-of-crisis function as an overexploitation indicator and irreversible configurations related to resource extinction (Béné, Doyen and Gabay, 2001); viable recovery paths toward sustainable fisheries (Martinet, Thébaud and Doyen, 2007); stochastic viability for ecosystem-based management under multiple conflicting objectives (Doyen et al., 2012); and viability as a “satisficing” alternative to optimal control, whose admissible policies are generically multiple and depend on fewer parameters than optimal or equilibrium strategies (Krawczyk and Pharo, 2013; Krawczyk et al., 2013). This paper sits in that lineage and uses its vocabulary throughout.

The production model. The governed object is a surplus-production model of Schaefer form, and the originals behind that form and behind the alternatives compared below should be named rather than assumed. The logistic surplus-production model, in which the population’s net increase is a concave quadratic in biomass with a maximum at intermediate stock size, is Schaefer’s (Schaefer, 1954); the exponential surplus-yield form fitted as a co-equal alternative is Fox’s (Fox, 1970). Both are standard, neither is contributed here, and the comparison between them is decided on residual MSE rather than on convention.

What is not standard, and is where this paper leaves the production-model literature, is the use to which the fitted form is put. That literature estimates the parameters and then asks what harvest maximises yield or profit. Here the fitted surplus enters a viability problem, and the quantity extracted from it is not an optimal effort but a *kernel*: on the increasing branch, two

constants reproduce the whole kernel table in closed form. The Schaefer-versus-Fox comparison therefore matters here for a reason it does not usually matter — the two forms give different kernels, not merely different optimal efforts — which is why it is reported as a form comparison rather than a sensitivity check.

Robust viability kernels. The set whose computation is collapsed here is the robust viability kernel: the set of initial biomasses from which at least one harvesting strategy guarantees the required production and preservation levels for all times, *whatever* the uncertainties, these being disturbances taking values in a known given set (Regnier and De Lara, 2015). Their central finding — that accounting for uncertainty sensibly shrinks the deterministic viability kernel — is the phenomenon this paper is organised around, and nothing here contradicts it.

What differs is how the set is obtained. In that literature the kernel is computed numerically, by a descending set iteration on a discretised state space. On the increasing branch of the surplus curve the computation here **collapses to algebra** instead: two constants, $C^* = g(\text{LRP}) - |e| = 91.59$ kt and $g_{\max} - |e| = 215.2$ kt, reproduce the kernel table in closed form, and every rule’s protection margin is exactly C^* minus its catch at the reference point. At that point harvest and protection are one budget. A closed-form collapse is not a faster numerical method; it is a different kind of result, and it is what makes the no-dominance verdict below structural rather than empirical.

Harvest control rules and management strategy evaluation. The standard instrument for choosing among harvest control rules is management strategy evaluation: simulate the entire management system — operating model, observation, assessment and rule — under specified uncertainties, and compare candidate rules on pre-specified performance metrics (Butterworth, 2007; Punt et al., 2016). It is a large, mature and practically indispensable machinery. The opening remark of this paper, that reference points are usually defended by simulation, is a description of that literature rather than a criticism of it.

The distinction is in what the two deliver, and it can be stated without rhetoric. An MSE verdict is *conditional on the operating model and on the ensemble simulated*: a no-dominance finding holds for the rules, the uncertainties and the replicates that were actually run. The no-dominance verdict here falls out of an identity — protection margin equals C^* minus catch — and therefore holds for *every* rule in the class at once, including rules that were never simulated. That is a genuine difference in kind.

It is also a narrower one, and the narrowing should be conceded. MSE routinely handles multiple conflicting objectives, implementation error, observation error and institutional constraints simultaneously; the scalar reduction here buys its closed form by giving most of that up. The two are complements rather than rivals: the characterisation says what cannot be bought at any price within the class, while the simulation says which rules do best on the metrics that were specified.

2. Methods

2.1 Operating model and data

All inputs are public and the analysis is reproducible: every stochastic layer uses a fixed seed and a fixed draw budget (Sections 3.8 and 3.10), and every deterministic layer is a closed-form or fixed-point computation. The stock series is the Northern cod (NAFO 2J3KL) spawning-stock biomass of DFO (2016), Table A2. Annual landings are Schijns et al. (2021), Table 1. The fit window is 1983–2007 (24 one-step transitions; the out-of-sample audit runs 2008–2015), and the protocol was frozen on 2026-08-26, before any kernel, boundary, replay, or selection score was computed. The training-window residuals of the fitted map have mean -10.9 kt, SD 114.9 kt, range $[-329.0, +206.6]$ kt, and lag-1 autocorrelation 0.55 . The 1992 transition residual is the minimum (-329.0 kt). The training-window maximum stock is 940.75 kt, which is why the fitted carrying capacity, pinned at its 5000 kt optimization bound, sits above the observed

range (the declared optimization box runs from just above the training maximum to 5000 kt). A companion forecast-evaluation study scored the same point-estimate map against persistence on the same series and issued a negative certificate (persistence unbeaten out of sample). That certificate is scored there, not here, and the closed-loop object below is the same fitted map by construction.

All analyses beyond the frozen kernel tables — the carrying-capacity grid, the stochastic viability layers, the finite-duration floors, and the bootstrap bands of Sections 3.7–3.10 — are additional scored objects executed after the freeze on the declared machinery. None of them replaces or alters the frozen family, the frozen floor classes, or the dominance rule of Definition 2.6.

2.2 The governed object and the scoring rule

The governed object is the companion forecast-evaluation study’s own stock-flow class.

$$S_{t+1} = [S_t + rS_t(1 - S_t/K) - C_t + e_t]_+$$

The Allee term is off. The map is fitted by one-step least squares on 1983–2007 with annual landings (Schijns et al., 2021), giving $r = 0.2369$ and $K = 5000$ kt (pinned at its optimization bound; the series never approaches carrying capacity, a declared defect; the LRP-boundary results depend chiefly on the identified r). Fit residual SD is 114.9 kt. The defect declaration is $\varepsilon = 329.0$ kt yr⁻¹ (the 1992 collapse transition). The out-of-sample audit on 2008–2015 gives a maximum of 47.1 kt, which does not exceed the declared defect (by contrast, the groundwater object of the companion intervention study exceeds its declared defect out of sample).

The declared formal objects are stated as definitions and propositions so that the empirical content can be read off cleanly.

Definition 2.1 (Governed surplus-production map). The governed object is the one-step map above with parameters $(r, K) = (0.2369, 5000 \text{ kt})$ fitted by one-step least squares on 1983–2007, Allee term off, and the declared defect declaration $\varepsilon = 329.0$ kt yr⁻¹.

Definition 2.2 (Safe set). The single declared threshold is $K^* = \text{LRP} = 884.6$ kt (the 1983–1989 mean of Table A2). The safe set is $[K^*, \infty)$. No row on the second assessment specification (the 1954–2024 extended xteNCAM series, LRP 276 kt) enters the frozen primary protocol. The labelled post-freeze row of Section 3.11 reports that specification separately, unpooled, against its own reference point, and no verdict transfers between the two.

Definition 2.3 (Disturbance classes). The disturbance classes are persistent additive productivity floors from the fit-window residual distribution: (i) the perpetual worst observed one-step shock (-329.0 kt yr⁻¹), (ii) the 5th-percentile class (-287.4 kt yr⁻¹), and (iii) the 10th-percentile class (-80.9 kt yr⁻¹). Because this object has no independent input channel (unlike the recharge series of the companion groundwater study), the disturbance classes and the defect declaration are the same measured quantity in two roles.

Definition 2.4 (Governance family). The governance family consists of the following declared policies. BAU has $C \equiv 5$ kt (moratorium-level inshore removals, the declared implementable use post-1992). Flat caps are $\rho \cdot 240$ kt with $\rho \in \{1.0, 0.75, 0.5, 0.25, 0.0\}$ (i.e. 240, 180, 120, 60, 0 kt; every member is scored). S1 is the DFO-2009 critical-zone rule (DFO, 2009) at a declared 60 kt cap (60 above the LRP, 0 below). The cascade is 60/30/5/0 kt at LRP/0.75·LRP/0.5·LRP/below; the sub-LRP stages are declared scenarios, not verified institutions. Two new families are reactive in the strict sense — their catch is a function of the current stock, switching above the LRP and vanishing at or below it. Family A is surplus-proportional: $C(S) = \phi g(S)$ for $S \geq \text{LRP}$ and 0 below, with $\phi \in \{0.25, 0.50, 0.75\}$; its catch scales with the harvested surplus $g(S) = rS(1 - S/K)$ and therefore feeds back on the stock. Family B is graded above the LRP: graded2 is 0 / LRP–1.25·LRP: 60 / >1.25·LRP: 90 kt, and graded3 is 0 / LRP–1.15·LRP: 30 / 1.15·LRP–1.35·LRP: 60 / >1.35·LRP: 90 kt. Families A and

B are post-freeze scored objects added to address the reactive-management question directly; they are not declared institutions.

Two further reactive designs were scored and are not reported as a family because they are dominated. The surplus-harvest fraction $C(S) = \alpha \cdot \max(0, g(S) - g(K^*))$ removes only the surplus above the LRP’s own regeneration, and the reactive ramp $C(S) = \min(C_{\max}, q(S - K^*))$ ramps linearly above the LRP. Under the declared source-year classes, every member of both was empty at the 5th-percentile $T = \infty$ reading (the clause-H1 failure of Result 3.1 applies to them as to every positive-catch rule) and, on the training-supply replay, harvested at or below the moratorium level (0.18–0.71 kt mean catch for the surplus-harvest fractions, 0.64 kt for the ramp; 5 kt for BAU), so neither family achieves the “harvest more than the moratorium” property that lets Family A register a genuine trade-off. They are recorded as rejected designs and omitted from the kernel tables below.

Definition 2.5 (Robust viability kernel, closed-loop reading). For each policy and disturbance class, the robust viability kernel is the set of initial SSB values from which the closed-loop path keeps the stock at or above the LRP under the persistent floor. Table 1 reports kernel lower boundaries at $T = 1$ and $T = \infty$, with “empty” denoting that no state is robustly viable. The term is used in its closed-loop reading — the robust positively invariant set of a fixed declared policy — not in the classical viability reading of Aubin (1991), which is existential over controls. No control choice enters the kernel computation; only the evaluation of declared rules under the disturbance floor enters. The word is retained because the object answers the same question (“from which states can the constraint be held”) in the policy-fixed form.

Definition 2.6 (Dominance partial order). The rule the frozen protocol calls its retention rule is restated here, clauses verbatim, as a partial order on policies rather than a filter that selects an adopted policy: a non-BAU policy *dominates* BAU only if all three of the following hold.

- (H1) Its kernel is at least as protective as BAU’s at every reading (compared on the kernel lower boundary; empty = worst).
- (H2) It improves on BAU somewhere.
- (H3) At some reading where it improves, its mean allowed catch exceeds that of every at-least-as-protective flat cap.

Why the filter vocabulary is not used. Clause (H1), read at every reading with empty scored as worst, is a structural block rather than a comparison that could go either way: under the 5th-percentile class at $T = \infty$ every declared positive-catch rule has an empty kernel while BAU’s is nonempty (2219.6 kt), so within the declared family no positive-catch rule can satisfy (H1) at all — the rule’s outcome is fixed by the declared classes and BAU’s marginal nonemptiness there. Presenting that outcome as a “retention” decision would overstate it; the vocabulary here is dominance, the outcome is stated as a selection property of the declared partial order (Result 3.1), and the two identity-level statements of the construction (the vacuous-class identity and the structural block itself) are stated as definitional notes.

Definition 2.7 (Certified layer conversion). The certified layer applies the defect-to-margin conversion in the form the fitted map admits. If the closed loop contracts with rate $a < 1$ on the safe domain, the erosion margin is $r_T = \varepsilon(1 - a^T)/(1 - a)$. Otherwise the expansive form $r_T = \varepsilon(a_{\max}^T - 1)/(a_{\max} - 1)$ applies with $a_{\max} = \sup |F'|$ over the safe domain, and the certified kernel is the nominal kernel of $K^* + r_T$.

2.3 The catch-timing convention

One modelling choice has to be declared explicitly, because it moves the reported numbers. Write the one-step residual of the governed map as

$$e_t = S_{t+1} - (S_t + g(S_t) - C_{t+\delta}), \quad \delta \in \{0, 1\},$$

so that $\delta = 0$ pairs the transition $t \rightarrow t + 1$ with the catch of the year the transition leaves (*source-year*) and $\delta = 1$ pairs it with the catch of the year it enters (*destination-year*). The two are different residual distributions on one and the same fitted map: at the committed (r, K) the source-year pool has mean -10.9 kt, SD 114.9 kt and lag-1 autocorrelation 0.554 , against -20.4 kt, 135.0 kt and 0.652 for the destination-year pool, and the classes of Definition 2.3 move with them ($-329.0/-287.4/-80.9$ kt against $-460.0/-318.8/-114.9$ kt).

Convention (adopted). Every kernel, boundary, floor, replay, figure and table of this paper is computed in the source-year convention ($\delta = 0$).

It is adopted on three grounds, none of them provenance. (i) *Coherence*. The committed parameters are the minimisers of the source-year loss. The estimation routine regresses the increment $S_{t+1} - S_t$ on the catch of year t , so $r = 0.2369$ and $K = 5000$ kt minimise the mean squared source-year residual: $12,772$ kt², against $17,873$ kt² for the destination-year pool at those same parameters (28.5% lower). Scoring those parameters against destination-year residuals would report classes drawn from a loss the fit never minimised. (ii) *Fit*. The source-year pool is the better-fitting and the less serially correlated of the two, as tabulated above. (iii) *No interior alternative*. Refitting (r, K) freely against the blended catch $wC_t + (1 - w)C_{t+1}$ gives a mean squared error monotone decreasing in w over the whole unit interval, with the optimum at the boundary $w = 1$: $17,713$ kt² at $w = 0$ (where the refit gives $r = 0.2084$), $15,026$ kt² at $w = 0.5$, and $12,772$ kt² at $w = 1$; the out-of-sample error over 2008–2015 moves the same way (834.9 to 783.1 kt², eight one-step transitions). There is no interior optimum, so a half-and-half compromise is not supported by the data.

Limitation, stated because a referee will ask. Nothing in the archived source documentation records whether the spawning-stock column of DFO (2016) Table A2 is measured before or after the calendar year’s removals, and the biology does not settle it either: Northern cod spawn in March–May, so an SSB observation sits mid-year while landings are calendar-year totals, and the interval between two SSB observations spans the back half of one year’s catch and the front half of the next. On that reading neither pure convention is mechanistically exact and the truth is a blend — which is exactly why ground (iii) is needed. The convention is therefore declared on statistical and coherence grounds rather than derived from provenance, and it is stated here rather than left implicit. The sensitivity is not suppressed: the primary kernels and their boundary tables were also computed under $\delta = 1$, and those artifacts are archived with the superseded runner rather than overwritten, so every affected number can be recomputed from the archive instead of taken on trust. Four quantities move materially — the constructive bound (91.59 kt against 57.61 kt), the number of vacuous floor classes (one against two), the certified horizon ($6/7$ years against $5/6$), and the 60-kt rules’ $T = \infty$ boundary (884.6 kt against 900.3 kt). No parameter estimate, no form comparison and no certificate direction changes.

2.4 What the kernel problem reduces to

The kernels of Section 3 are computed by exact backward recursion on the declared interval engine, but they are not merely numerical. On the increasing branch of the surplus curve — the only branch the safe set occupies here, since $K^* < K/2$ — the whole table reduces to three constants. Write g for the surplus, $|e|$ for the persistent floor, $C(S)$ for the policy’s catch and $F(S) = S + g(S) - C(S) - |e|$ for the worst-case closed loop. Every statement below is verified cell by cell against the committed kernels by `campaign_e2_structure_v3.py` (23 checks).

Proposition 2.1 (Constructive bound and the protection–supply budget). For any policy whose catch at the reference point is $C(K^*)$, the safe set is held from itself if and only if $C(K^*) \leq C^* := g(K^*) - |e|$, and the margin by which it is held is exactly $C^* - C(K^*)$.

Proof. One step from K^* gives $K^* + g(K^*) - C(K^*) - |e|$, which is at least K^* under exactly the stated inequality, and the margin is the difference of the two sides. It remains only that the reference point is the binding state, i.e. that $S \mapsto S + g(S) - C(S)$ attains its minimum over the safe set at K^* : this holds on the increasing branch for every declared policy, the graded rules included, because their catch is piecewise constant and each upward jump (30 kt at most) is smaller than the surplus gained across the jump. The campaign verifies it directly for all twelve policies. $\square$

The largest catch that holds the reference point from itself is therefore $C^* = 172.46 - 80.87 = 91.59$ kt (Section 3.3). The identity is the substantive part: at the reference point harvest and protection are one budget, and every kilotonne taken there is a kilotonne of margin surrendered. Section 3.2 shows that this, and not the tabulation, is why no declared rule dominates the moratorium.

Proposition 2.2 (Perpetual bound and the infinite-horizon boundary). For a constant catch C , the $T = \infty$ kernel is $[b_\infty, \infty)$ with $b_\infty = \max(K^*, s_-(C))$, where $s_-(C)$ is the smaller root of $g(S) = C + |e|$; it is empty if and only if $C > C_{\text{vac}} := g_{\text{max}} - |e|$.

Proof. $F(S) - S = g(S) - C - |e|$ is positive exactly between the two roots of $g(S) = C + |e|$. From any state above the smaller root the trajectory increases towards the larger and so never falls below the smaller; from any state below it the trajectory declines. If $C + |e| > g_{\text{max}}$ there is no root and the trajectory declines everywhere. $\square$

Two constants therefore organise the last column of Table 1: below $C^* = 91.59$ kt the kernel is the whole safe set; between C^* and $C_{\text{vac}} = 296.09 - 80.87 = 215.2$ kt it is a proper half-line strictly above the reference point; above C_{vac} it is empty. The column is reproduced in closed form to 0.1 kt for all six flat caps.

Proposition 2.3 (Reactive families). For $C(S) = \phi g(S)$ above the LRP and 0 below, the closed loop is $F(S) = S + (1 - \phi)g(S) - |e|$ and there are three regimes: the kernel is the whole safe set iff $\phi \leq 1 - |e|/g(K^*)$; it is a proper half-line iff that fails but $\phi < 1 - |e|/g_{\text{max}}$; and it is empty iff $\phi \geq 1 - |e|/g_{\text{max}}$.

Proof. Proposition 2.1 with $C(K^*) = \phi g(K^*)$, and Proposition 2.2 with the effective surplus $(1 - \phi)g$. $\square$

Under the declared 10th-percentile floor the two thresholds are 0.531 and 0.727. The members $\phi = 0.25$ and 0.50 hold the entire safe set — the second by 5.4 kt, not by the wide margin an earlier version of this paper reported (Section 3.4) — $\phi = 0.75$ is empty, and the band $0.531 < \phi < 0.727$ is the regime in which the rule is viable but only from above the reference point. No member of the originally declared family fell in that band, which is why the erroneous criterion went unnoticed; the family has been extended by $\phi = 0.60$ so that Table 1 exhibits it.

Proposition 2.4 (The certified horizon is a crossing). For a constant catch — the case the certified layer is computed on — let S_{hi} be the domain ceiling and $r_T = \varepsilon(a_{\text{max}}^T - 1)/(a_{\text{max}} - 1)$ the erosion margin of Definition 2.7. The certified kernel at horizon T is nonempty iff $F^T(S_{\text{hi}}) \geq K^* + r_T$, and since $F^T(S_{\text{hi}})$ is nonincreasing in T while r_T increases, the certified horizon is the unique crossing $T^* = \max\{T : F^T(S_{\text{hi}}) \geq K^* + r_T\}$. It is finite whenever the loop is expansive.

Proof. For a constant catch $F'(S) = 1 + g'(S) > 0$ throughout $[K^*, S_{\text{hi}}]$ — the surplus slope reaches -1 only at $S = \frac{K}{2}(1 + 1/r) \approx 13,000$ kt, beyond the domain — so F is increasing and the supremum over initial states of the T -step minimum is attained at the ceiling; the rest is the definition of the certified kernel. Finiteness follows because r_T grows geometrically in T while $F^T(S_{\text{hi}})$ is bounded below (it converges to the attracting fixed point when one exists, and declines otherwise). $\square$

Nothing in this subsection uses the Schaefer form. The only properties invoked are that the surplus is unimodal with its peak above the reference point and that the safe set lies on the increasing branch, so the constants travel to any surplus function with those two properties. They do: on the Fox row of Section 3.6 the formula $C^* = g(K^*) - |e|$ gives 79.1 kt against the tabulated 79.05, with $C_{\text{vac}} = 111.2$ kt and $g_{\text{max}} = 192.0$ kt; on the two Allee rows it gives 123.27

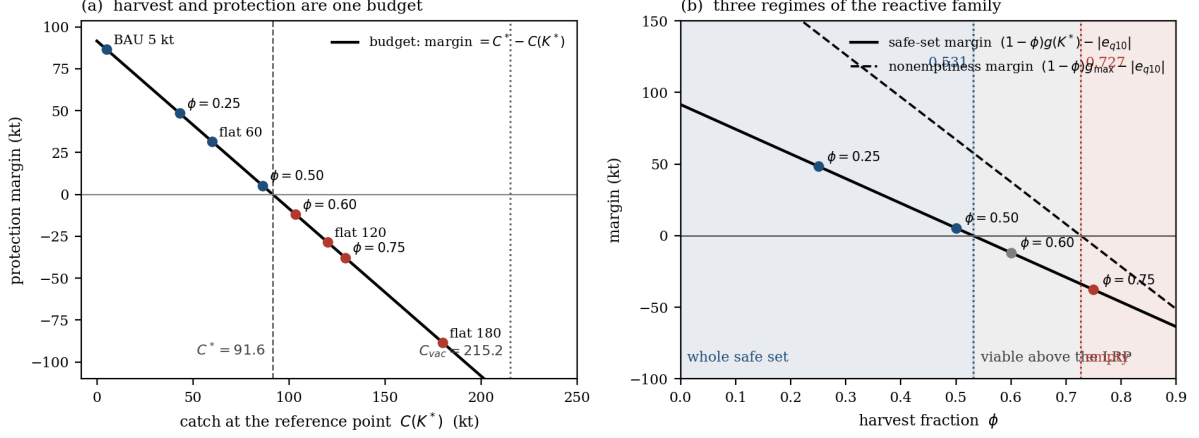


Figure 1: The two structural results of Section 2.4. (a) The protection–supply budget of Proposition 2.1: every policy lies on the line, so a rule’s harvest at the reference point and the margin by which it holds that point sum to $C^* = 91.59$ kt; the flat 60-kt cap, BAU and $\phi \leq 0.50$ sit on the protective segment, the 120- and 180-kt caps and $\phi = 0.75$ below it, and the dotted line at $C_{\text{vac}} = 215.2$ kt is where the kernel becomes empty. (b) The three regimes of Proposition 2.3: the safe-set margin $(1 - \phi)g(K^*) - |e|$, zero at $\phi = 0.531$, and the nonemptiness margin $(1 - \phi)g_{\text{max}} - |e|$, zero at $\phi = 0.727$, with the four declared members marked.

kt (declared s_0) and 115.19 kt (data-preferred), both reproducing the tabulated constructives exactly. The Schaefer numbers of Section 3.3 are one instance of a statement that holds across the forms the paper fits.

The crossing reproduces the computed horizons exactly — $T^* = 6$ under the perpetual-worst and 5th-percentile floors and $T^* = 7$ under the informative one, for both BAU and zero catch (Section 3.4). Where the loop has an attracting fixed point $S^\dagger$ the crossing is bounded below by $\lfloor \log(1 + (S^\dagger - K^*)(a_{\text{max}} - 1)/\varepsilon) / \log a_{\text{max}} \rfloor$, which is attained once the ceiling trajectory has converged: it gives 7 against 7 under the informative floor, and 4 against 6 under the 5th-percentile floor, where at six years the ceiling trajectory is still 1,500 kt above the attractor.

3. Results

3.1 Robust kernels (nominal)

Table 1. Lower boundaries of the robust viability kernels of the LRP (kt) under the declared catch-policy family (frozen rules plus the two post-freeze reactive families) and disturbance classes, in the source-year convention. “empty” denotes that no state is robustly viable at that horizon. The infinite-horizon emptiness of Section 3.2 is recorded in the table, not only in prose.

Schaefer, M.B., 1954. Some aspects of the dynamics of populations important to the management of the commercial marine fisheries. <i>Bulletin of the Inter- American Tropical Tuna Commission,</i> 1(2), 27–56. Reprinted in <i>Bulletin of Mathematical Biology,</i> 53, 253–279, 1991, doi:10.1007/bf02464432.						
Policy	Worst, $T = 1$	$T = \infty$	q05, $T = 1$	$T = \infty$	q10, $T = 1$	$T = \infty$
BAU (5 kt)	1025.5	empty	989.0	2219.6	884.6	884.6
flat 240 kt	1233.5	empty	1196.4	empty	1014.0	empty
flat 180 kt	1180.0	empty	1143.1	empty	961.5	1637.8
flat 120 kt	1126.8	empty	1090.1	empty	909.3	1082.3
flat 60 kt	1073.8	empty	1037.2	empty	884.6	884.6
S1 / cascade	1073.8	empty	1037.2	empty	884.6	884.6
flat 0 kt	1021.1	empty	984.7	2070.9	884.6	884.6
Family A, $\phi=0.25$	1064.7	empty	1027.3	empty	884.6	884.6
Family A, $\phi=0.50$	1111.3	empty	1071.7	empty	884.6	884.6
Family A, $\phi=0.60$	1131.2	empty	1091.0	empty	895.2	1074.8
Family A, $\phi=0.75$	1161.0	empty	1119.9	empty	920.2	empty
Family B, graded2	1073.8	empty	1037.2	empty	884.6	884.6
Family B, graded3	1073.8	empty	1010.9	empty	884.6	884.6

Under the 10th-percentile class the moratorium, zero catch, the flat 60-kt cap, the critical-zone rule, the cascade, both graded rules, and the surplus-proportional family at $\phi \leq 0.50$ all hold the entire safe set $[884.6, 10^4]$ kt at every horizon. Every higher nonzero cap lifts the kernel’s lower edge above the LRP, and the column is not a list of accidents: by Proposition 2.2 the boundary is $\max(K^*, s_-(C))$ with $s_-(C)$ the smaller root of $g(S) = C + |e_{q10}|$, so the catches divide at $C^* = 91.59$ kt (whole safe set), $C_{\text{vac}} = 215.2$ kt (nonempty) and beyond it

(empty) — which is exactly what the last column shows, to 0.1 kt, for all six flat caps. Under the perpetual-worst floor the infinite-horizon kernel is empty for every policy, because the maximum surplus $g_{\max} = rK/4 = 296 \text{ kt yr}^{-1}$ lies below that floor. Under the 5th-percentile class the infinite-horizon kernel is nonempty for the moratorium (2219.6 kt) and zero catch (2070.9 kt) but empty for every positive-cap rule.

The critical-floor axis makes the qualification explicit. At zero catch, $\bar{e} = g_{\max} = 296.1 \text{ kt yr}^{-1}$ separates vacuous from informative classes. Only the perpetual-worst floor (-329.0 kt) sits beyond it and is vacuous. The 5th-percentile (-287.4 kt) and 10th-percentile (-80.9 kt) floors both lie on the informative side, which is why the constructive boundary of Section 3.3 exists for the 10th-percentile class and the 5th-percentile class now carries informative, non-vacuous content.

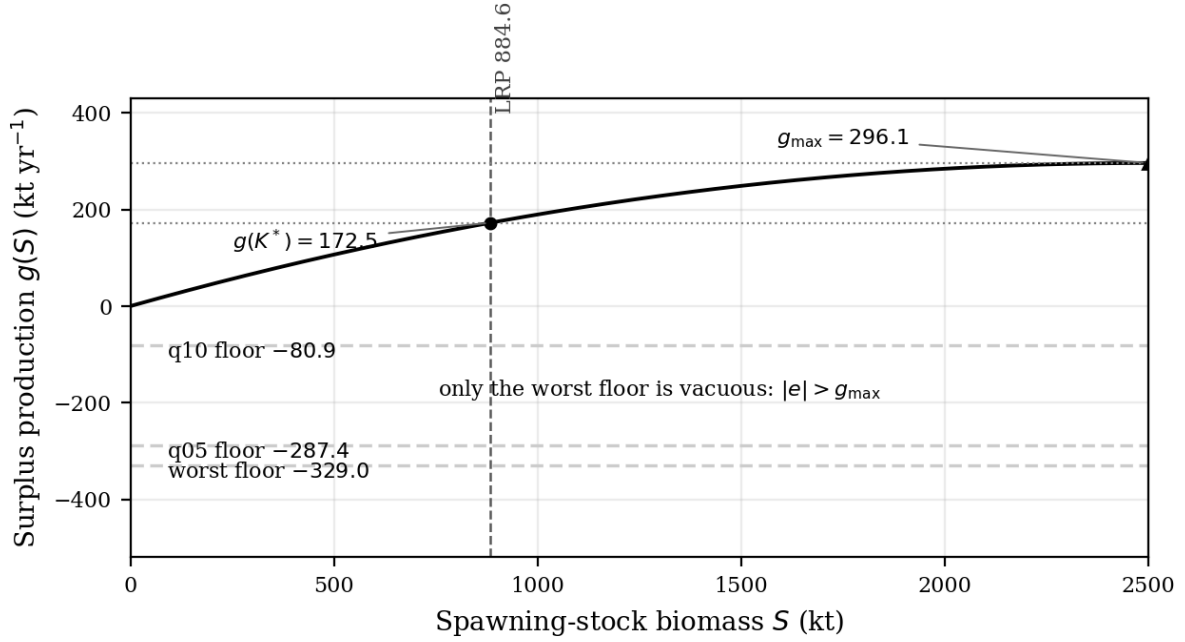


Figure 2: Surplus production of the registered fit with the three persistent floors; only the perpetual-worst floor sits beyond $g_{\max} = 296.1 \text{ kt yr}^{-1}$, so only that floor’s emptiness is vacuous.

3.2 The no-dominance verdict and the vacuous-class identity

The first result concerns dominance; the second is the vacuous-class identity, stated as a definitional note. Both are scoped to the declared rule and disturbance classes.

Result 3.1 (No dominance). Under the dominance partial order of Definition 2.6 (the frozen rule; clauses verbatim), no non-BAU policy dominates BAU.

Reason. The binding failure is clause (H1), read at its stated scope: “at every reading,” with empty scored as worst. Under the 5th-percentile class at $T = \infty$ every positive-catch rule — the flat caps, the critical-zone rule, the cascade, both graded members of Family B, and every member of the surplus-proportional Family A — has an empty kernel, while BAU’s kernel is nonempty there (2219.6 kt). Because empty is worst, none of them is at least as protective as BAU at that reading, so clause (H1) fails for every positive-catch rule. The informative 10th-percentile class is not where the verdict is decided: under that class the flat 60-kt cap, the critical-zone rule, the cascade, and both graded rules hold the LRP from itself at every horizon (884.6 kt at $T = 1$ and $T = \infty$), exactly matching BAU, and the surplus-proportional family at $\phi \leq 0.50$ holds it too while harvesting more than the moratorium — so the informative class is where the reactive family *almost* passes, not where it fails. Clause (H3) is therefore a second,

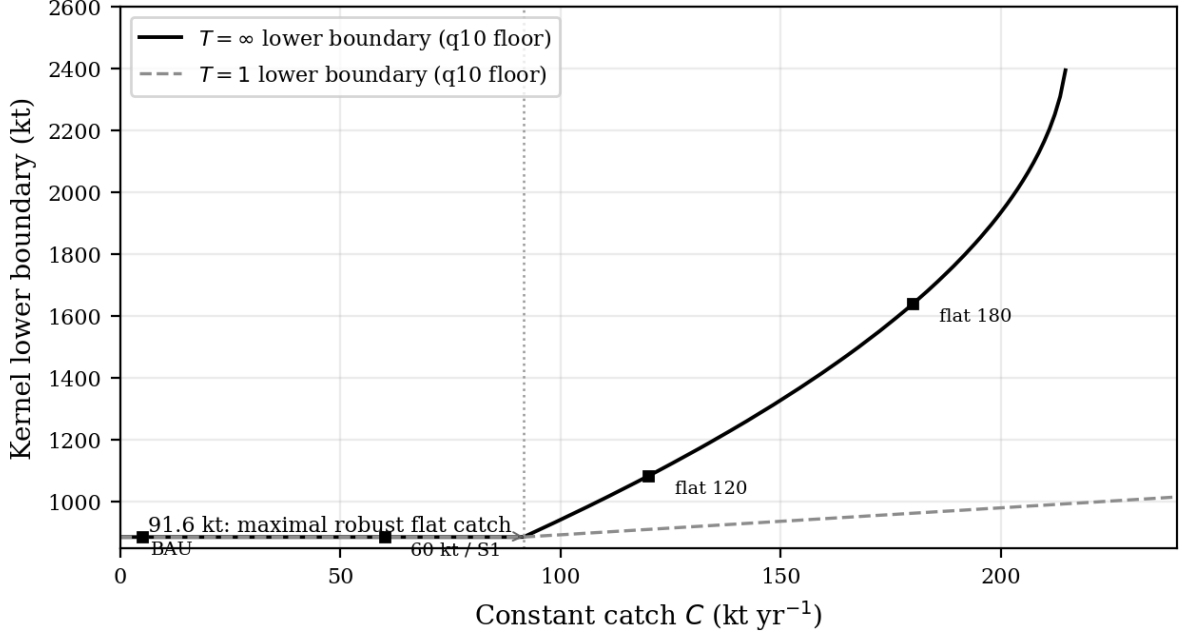


Figure 3: Kernel lower boundary versus constant catch under the 10th-percentile class: the $T = \infty$ boundary leaves the LRP at the constructive 91.59 kt.

independent failure rather than the mechanism: the equal-protective flat-60 cap (mean allowed catch 60 kt) exceeds the training means of the critical-zone rule (10.0 kt), the cascade (16.3 kt), graded2 (10.0 kt), and graded3 (5.0 kt). At $\phi = 0.75$ Family A also fails on the informative class itself, its $T = \infty$ kernel empty (920.2 kt at $T = 1$ against 884.6; see Result 3.4). No non-BAU policy therefore dominates BAU, and the mechanism is the empty-at-the-5th-percentile-class reading of clause (H1), with the supply comparison as a second failure. The verdict’s status is selection-theoretic, not empirical: given the declared classes, clause (H1) blocks every positive-catch rule by construction (Definition 2.6’s note), so the no-dominance outcome is a property of the declared partial order and classes as much as of the policies. The companion groundwater evaluation operates under a different constraint structure — not a single fatal floor — so its retained reactive rules and this no-dominance outcome are not comparable selections. $\square$

Which governance architecture justifies its additional structure is system-dependent. The framework’s deliverable is the scored comparison, not a universal architecture verdict.

Definitional note 3.2 (The vacuous-class identity). Under the perpetual-worst floor, no catch policy — zero catch included — holds the LRP. This statement is a definitional note rather than a numbered result: it is an arithmetic identity of the declared class, not an empirical finding, and it is not numbered on a par with the constructive boundary or the stochastic layer. Under the 5th-percentile and 10th-percentile classes the statement is not vacuous: those floors lie below the map’s maximum surplus and the classes carry informative content.

Statement. The perpetual-worst floor (-329.0 kt yr $^{-1}$) exceeds the map’s maximum surplus ($g_{\max} = 296$ kt yr $^{-1}$). Every trajectory therefore declines for every catch, zero included: $|e| > g_{\max}$ implies monotone decline for every $C \geq 0$ — an identity of the map’s algebra, not a measurement. The 5th-percentile class (-287.4 kt yr $^{-1}$) and the 10th-percentile class (-80.9 kt yr $^{-1}$) both lie below $g_{\max}$ and are not vacuous: their kernels carry substantive content (Table 1, Result 3.3), and the 5th-percentile moratorium kernel is nonempty (2219.6 kt). The correction of the residual convention therefore reduces the vacuous family from two classes to one. The analogy to the companion groundwater institutional certificate is one of form only; that certificate is institutional, this one is a floor-above-surplus identity, and the two are not pooled. $\square$

On this map the reference point is protected by good years rather than by demand management — but under the corrected class the margin that good years must supply is smaller than the frozen reading implied.

3.3 Constructive boundary

The maximal robust flat catch is the largest constant catch whose worst-case low equilibrium stays at or above the LRP. Under the 10th-percentile class this constructive boundary is **91.59 kt**.

Result 3.3 (Constructive boundary). Under the 10th-percentile class, the maximal robust constant catch is 91.59 kt yr⁻¹. Under the 5th-percentile and perpetual-worst classes no positive catch is robust.

Reason. The value is $g(K^*) - |e_{q10}| = 172.46 - 80.87 = 91.59$ kt yr⁻¹. It is independent of the family scaling and rests only on the identified r and the corrected 10th-percentile floor. Under the 5th-percentile floor the corresponding quantity is $172.46 - 287.36 = -114.9$ kt, so no constant catch is robust there; under the perpetual-worst floor the same holds ($172.46 - 328.97 = -156.5$ kt). $\square$

Proposition 2.1 gives the bound a second reading: the margin by which any policy holds the reference point is exactly 91.59 kt minus its catch there, so the bound is the whole budget and harvest and protection are the same quantity seen from two sides. This is certification geometry at one declared shock class, not a harvest rule. The stochastic analogue of the bound is reported in Section 3.8, and its sampling distribution in Section 3.10. The corrected bound is read as order 70–90 kt, not the 40–60 kt of the frozen convention.

Result 3.4 (Reactive rules above the LRP). The surplus-proportional family is genuinely reactive: at $\phi = 0.25$ and 0.50 it holds the entire safe set under the 10th-percentile class at every horizon, including $T = \infty$, and its catch feeds back on the stock (it rises with the surplus and vanishes below the LRP). It is nonetheless empty at the 5th-percentile $T = \infty$ reading, so its protection is a property of the informative class only — a trade-off, not a strict improvement over BAU, as set out in Result 3.1. The graded family is protective on the informative class but not supply-superior. The two families are compared against the frozen rules in Table 1.

Reason. For a surplus-proportional rule $C = \phi g(S)$ the closed loop is $F(S) = S + (1 - \phi)g(S) + e$, a single concave quadratic, and Proposition 2.3 gives three regimes rather than two. The safe set is held from itself iff the scaled surplus at the reference point covers the floor, $(1 - \phi)g(K^*) \geq |e_{q10}|$, i.e. $\phi \leq 1 - 80.87/172.46 = 0.531$; at $\phi = 0.25$ (0.50) the margin is 48.5 (5.4) kt, so both qualify, the second only just. Nonemptiness is the weaker condition $(1 - \phi)g_{\max} > |e_{q10}|$, i.e. $\phi < 1 - 80.87/296.09 = 0.727$, and it fails at $\phi = 0.75$ (margin -37.8 kt), so that member's $T = \infty$ kernel is empty (920.2 kt at $T = 1$). Between the two thresholds the rule is viable but only from above the reference point: at the added member $\phi = 0.60$ the informative kernel is $[1074.8, 10^4]$ kt, not the safe set, and it is this regime that an earlier version of this paper missed by quoting 0.727 as the threshold for holding the safe set. The three tabulated verdicts were unaffected, which is why the error survived every numeric check. Under the 5th-percentile floor the criterion is $\phi < 0.029$, so no surplus-proportional rule with positive ϕ holds that class — which is exactly why the same family is empty there, and why it does not dominate BAU under the frozen rule. Under the perpetual-worst floor the loop declines for every ϕ , so the kernel is empty. The graded members (Family B) are piecewise-constant and are scored exactly as the frozen rules: under the 10th-percentile class both hold the LRP from itself at every horizon, and both are empty at the 5th-percentile $T = \infty$ reading. The rejected-designs note following Definition 2.4 reports the same clause-H1 emptiness for the surplus-harvest and ramp designs, which moreover do not improve supply. Supply replays (training mean allowed catch over the observed states) give BAU 5 kt, S1 10.0 kt, the cascade 16.3 kt, flat-60 60 kt, flat-0 0 kt, graded2 10.0 kt, graded3 5.0 kt, and the surplus-proportional family 7.4/14.7/22.1 kt at $\phi = 0.25/0.50/0.75$. The critical-zone rule's low mean is a property of the window, not of

the rule: the cut is active in 20 of the 24 replay years (the stock sat below the reference point in all but the 1985 and 1987–1989 states, 83% of the window), a fact about the collapsed-era estimation window rather than about the rule’s post-recovery supply properties, and the two regimes are not mixed. $\square$

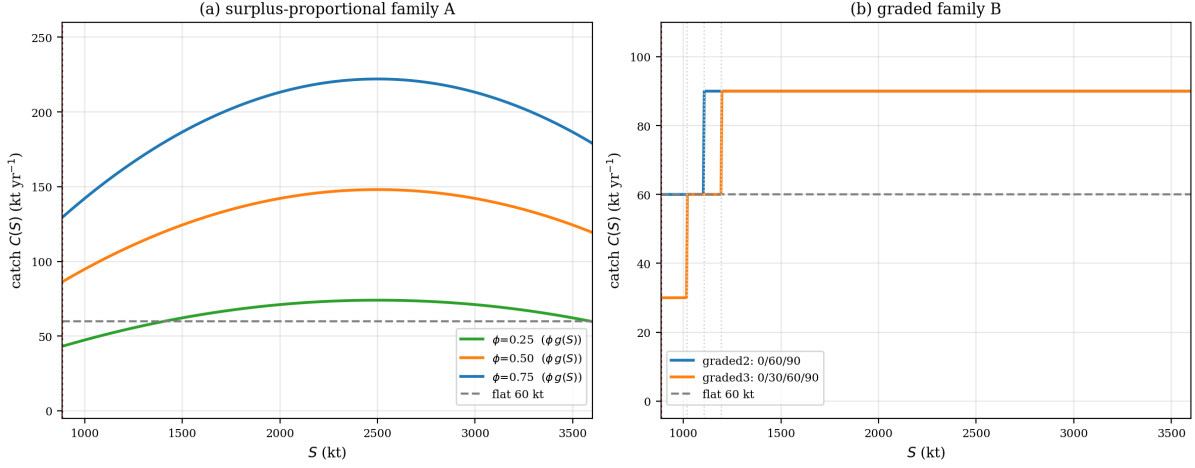


Figure 4: The two reactive families: (left) the catch schedules of the surplus-proportional Family A ($C = \phi g(S)$, $\phi = 0.25, 0.50, 0.75$) against the flat 60-kt cap, showing that the rule harvests less than the cap near the LRP while scaling with the surplus; (right) the catch schedules of the graded Family B (graded2 and graded3) against the flat 60-kt cap. Both families vanish at or below the LRP. The dominated surplus-harvest ($\alpha \cdot \max(0, g(S) - g(K^*))$) and reactive-ramp designs are recorded in the rejected-designs note following Definition 2.4 and not plotted.

3.4 Certified layer: the expansion obstruction

The conversion of Definition 2.7 needs the closed loop’s contraction rate. The next result shows that no such contraction is available at the declared safe set, and that the certified layer therefore has a finite horizon: six years under the two harsher floors, seven under the informative one.

Result 3.5 (Expansion obstruction). On the governed surplus-production map of Definition 2.1 the certified kernel is empty from $T = 7$ under the perpetual-worst and 5th-percentile floors, and from $T = 8$ under the 10th-percentile floor, for every declared policy, zero catch included. At the last nonempty horizon the certified sets lie far above the observed range of the stock: $[5132.9, 10^4]$ kt (perpetual worst, BAU, $T = 6$), $[4593.2, 10^4]$ kt (5th percentile, BAU, $T = 6$), and $[4560.3, 10^4]$ kt (10th percentile, BAU and zero catch, $T = 7$); under the informative floor the 180- and 240-kt caps are already empty at $T = 7$.

Reason. Here $F'(S) = 1 + r(1 - 2S/K)$ is increasing as S falls. At the LRP, $F'(K^*) = 1.153 > 1$. The governed surplus map is therefore expansive at the declared safe set, contracting only above $K/2 = 2500$ kt — a stock level the series never approaches. The contraction form of the conversion is therefore inapplicable. The expansive form $r_T = \varepsilon(a_{\max}^T - 1)/(a_{\max} - 1)$ grows without bound; with the committed $\varepsilon = 328.9725$ kt — the perpetual-worst class of Section 2.3, printed there to one decimal as 329.0 — and $a_{\max} = 1.1531$: $r_1 = 329.0$, $r_2 = 708.3$, $r_3 = 1145.7$, $r_4 = 1650.1$, $r_5 = 2231.7$, $r_6 = 2902.4$, $r_7 = 3675.7$, $r_8 = 4567.4$ kt, so the shifted threshold $K^* + r_T$ runs 1213.6, 1592.9, 2030.3, 2534.7, 3116.3, 3787.0, 4560.3, 5452.0 kt at $T = 1, \dots, 8$.

Emptiness is not simply the shifted threshold outgrowing the carrying capacity: at $T = 7$ the threshold (4560.3 kt) is still below $K = 5000$ kt, and the certified kernel is empty under the two harsher floors nevertheless. What fails is the trajectory, not the arithmetic of the shift. Under the perpetual-worst floor the BAU closed loop has no worst-case fixed point at all, so every trajectory declines; started from the domain ceiling of 10^4 kt it reaches 4689.6 kt after five steps and 4424.5 kt after six, which clears the six-year threshold (3787.0 kt) but not the

seven-year one (4560.3 kt). Under the informative 10th-percentile floor the loop does have an attracting worst-case fixed point (4606.5 kt for BAU, above the seven-year threshold and below the eight-year one), which is why the horizon is one year longer there. The certified sets at those horizons lie above the entire observed range of the stock. $\square$

Proposition 2.4 names the mechanism precisely: the certified horizon is the crossing at which the geometrically growing margin $K^* + r_T$ overtakes the worst-case trajectory from the domain ceiling, and it reproduces the computed horizons exactly ($T^* = 6, 6$ and 7). On this object the binding obstruction to certified intervention claims is the expansion rate itself, not the defect magnitude. This is a failure mode qualitatively different from the companion groundwater object, where the governed map is contracting and the certified horizon is defect-bound to $T \leq 3$ years. The source-year convention lengthens the horizon by one year at every class ($T = 5 \rightarrow 6$ under the two harsher floors, $T = 6 \rightarrow 7$ under the informative floor), because it replaces the registered defect magnitude ($\varepsilon = 460.0$ kt) by the source-year one (329.0 kt) while leaving $a_{\max}$ unchanged.

The horizon is not a property of the policy, and that is the second thing the crossing shows. Table 2 gives T^* against the constant catch. It is 6 years under the perpetual-worst and 5th-percentile floors and 7 under the informative one for every catch from zero to 150 kt — the moratorium included — and for every declared reactive and graded rule as well; the crossing agrees with the committed kernel computation at all 48 (rule, class) pairs the campaign tries — 30 constant catches and 18 declared rules. Only catches outside the constructive bound shorten it: 6 years under the informative floor from 180 kt and 5 under the perpetual-worst floor from 200 kt. No declared policy lengthens it. The reason is the one the formula exposes: the margin’s year-on-year increments are 379, 437, 504, 582, 671 and 773 kt, so every one of them exceeds the entire admissible catch range $[0, 91.59]$ kt — the smallest by a factor of four — and no reduction inside that range can add a certified year. Certified years cannot be bought with conservation; the horizon is set by the expansion rate at the reference point and by nothing the manager chooses.

Table 2. The certified horizon T^* by constant catch, under each declared floor class.

Constant catch (kt)	perpetual-worst	5th pct.	10th pct.
0 (moratorium)	6	6	7
5 (BAU)	6	6	7
60 (flat cap)	6	6	7
91.59 (C^*)	6	6	7
120	6	6	7
150	6	6	7
180	6	6	6
200	5	6	6
215.2 (C_{vac})	5	6	6

3.5 Stress replay and classification

The stress replay runs the closed loop from the observed 1990 SSB (861.9 kt — already below the LRP, so the replay starts outside the safe set and is uncontrolled shock accounting rather than a kernel-membership test) with the observed 1991–1995 source-year residuals. In 1991 every policy holds the stock above the LRP: flat-0 and S1 reach 953.9 kt (the S1 switch rule observes the sub-LRP 1990 stock and prescribes zero catch, so it coincides with flat-0 in that year), business-as-usual 948.9 kt, the cascade 923.9 kt, and flat-60 893.9 kt. Yet by 1992 every policy has fallen below the LRP — flat-60 to 678.9 kt, S1 to 747.8 kt, the cascade to 713.4 kt, and BAU to 797.1 kt — because the 1992 transition carries the perpetual-worst observed shock (−329.0 kt). The sharpest drawdown is under flat-60 (to 434.3 kt in 1993), while the

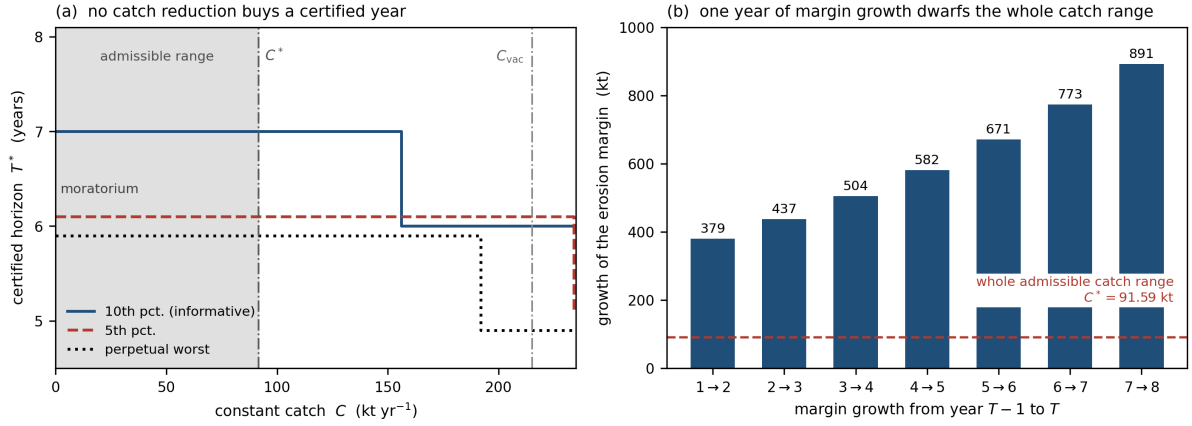


Figure 5: The certified horizon is not a property of the policy. (a) The exact crossing of Proposition 2.4 against the constant catch, under each declared floor class, at 1-kt resolution: the horizon is 6 years under the perpetual-worst and 5th-percentile floors and 7 under the informative one for every catch in the shaded admissible range, a moratorium included, and it shortens only outside the constructive bound ($C_{\text{vac}} = 215.2$ kt). Every declared reactive and graded rule lies on the same plateau. Grey band: the admissible range $[0, C^*]$. (b) Why: the erosion margin grows by 379–773 kt a year, so a single year’s growth exceeds the whole admissible catch range ($C^* = 91.59$ kt, dashed) by a factor of four at its smallest. Bars are the growth from year $T - 1$ to year T , computed from the same closed form as (a); the horizon curve reproduces all 30 archived (catch, floor) pairs exactly.

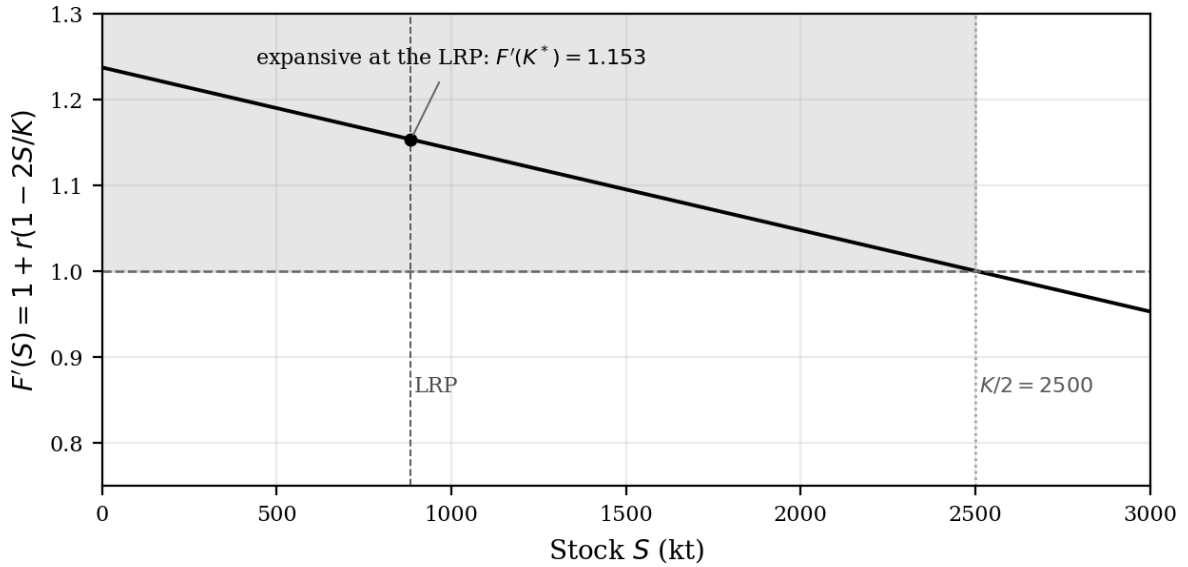


Figure 6: The closed loop’s slope $F'(S)$; the map is expansive at the LRP and contracts only above $K/2 = 2500$ kt. The expansion classification is not an artifact of the particular pinned value: the grid of Section 3.7 shows $F' \geq 1.000$ at every admissible $K \geq 2K^* = 1769.2$ kt. The classification is conditional on $K \geq 2K^*$: contraction ($F' = 0.61\text{--}0.93$) is restored only below $2K^*$ — exactly where the informative certificates of Sections 3.2–3.3 collapse and the fit cost rises — and the residual bootstrap of Section 3.10 gives F' a 90% interval of $[1.010, 1.179]$.

switch rules lose less (BAU 669.7 kt in 1994); no policy avoids the post-1992 decline. The crash is a productivity event, not a catch event — exactly the catch-insufficiency certificate of the companion forecast-evaluation study.

At the $T = 5$ classification under the 10th-percentile class, only the 1980s peak years of the 33 observed states lie inside the nominal kernels — {1985, 1987, 1989} for the 60-kt rules and additionally 1988 for BAU. The entire post-1990 history and most of the 1980s are outside. Under the perpetual-worst class all 33 are outside; under the 5th-percentile class the moratorium’s kernel is nonempty (its $T = \infty$ boundary is 2219.6 kt, above the observed range), so the classification there follows the informative geometry rather than universal emptiness.

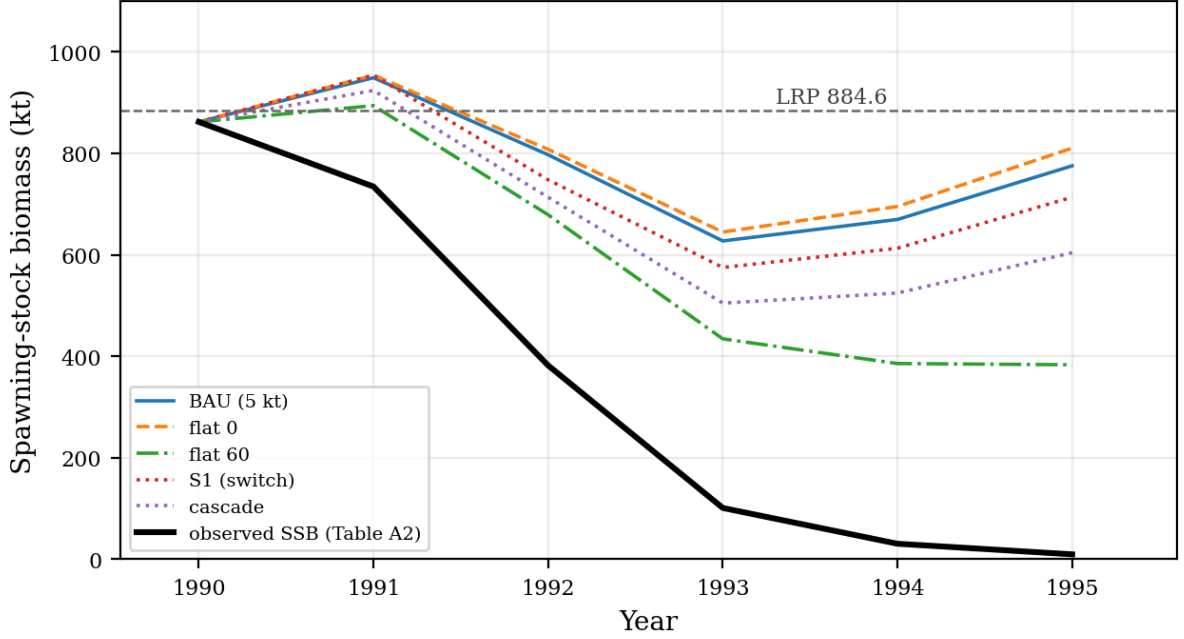


Figure 7: Closed-loop replay from the observed 1990 stock with the observed residuals; BAU, flat-0, flat-60, the S1 switch rule, and the cascade are plotted separately (the S1 rule differs from the flat-60 cap below the LRP).

3.6 Model-form comparison: the depensatory refit and the Fox form as co-equal specifications

The primary kernels are Schaefer-form (Allee term off). The depensation sensitivity refits the same object with the Allee term on, on the same 1983–2007 window with the same annual landings. The refit gives $r = 2.0$ (pinned at its optimization bound, the mirror image of the registered K), $K = 1671.7$ kt, and $s_0 = 642.3$ kt — 242 kt below the LRP, inside the observed range — with residual MSE 7690.1 kt² against 12,772.2 kt² for the registered Schaefer form. The bare fit parameters and the MSE are convention-independent (the fit already pairs the year- t catch with the $t \rightarrow t + 1$ transition), so they are unchanged from the frozen reading. Only the fitted form’s own floor classes and its kernels depend on the residual convention, and those are reported here in the source-year convention. The depensatory form is the better-fitting of the two, under identification caveats of the same kind as the registered fit.

The two forms are reported side by side as co-equal primary specifications in Table 3. The identification caveats apply symmetrically: the Schaefer form pins K at its bound, the depensatory form pins r at its bound (2.0), and neither pin is a biological finding. Table 3 reports the rows at the declared class endpoints, held frozen across forms as the corrected source-year classes (−287.4, −80.9, −329.0 kt) so that only the form varies.

Result 3.6 (Form sensitivity). The certificate directions survive the data-preferred depensatory refit. The Allee form removes the vacuity of the perpetual-worst floor — the one class vacuous on the registered form — because its maximum surplus (372.4 kt) exceeds both class magnitudes (-329.0 and -287.4 kt yr $^{-1}$).

Reason. Under the 10th-percentile class the BAU and zero-catch certificates are exactly unchanged — lower boundary 884.6 kt at every horizon, $T = \infty$ included — and the critical-zone and cascade rules are protective of the entire safe set at that class (884.6 kt). The constructive boundary of Section 3.3 and the dominance verdict of Section 3.2 are therefore unaffected — indeed the constructive bound rises, since at the declared 10th-percentile floor the Allee form gives $g(K^*) - |e_{q10}| = 196.06 - 80.87 = 115.2$ kt (123.3 kt on the declared-strength row), so the 91.59 kt certificate holds a fortiori. The clause-(H1) block itself changes shape without changing the verdict: under the Allee form the 60-kt rules are no longer empty at the 5th-percentile $T = \infty$ reading (their boundary is 1131.1 kt against BAU’s 1020.9 kt), so the block operates through a margin rather than through emptiness at that reading. Under the Allee form, whose maximum surplus is 372.4 kt, both the perpetual-worst floor (-329.0 kt) and the 5th-percentile floor (-287.4 kt) lie below it, so the BAU kernel is nonempty at those classes with infinite-horizon boundaries 1098.7 and 1020.9 kt respectively (against the empty kernels of the registered Schaefer form under the perpetual-worst floor). The expansion obstruction is not a Schaefer-form artifact: the Allee refit raises the expansion rate at the LRP from 1.1531 to 1.7818. A declared-strength alternative ($s_0 = 0.5K^* = 442.3$ kt) refits to $r = 2.0$, $K = 3223.7$ kt (MSE 9330.0 kt 2), with maximum surplus 883.6 kt; under the source-year classes its kernels are nonempty at every reported cell and lie within 12 kt (about 1%) of the identified row’s at all of them, so the depensation row does not hinge on the identification of s_0 .

Reproducibility note (two declared sensitivities). First, the manuscript’s $s_0 = 642.3$ kt is a rounding of the fitted 642.3296 kt, and one printed cell moves by a single grid step between them: the BAU worst-class $T = \infty$ boundary is 1098.75 kt at the rounded s_0 and 1098.8 kt at the fitted one (the q05 $T = \infty$ boundary moves $1020.9 \rightarrow 1020.95$ kt, which prints identically). Every other cell is unchanged, and the published cells are those of the rounded s_0 . Second, the Allee and Fox rows are computed on a 0.05 kt state grid, while the registered row comes from the committed interval-arithmetic engine; the campaign reproduces the registered row on that same grid, so the two engines can be read against each other. At finite horizons they agree to within about one grid step — of the 81 cells the campaign checks, the 72 finite-horizon ones differ by at most 0.06 kt — but adjacent to the repelling boundary, where $F' > 1$, the grid’s rounding compounds along the orbit: at $T = \infty$ five of the nine cells agree exactly and the other four differ by 0.1, 0.3, 0.6 and 0.9 kt, the largest being the BAU q05 boundary, 2218.75 against 2219.649 kt (and 2070.30 against 2070.884 kt for zero catch). The two engines are therefore not to be compared at 0.1 kt resolution, and the tolerance the campaign declares is 1.0 kt.

The third co-equal form is the Fox surplus law $g(S) = rS \ln(K/S)$, refitted one-step on the same window with the same landings and the same declared box. The refit gives $r = 0.1044$, $K = 5000$ kt pinned at its bound, residual MSE 13,873.1 kt 2 (Schaefer 12,772.2; Allee 7690.1). The Fox form is the worst-fitting of the three, and the Allee form remains the data-preferred one. The certificate directions survive the form. The expansion at the LRP is preserved ($F' = 1.0764 > 1$, weaker than the registered 1.1531). The constructive 10th-percentile bound is $g(K^*) - |e_{q10}| = 159.92 - 80.87 = 79.05$ kt, under the source-year 10th-percentile floor (-80.87 kt); the Fox form’s own refit-boundary floor is not used so that only the form varies across Table 3. The BAU and zero-catch certificates under the informative class are exactly unchanged (lower boundary 884.6 kt at every horizon, $T = \infty$ included). Three further Fox cells are computed by the campaign but not carried in Table 3: the 60-kt rule’s infinite-horizon lower boundary is empty under the 5th-percentile floor, and the 120-kt rule is empty under the 5th-percentile and perpetual-worst floors and, under the informative floor, at $T = \infty$ (its $T = 1$ boundary there is 922.7 kt). Table 3 carries the Fox row.

Table 3. Depensation and form sensitivity: kernel lower boundaries (kt) at the declared source-year class endpoints.

Form	BAU, q05 $T=1$	BAU, q05 $T=\infty$	BAU, q10 $T=\infty$	BAU, worst $T=\infty$	S1, q10 $T=\infty$
Registered Schaefer	989.0	2219.6	884.6	empty	884.6
Allee refit ($s_0 = 642.3$)	939.4	1020.9	884.6	1098.7	884.6
Declared $s_0 = 0.5K^*$ (442.3), refit	939.2	1024.1	884.6	1086.9	884.6
Fox refit ($r = 0.1044$, K pinned)	1008.4	empty	884.6	empty	884.6

3.7 Carrying-capacity sensitivity

The registered fit pins K at its optimization bound, and the expansion classification of Section 3.4 inherits the question of how much of the result is carried by that pin. The sensitivity refits r by one-step least squares at each K on the same window, in the source-year convention, with the kernel scored against each refit’s own source-year 10th-percentile floor.

Table 4. Carrying-capacity sensitivity in the source-year convention: r refit at fixed K , kernel scored against that refit’s own corrected 10th-percentile floor. Values are the BAU lower boundary of the robust viability kernel at $T = 1$ and $T = \infty$.

K (kt)	r	$g_{\max}$	$F'(K^*)$	Constr. bound (kt)	BAU q10, $T = 1$	BAU q10, $T = \infty$	q05 vacuous
1000	0.5094	127.4	0.608	−28.87	943.2	empty	yes
1200	0.5248	157.4	0.751	41.14	884.6	884.6	yes
1500	0.4099	153.7	0.926	67.9	884.6	884.6	yes
1769.2 ($= 2K^*$)	0.3559	157.4	1.000	76.55	884.6	884.6	yes
2000	0.3273	163.6	1.038	80.6	884.6	884.6	yes
2500	0.2908	181.7	1.085	85.33	884.6	884.6	yes
3000	0.2704	202.8	1.111	87.79	884.6	884.6	yes
4000	0.2485	248.5	1.139	90.31	884.6	884.6	yes
5000 (registered)	0.2369	296.1	1.153	91.59	884.6	884.6	no
7000 (out of box)	0.2248	393.5	1.168	92.89	884.6	884.6	no

Result 3.7 (Carrying-capacity grid: three readings). The grid of Table 4 supports three readings.

- (i) The expansion obstruction holds for every admissible $K \geq 2K^* = 1769.2$ kt and is the data-selected regime within that range — not an artifact of the particular pinned value $K = 5000$ — while remaining conditional on $K \geq 2K^*$: $F' \geq 1.000$ at every $K \geq 2K^*$ — exactly 1.0000 at $K = 2K^*$, rising to 1.1531 at the registered K — and the informative constructive bound rises monotonically toward the registered end of the box (−28.87 kt at $K = 1000$ kt to 91.59 kt at $K = 5000$ kt), with the fit cost rising as K falls below $2K^*$.

- (ii) Contraction is restored only below $2K^*$ ($F' = 0.61\text{--}0.93$), exactly where the informative certificates collapse: the constructive bound is negative at $K = 1000$ kt, and the BAU kernel under the 10th-percentile class is empty at $T = \infty$ with the $T = 1$ boundary raised to 943.2 kt — the moratorium itself no longer holds the reference point.
- (iii) The vacuity structure is K -dependent in the source-year convention. The perpetual-worst floor is vacuous at every in-box K , but the 5th-percentile class is vacuous only at $K \leq 4000$ kt and becomes informative at $K \geq 5000$ kt ($g_{\max} = 296.1$ kt at the registered K), which is itself the registered point.

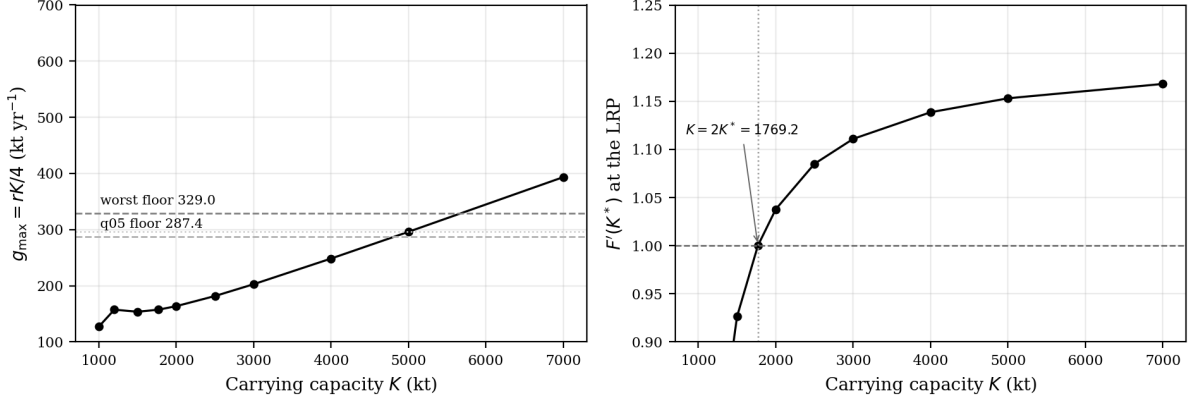


Figure 8: The two panels of the carrying-capacity sensitivity of Section 3.7.

How much of that is carried by the pin? The question can be answered without resolving it. Profiling K — refitting r in closed form at each fixed K on the same window, with the grid extended to 50,000 kt precisely because the committed value sits on the box — gives a profile-likelihood 95% set running from 1500 kt to the top of the grid: the series never approaches carrying capacity, so K is not identified from above at all. The pin itself buys little: the sum of squares at $K = 5000$ kt is 1.027 times the profiled minimum. What is identified is everything the results depend on. Across the profile set $g(K^*)$ lies in $[148.8, 176.1]$ kt and C^* in $[67.9, 95.2]$ kt, so the constructive bound is pinned to within 15% by data that do not determine K at all; and $F'(K^*) > 1$ for every profile-set K at or above $2K^* = 1769.2$ kt — the smallest expansive value on the grid is 1775 kt — which is exactly the condition reading (i) declares. The expansion obstruction is conditional on $K \geq 2K^*$; it is not conditional on the pinned value.

3.8 Stochastic viability

Persistent floors are a worst-case layer. The empirical residual pool supplies the stochastic counterpart, with the 1992 draw treated both as a member of the pool and as a removed one-off. For each policy and initial stock, 20,000 trajectories resample the 24 source-year training residuals (i.i.d., and in moving blocks of four, respecting the residual autocorrelation of 0.55). Viability is scored as the probability of remaining at or above the LRP.

Table 5. Probability of remaining at or above the LRP over 20 years ($N = 20,000$ per cell; seed fixed; draws shared across policies), source-year pool.

Policy	from the LRP, i.i.d.	from the LRP, blocks-4	from the LRP, i.i.d. without 1992	from 1500 kt, i.i.d.
zero catch	0.906	0.852	0.954	1.000
BAU (5 kt)	0.903	0.852	0.954	1.000

Policy	from the LRP, i.i.d.	from the LRP, blocks-4	from the LRP, i.i.d. without 1992	from 1500 kt, i.i.d.
60 kt / S1 / cascade	0.835	0.806	0.917	1.000
120 kt	0.647	0.650	0.769	0.999

From 1980s-peak starts (1500 kt and above) survival is near certain under every scheme — the peaks sit far above the LRP. The corrected source-year pool is marginally less severe than the frozen destination-year pool, so the survival probabilities are higher across the board.

The constructive boundary of Section 3.3 carries a stochastic reading. At the grid catch nearest the bound (92.5 kt on the campaign’s 2.5 kt grid) the 20-year survival probability from the LRP is 0.74 under i.i.d. draws, 0.73 under block resampling and 0.84 when the 1992 draw is removed — roughly three-quarters, not the near-certainty that holds at moratorium-level removals.

The $P \geq 0.9$ bar separates the two resampling schemes. Under i.i.d. resampling it is attained, but only by near-moratorium catches: the interpolated crossing is 13.5 kt against a ceiling of 0.906 at zero catch (Table 5), and by 60 kt survival has already fallen to 0.835. Under block resampling it is not attained at all, the ceiling there being 0.852 (Table 5; 0.849 in the crossing sweep, which carries its own fixed seed — a Monte-Carlo difference of 0.003 at $N = 20,000$); blocks of four preserve the residual autocorrelation of Section 2.1, so a poor year tends to be followed by another. Removing the 1992 draw lifts that crossing to 78.9 kt. The $P \geq 0.8$ crossings are 81.2 kt (i.i.d.), 72.3 kt (blocks) and 105.2 kt (no-1992). The one-off treatment of 1992 is a declared sensitivity, not an identified break.

Two properties of the pool explain the ceiling. First, failures are concentrated at the start of the horizon: the chance of an immediate breach is $2/24 = 0.083$ against a total failure probability of 0.094, and 99% of failures occur within the first three years. The reason is that from the reference point the stock is one year’s growth from safety — under zero catch a draw worse than -172.5 kt breaches the LRP at once, and two of the twenty-four residuals are worse than that, the 1992 draw (-329.0 kt) and one other (-323.5 kt). Second, a draw of that magnitude is harmless once the stock has grown past about 1020 kt, roughly a year later, which is why the ceiling is 0.906 and not the 0.43 that the $1/24$ recurrence alone would suggest: the probability of escaping the 1992 draw for twenty years is $(23/24)^{20} = 0.427$, and that would be the ceiling only if every 1992 draw were fatal.

The constructive bound of 91.59 kt sits above the 60-kt flat-cap family, so those rules are robust in the worst case, while the stochastic layer shows the survival probability falling steadily across that range (0.91 at zero-to-moratorium removals to 0.65 at 120 kt).

3.9 Finite-duration floors

A perpetual floor is a new stationary climate. The finite-duration variant asks how long a poor-productivity episode must last before the safe set contracts, with the floor active for n years and zero residual thereafter. The boundaries are computed by exact backward recursion on the monotone map, in the source-year convention (the $n = 5$ worst-floor BAU boundary reproduces the registered $T = 5$ boundary of the archived kernel table as a built-in check).

Table 6. Infinite-horizon lower boundary (kt) after a finite floor of n years followed by zero residual (source-year); empty = no state is robustly viable.

Policy	q05, $n = 5$	q05, $n = 10$	q05, $n = 15$	worst, $n = 5$	worst, $n = 10$	worst, $n = 15$
zero catch	1280.8	1510.5	1657.6	1431.6	1770.9	2015.3
BAU (5 kt)	1298.7	1540.7	1697.8	1450.0	1803.7	2062.3

Policy	q05, $n = 5$	q05, $n = 10$	q05, $n = 15$	worst, $n = 5$	worst, $n = 10$	worst, $n = 15$
60 kt / S1	1499.6	1893.2	2193.4	1656.7	2188.8	2657.5
/ cascade						
120 kt	1727.7	2329.0	2898.1	1891.7	2671.8	3562.3
180 kt	1999.0	2869.7	3986.2	2172.3	3285.8	5180.7
240 kt	2623.1	4013.4	8790.9	2825.0	4687.8	empty

Result 3.8 (Finite-duration floors). Two readings hold for the finite-duration grid.

- (i) Duration is informative. A five-year 5th-percentile episode already lifts the BAU boundary from the LRP to 1298.7 kt, and a fifteen-year episode lifts it to 1697.8 kt. The safe set contracts as the episode lengthens, without the arithmetic vacuity of the perpetual classes. The source-year floors are milder than the frozen ones, so the boundaries sit below the earlier reading at every duration.
- (ii) The ordering across policies is the same as in Table 1. Moratorium-level removals dominate every harvesting rule at every duration, and the reactive rules coincide with their 60-kt cap throughout because on the kernel domain their catch is exactly the cap's.

3.10 Uncertainty bands

The 91.59 kt constructive boundary is an analytic limit. Its sampling distribution is reported because a point would overstate the precision of the identified r . A parametric residual bootstrap ($B = 2000$, seed fixed) generates synthetic one-step series on the registered map with resampled source-year training residuals and refits r at $K = 5000$ kt on each. The refits give r a median of 0.219 (90% interval $[0.016, 0.277]$), $g(K^*)$ a median of 159.6 kt yr⁻¹ ($[11.6, 201.9]$), and the constructive bound a median of 78.7 kt with 90% interval $[0.0, 121.1]$ kt. Of these refits, 88.2% retain a positive constructive bound. The expansion classification survives the resampling: $F'(K^*)$ has median 1.142 and 90% interval $[1.010, 1.179]$. The constructive boundary is therefore best read as order 70–90 kt under the declared 10th-percentile class, not as a quota figure.

Refitting r alone at $K = 5000$ kt is the narrower of the two available bands, because it conditions the pin away rather than propagating it. Refitting both parameters on every replicate gives r a median of 0.261 (90% interval $[0.092, 0.500]$), K a median of 4191 kt with the pin active in 41.7% of replicates, and the constructive bound a median of 73.7 kt with 90% interval $[-89.4, 125.7]$ kt. The interval's negative tail is made of replicates in which the refit places K below the reference point — admissible under the declared box, and excluded by construction when K is held fixed. Restricted to the expansive regime $K \geq 2K^*$ on which the results are conditional — 72.9% of the 2000 replicates — the bound has median 88.1 kt (90% interval $[-5.6, 130.6]$ kt) and $F'(K^*)$ has median 1.137 with 90% interval $[1.030, 1.178]$: expansion holds throughout.

Two intervals for one quantity, a factor of eight apart in width, need the difference said out loud rather than left for a reader to reconcile. The profile range of Section 3.7 is an F -based interval: it holds the fit inside the 95% likelihood cut, lets K run the whole ridge from 1500 to 50,000 kt with r reprofiled at each step, and asks how far C^* can move. It answers an identification question — is the functional pinned, given the model — and the answer is yes, to within 27 kt, even though K is not identified at all. The bootstrap answers a different question: how far the estimate itself moves when the 24 transitions are resampled and both parameters are refitted freely. It is wider for two separable reasons. Its lower tail comes from replicates in which the refit places K below the reference point (7.4% of them) — admissible under the declared box, excluded by construction when K is held fixed, and not admissible as a description of a limit reference point. Removing those and every replicate outside the expansive regime still leaves $[-5.6, 130.6]$ kt, so the width is not an artefact of degenerate refits: the

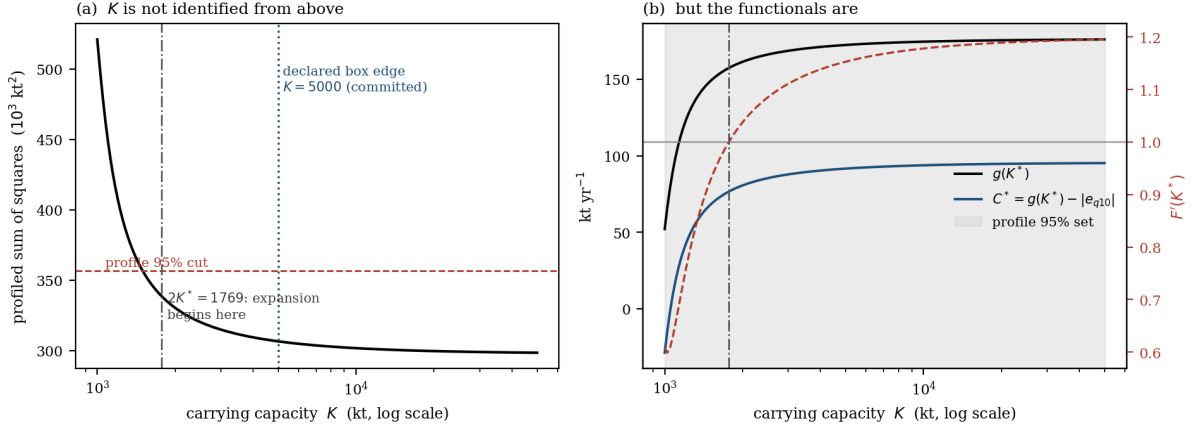


Figure 9: Identification under the pinned carrying capacity. (a) The profile sum of squares over K , refitting r in closed form at each K : the profile is flat from about 1500 kt to the top of the grid, so K is not identified from above and the committed value (dotted) buys 2.7% of the criterion. (b) The functionals the results depend on are identified nevertheless: across the profile 95% set (shaded), $g(K^*)$ and $C^* = g(K^*) - |e_{q10}|$ vary by 15%, and $F'(K^*)$ (right axis) exceeds 1 exactly above $2K^*$.

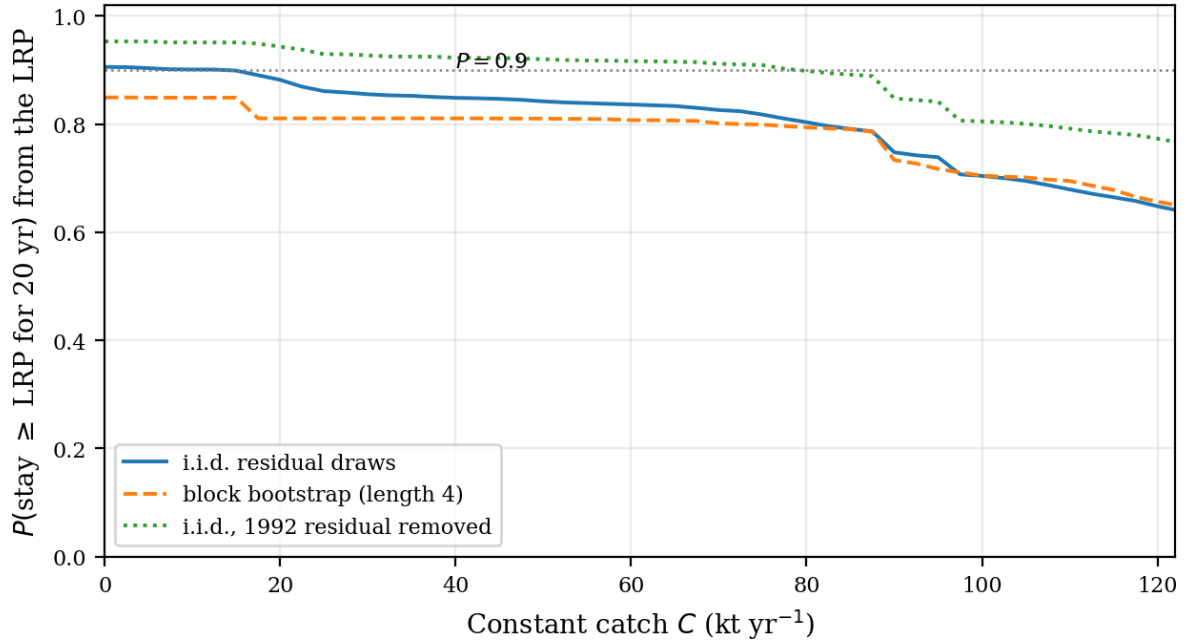


Figure 10: The three trade-off curves: 20-year survival probability from the LRP against constant catch, by resampling scheme (i.i.d. draws, blocks, no-1992).

bootstrap is wider at both ends, 35 kt above the profile ceiling and 74 kt below its floor. What remains is small-sample: with 24 transitions the resampling dispersion of a two-parameter refit exceeds what an F -based cut implies, and the profile range should not be read as a statement of precision. Neither interval is in error. They are evidence for different claims, and the claim this paper makes is the identification one.

One further declared sensitivity is made quantitative here. The residual of Section 2.1 conflates process noise and observation error, and only the process part can act as a persistent floor. If a fraction λ of the residual variance is observation error, rescaling the empirical pool about its mean by $\sqrt{1-\lambda}$ — its shape preserved, only its scale changed — moves the constructive bound from 91.6 kt at $\lambda = 0$ to 112.1 kt at $\lambda = 0.5$, and the vacuous class disappears from $\lambda \geq 0.2$ onward. The reported bound is the conservative end of that range.

3.11 Second specification (xteNCAM): a labelled sensitivity row

The registered object is the NCAM series. The second, unpooled specification is the xteNCAM series (Regular et al., 2025, Table 17; 1954–2024; LRP = 276 kt; 2024 landings persisted from 2023), refitted in the same source-year convention as the primary object (Section 2.3) on 1954–2007, with the same box rule and the safe set written against its own reference point. The floor classes are not transferred from the NCAM fit (different series scale): the row declares its own classes from its own training residuals, and no verdict transfers between the specifications. The fit gives $r = 0.5023$, $K = 4812.9$ kt (unpinned), MSE 18,028 kt², and expansion at the LRP $F' = 1.4447$ — the expansion classification is preserved and stronger than on the registered series.

The informative certificate does not survive. Under the fit’s own 10th-percentile class the constructive bound is negative ($g(\text{LRP}) = 130.7$ kt against $|e_{q10}| = 178.7$ kt). The zero-catch kernel does not hold the LRP from the LRP itself ($T = 1$ lower boundary 309.4 kt against the 276-kt reference point; business-as-usual at 5 kt needs 312.9 kt). The infinite-horizon zero-catch boundary under that class is 386.9 kt — the reference point is not robustly viable from itself on the second specification, and every positive constant catch raises the $T = 1$ boundary further (351.2 kt at 60 kt, 393.4 kt at 120 kt). The harsh classes are not vacuous there ($g_{\max} = 604.4$ kt > 470.8 kt), and their $T = 1$ boundaries run 515.7 kt (zero catch) to 602.3 kt (120 kt).

The row is a labelled sensitivity with opposite structure at its own reference point (Table 7). The two specifications agree on the expansion classification and on the kernel ordering across the declared catches (zero catch $\leq$ business-as-usual $\leq$ 60 kt $\leq$ 120 kt at $T = 1$ on both). They disagree on the reference point’s self-viability. The 2024 xteNCAM stock (342 kt) sits between the second specification’s $T = 1$ (309 kt) and $T = 5$ (368 kt) 10th-percentile zero-catch boundaries.

Table 7. The two specifications compared at their own reference points (the registered NCAM row and the second, unpooled xteNCAM row of Section 3.11).

Object	r	K (kt)	$F'(\text{LRP})$	Constr. bound (kt)	zero-catch q10, $T = 1$	zero-catch q10, $T = \infty$
NCAM (regis- tered)	0.2369	5000 (pinned)	1.1531	91.59	884.6	884.6
xteNCAM (this row)	0.5023	4812.9	1.4447	−48.0	309.4	386.9

3.12 Is the reference point self-viable under current productivity?

The fit window 1983–2007 spans the 1992 collapse, so the identified r averages two productivity regimes and the floor classes average them with it. Refitting the same object on post-moratorium

windows — same source-year convention, same declared box, same construction throughout — asks whether the reference point is self-viable under recent productivity (Table 8). The rows are labelled sensitivities in the sense of Section 3.11: no verdict transfers between the two series, and none of these windows replaces the frozen protocol.

Table 8. The same construction on post-moratorium windows. K is reported with its bound status (ub: upper bound of the declared box, lb: lower bound, i.e. just above the largest observed stock on that window); C^* is the constructive bound at each window’s own 10th-percentile floor. The same quantity at each window’s perpetual-worst floor is given in the text; the residual SD, which falls from 114.9 kt on the frozen window to 18.9 kt on NCAM 1995–2015, is what drives the change.

Window	n	r	K (kt)	$g(\text{LRP})$	$ e_{q10} $	C^*	$F'(\text{LRP})$
1983–2007 (frozen)	24	0.2369	5000 (ub)	172.5	80.9	91.6	1.153
NCAM 1995–2015	20	0.2731	5000 (ub)	198.9	27.4	171.4	1.176
NCAM 1995–2007	12	0.3901	5000 (ub)	284.0	10.0	274.0	1.252
xteNCAM 1995–2024	29	0.4310	472 (lb)	49.4	41.4	8.0	0.927
xteNCAM 2005–2024	19	0.4439	472 (lb)	50.9	58.5	-7.7	0.925

On the assessment series the answer is yes, and more comfortably than on the collapse window. On 1995–2015 the fit gives $r = 0.273$, $g(\text{LRP}) = 198.9$ kt and a 10th-percentile floor of only 27.4 kt — the post-moratorium decline was smooth and therefore largely predictable one step ahead, the residual SD falling from 114.9 kt to 18.9 kt — so the constructive bound rises to 171.4 kt and stays positive under that window’s perpetual-worst floor (166.4 kt), with the expansion at the LRP intact ($F' = 1.176$). On the shorter 1995–2007 window the bound is larger again (274.0 kt).

On the second specification the answer is no. On xteNCAM 2005–2024 the fit places K at its lower bound, 472 kt, barely above the largest observed stock, so the fitted surplus at that specification’s reference point is 50.9 kt against a 10th-percentile floor of 58.5 kt: the constructive bound is -7.7 kt, and on 1995–2024 it is +8.0 kt. Under either window’s perpetual-worst floor it is negative (-37.9 and -38.8 kt). The classification reverses with it — $F'(\text{LRP}) = 0.925 < 1$ — so on the modern series in its recent state the closed loop contracts at the reference point and the contraction form of the certified-layer conversion, inapplicable on the frozen object, is the applicable one.

Two caveats are inseparable from these rows and are stated rather than buried. The windows are short (12–29 transitions), and in three of the four the carrying capacity is at a bound, so these are regime readings rather than estimates of productivity. What they establish does not depend on reading them as estimates: the reference point’s self-viability is regime-dependent — comfortable on the assessment series after the moratorium, and at best marginal on the modern series, at roughly 0 ± 8 kt and negative under that series’ worst observed floor — and the expansion classification moves with it. That is consistent with Section 3.11, where the same specification’s 1954–2007 fit also failed to make its reference point self-viable.

4. Discussion

The horizon has a management consequence that is easy to miss while it is being read as a limitation of the method. A robustness certificate for a reference point has a shelf life, and Table 2 and Figure 10 show the shelf life cannot be extended by fishing less: it is six or seven years for every admissible catch, including none at all. Any interval at which the reference point is re-assessed therefore has to sit inside it, or the certificate on which the reference point’s defence rests has already expired when it is consulted. Two real cadences bracket that. The

IWC schedules an Implementation Review of its Revised Management Procedure “normally” every six years, and its aboriginal subsistence strike limits are set in six-year blocks; NOAA’s management-track assessments run on cycles of one to three years for many stocks, but for others “a decade may pass before an assessment is conducted.” A decadal cadence is outside the horizon computed here; a six-year one sits exactly at it, and only a cycle shorter than six years is comfortably inside. The same arithmetic says what would move the horizon: it is set by the expansion rate $F'(K^*)$, so neither better data about the disturbance nor a more cautious catch extends it — only a reference point low enough that the governed map contracts there, or a policy that changes the map’s shape rather than its level.

Two layers of negative content must be kept distinct. The vacuous-class identity (Section 3.2, definitional note) is a robust-layer statement: under the perpetual-worst persistent floor, no catch policy — zero catch included — holds the LRP. Under the 5th-percentile floor the statement is no longer vacuous: in the source-year convention that floor lies below the maximum surplus, and the moratorium’s $T = \infty$ kernel is nonempty (2219.6 kt). The certified-layer emptiness beyond $T = 6$ years under the two harsher floors and beyond $T = 7$ under the informative one (Section 3.4) is a different statement, about the conversion’s expansive form: the governed map’s expansion rate ($F' = 1.153$ at the LRP) empties every certified kernel at those horizons. Neither result paraphrases the other. The vacuous-class identity’s perpetual-worst reading is form-sensitive — the data-preferred compensatory refit restores a nonempty kernel at that class, the one class vacuous on the registered form (Section 3.6) — and its 5th-percentile half is not, that class being informative on both forms.

Three classes of findings must also be kept distinct — the form and specification sensitivities, the geometric findings, and the identified findings. The form and specification sensitivities carry the same division. The Fox form preserves every certificate direction at a higher fit cost (Section 3.6), and the second specification preserves the expansion classification and the policy ordering while reversing the informative certificate at its own reference point (Section 3.11). The geometric findings are form- and specification-independent; the identified findings are not; the scope is exactly as stated.

Identified findings are properties of the fitted map: the constructive boundary, the expansion rate at the LRP, and the stochastic survival probabilities of Section 3.8 all move with r and K . The boundary-harvesting rules’ “less protective by declared geometry” reading under the frozen convention is not an identified finding but a consequence of that convention’s harsh floor classes: under the corrected source-year 10th-percentile class the critical-zone rule, the cascade, and the graded rules all hold the LRP from itself at every horizon, so the geometry-based verdict of the frozen reading no longer applies. What survives is the clause-(H1) structural block at the 5th-percentile $T = \infty$ class: no positive-catch rule dominates BAU because every such rule is empty there while BAU’s kernel is nonempty (2219.6 kt). The supply comparison is a second failure, not the mechanism. The surplus-proportional family is the closest to a dominance, in that it is both reactive and protective at $\phi \leq 0.50$ under the informative class while harvesting more than the moratorium — a genuine trade-off, not a strict improvement. It fails of dominance because, on the 5th-percentile class and at $T = \infty$, it does not improve on BAU at all, and the equal-protection flat cap yields a higher mean catch. *Form-sensitive* findings reverse under the data-preferred compensatory refit, and the carrying-capacity grid of Section 3.7 shows the identified findings survive K within the declared box while the geometric ones are K -independent by construction.

The expansion obstruction is also the contribution to the viability-methods record. The certified-layer machinery used in the companion studies assumes a contracting closed loop, and the by-catch fishery case that anchors kernel computation in the fisheries literature (Krawczyk et al., 2013) is likewise a contracting setting. This cod object is the first scored instance in which the contraction form is provably inapplicable at the declared safe set — the map’s steepest growth occurs exactly where the stock is scarcest, at the boundary the governance question

is about. For a collapsed stock below half its estimated carrying capacity, every governance statement that survives certification must therefore be time-bounded and expansion-bound, not defect-bound.

Four consequences for the methods record follow from the new layers. First, the carrying-capacity grid shows the expansion classification is the data-selected regime, not an artifact of the pinned bound, so the certified-layer emptiness is a property of the identified map rather than of the conversion’s tuning. Second, the finite-duration floors separate the perpetual-class vacuity from a duration-respecting robustness question that remains informative. Third, the stochastic layer converts the worst-case certificates into survival probabilities on the empirical residual pool — the quantity a harvest-control discussion can actually weigh. Fourth, the reactive families extend the declared policy family with genuinely state-feedback rules, so the “how do reactive rules fare” question is answered directly rather than by proxy; the surplus-proportional family registers a real trade-off (protective under the informative class, more harvesting than the moratorium) while the surplus-harvest and ramp designs are dominated and recorded as rejected.

The map is one-pool surplus production on annual means. There is no age structure, migration, or survey catchability (the model-type limitations of the companion forecast-evaluation study carry over). K is pinned at its optimization bound. The expansive classification at the LRP carries the K -pin’s declared defect with it, in one reading consistent with Section 3.7 and Figure 4: $F'(K^*) = 1 + r(1 - 2K^*/K)$ exceeds 1 exactly while $K > 2K^* = 1769.2$ kt. The expansion is conditional on $K \geq 2K^*$, and within that range it is not an artifact of the particular pinned value — it holds for every admissible $K \geq 2K^*$, and the carrying-capacity grid shows the data select that range (the fit cost and the informative certificates both deteriorate below $2K^*$). Any data-supported K below twice the LRP would make the closed loop contract at the boundary and restore the contraction form of the conversion; the condition is on K , not on the identified r . The residual conflates productivity shock and model error (no observation-model separation). The closed loop observes the stock exactly at the decision instant; real governance operates under assessment lags (the one-year delay module of the companion forecast-evaluation study), so the reported kernels are upper bounds for a perfect-observation controller, and delay-aware kernels would be smaller or empty. The persistent-shock classes are deliberately harsh (a perpetual floor, not an independent draw). The 10th-percentile class is the mildest with non-vacuous content, and only the perpetual-worst floor sits beyond the map’s maximum surplus (Section 3.2), which is what makes its emptiness vacuous as a productivity statement.

The dominance rule’s protective clause is structurally conservative toward the moratorium. Under the source-year informative class the critical-zone, cascade, and graded rules match BAU’s protection exactly, and the surplus-proportional family at $\phi \leq 0.50$ holds the LRP at every horizon while harvesting more than the moratorium; but under the 5th-percentile class at $T = \infty$ every positive-catch rule is empty while BAU’s kernel is nonempty (2219.6 kt). Because clause (H1) is read at every reading with empty scored as worst, that 5th-percentile emptiness is what blocks dominance, and only a rule that is at least as protective as BAU at every reading and improves on the moratorium somewhere — none here — could dominate BAU. The Allee term is off in the primary specification; the depensation sensitivity of Section 3.6 shows the certificate directions are stable under the data-preferred Allee refit, with the expansion obstruction worsening and class-vacuity narrowing. The safe set’s upper edge (10^4 kt, twice K) is a computational grid cap, never approached on any reported kernel path; kernels are written $[s, \infty)$ because the cap is never approached, and the positive-part floor $[\cdot]_+$ never binds on any reported kernel path. Sub-LRP cascade stages are declared scenarios, not verified institutions. The certified layer is vacuous at observed stock levels.

Nothing here promotes or demotes any forecast module, transfers numbers from an interval-verified linear template (a companion methodological study, under review), or pools the extended xteNCAM series. The 91.59 kt boundary is not a quota recommendation; it is the

analytic limit of robustness at one declared shock class. For management, the deliverable is the risk statement rather than the point: on this map the moratorium-level removals and zero catch are the only policies that hold the safe set under the informative class, the largest robust constant catch is of order 70–90 kt, and the stochastic layer puts a probability on holding the line (0.90–0.91 at zero-to-moratorium removals, falling with catch) that the persistent floors cannot express. The result does not transfer to Northern cod productivity or to harvest-control rules in general; it is a scored comparison on one fitted map, and the LRP’s protection in good years is the property of that map and those classes.

5. Conclusions

Scored intervention selection on the fitted Northern cod surplus-production map yields five findings and two definitional notes, stated at their actual strength.

- (1) Under the informative 10th-percentile productivity class, moratorium-level removals and zero catch hold the entire safe set at every horizon, and the largest robust constant catch is 91.59 kt yr^{-1} — an analytic limit whose bootstrap 90% interval is $[0, 121.1]$ kt and whose 20-year survival probability from the LRP is 0.74 under i.i.d. resampling (0.73 under block resampling, 0.84 with the 1992 draw removed), so the operational reading is order 70–90 kt, not a quota.
- (2) No non-BAU policy dominates BAU, and the mechanism is the clause-(H1) reading at the 5th-percentile $T = \infty$ class rather than boundary geometry or, alone, supply: every positive-catch rule is empty there while BAU’s kernel is nonempty (2219.6 kt). Under the informative 10th-percentile class the critical-zone, cascade, and graded rules match BAU’s protection exactly, and the surplus-proportional family at $\phi \leq 0.50$ holds the LRP at every horizon and harvests more than the moratorium — so the reactive family registers a genuine trade-off, not a strict improvement. It still does not dominate BAU: it does not improve on BAU at the harsher-informative class, and an equally protective flat 60-kt cap supplies more.
- (3) The certified layer has a finite horizon — six years under the two harsher floors, seven under the informative one — because the map is expansive at the reference point — for every admissible $K \geq 2K^* = 1769.2 \text{ kt}$, so not an artifact of the particular pinned value, though conditional on $K \geq 2K^*$ — and the carrying-capacity grid shows this is the data-selected regime within that range: contraction is restored only below twice the LRP, where the informative certificates themselves collapse.
- (4) The certificates survive the data-preferred depensatory refit, and the class-vacuity reading narrows under the Allee form.
- (5) The certificate has a shelf life, and it cannot be bought. The certified horizon is six or seven years for every declared policy and every constant catch from zero to 150 kt — the moratorium included — because the erosion margin’s annual growth is 379–773 kt, four times the whole admissible catch range at its smallest. A review interval longer than that horizon is consulting an expired certificate.
- (6) The comparison is governed by an identity rather than by a tabulation: at the reference point a rule’s protection margin plus its catch equal 91.59 kt exactly (Proposition 2.1), so no reactive rule can out-supply a flat cap it matches in protection and the no-dominance verdict of finding (2) is structural. The certified horizon is likewise a crossing, not a search: six or seven years is where the geometrically growing erosion margin overtakes the worst-case trajectory.

- (7) The results are convention-dependent in the residual convention only in the sense that the source-year convention sets the constructive bound at 91.59 kt, brings the 60-kt rules onto the reference point itself (their $T = \infty$ boundary is 884.6 kt, against 900.3 kt under the registered convention), lengthens the certified horizon (to $T = 6$ under the two harsher floors and $T = 7$ under the informative one), and reduces the vacuous family to one class; every parameter and every certificate direction is otherwise stable.

Definitional note A (identity). Only the perpetual-worst floor is vacuous, and that vacuity is an arithmetic identity of the declared class — the floor exceeds the map’s maximum surplus, so every trajectory declines for every catch, zero included — not an empirical finding about Northern cod productivity (Section 3.2’s note). The 5th-percentile and 10th-percentile classes are informative, and the 5th-percentile moratorium kernel is nonempty (2219.6 kt).

Definitional note B (rule-level). The no-dominance outcome of finding (2) is a selection property of the declared partial order as much as of the policies: clause (H1), read at every reading with empty scored as worst, blocks every positive-catch rule at the 5th-percentile $T = \infty$ class given BAU’s nonempty kernel there, so within the declared family the outcome could not have gone the other way (Definition 2.6’s note). It is recorded as a note so that it is not read as an empirical selection finding on a par with the constructive bound and the stochastic layer.

The methodological content is the protocol itself — frozen scoring, layered certificates, the separation of geometric from identified findings, and the extension of the family to genuinely reactive rules — applied to a collapsed-stock reference point. The empirical content is scoped to the map, the classes, and the rule declared above. On that object the LRP is protected by good years, but the margin good years must supply is smaller than the frozen convention implied, and reactive rule design can protect the LRP but not out-supply a flat 60-kt cap. Across assessment specifications the findings are specification-conditional: on the second, unpooled xteNCAM specification (LRP 276 kt) the expansion classification and the kernel ordering survive while the reference point’s self-viability does not — the current stock sits between that specification’s one- and five-year zero-catch boundaries (Section 3.11). Across productivity regimes they are regime-conditional in the same way: the constructive bound is 171 kt on the post-moratorium assessment window and 0 ± 8 kt on the modern series (Section 3.12), so the reference point’s self-viability is a property of the regime as much as of the rule.

Part II — Regime viability on the Northern cod stock

Abstract

The viability machinery of the obstruction calculus has been developed and validated on deliberately small exact instances. This paper applies it to a real assessment record — the Northern cod stock (NAFO 2J3KL), read from the published assessment input series, the 2025 reassessment series, the official removals tables, the long-duration catch reconstruction, the research-vessel survey index, and an independently monitored forage-fish covariate — and certifies, in exact rational arithmetic, the obstruction structure that the historical collapse exhibited. The certified object is a *harvest-free multiplier bracket*: for each annual step, the interval’s per-capita non-fishing growth factor lies in $[\rho, \max\{\rho + r, \rho/(1 - r)\}]$, where ρ is the realized net multiplier and r the removals fraction, under either timing convention for the removals accounting. Both collapse steps certify as harvest-free contractions under every convention (upper bounds 0.754 and 0.372); the reference window’s worst step and the two prelude steps do not certify (brackets straddle unity) — a typology, not a blanket verdict. Under the post-moratorium removals rows (1,314 and 413 t, at most 1.4% of the stock), the brackets certify continued contraction, and the stock fell by a factor of $\frac{10105}{968} \approx 10.4$ in two years to its series minimum. The 276 kt auxiliary floor separates the reference window from

the collapse era without exception, while the 884.6 kt reference point — the window’s own mean — has three of the seven window readings below it in sample and therefore separates nothing. Three extensions certify robustness and scope: on the reassessment vintage’s published 95% bands, exactly the four collapse-era steps 1991–1995 certify as declines and the 1993 floor breach holds at the band’s upper edge ($115 < 276$), while no reference-decade step certifies; the reference decade’s window profile shows every subwindow of horizon four or more growing (the sustained-decline reading holds only at horizons one to three); and the vintage’s own 1954–1982 record has the window opening at 28.3% of the 1962 peak, three years after a prior floor episode (1977–1979). The collapse is present in the raw survey index (a certified factor of $\frac{21797}{2127417}$ over 1989–1994, falling 2.6 times further than the assessment series over their common window); the forage-fish covariate collapses in the same annual step in which the mortality diagnostic crosses $M = 1$, two steps before the floor breach; and both assessment vintages re-cross the floor at the 2015 reading, with outcome years at most $\frac{44000}{88460} \approx 49.7\%$ of the reference point. Every quantity is regenerated by an accompanying verification script in exact rational arithmetic.

Keywords: viability; obstruction; regime shift; Northern cod; exact computation; monitoring design; limit reference points

Introduction

The viability and obstruction machinery of the calculus — survivable sets, robust multipliers, breach-time certificates, and probe design — is developed in Abaee (2026, An obstruction calculus) and validated on deliberately small exact instances. This paper applies it to the Northern cod stock (NAFO divisions 2J3KL) — the running record of this paper — and certifies the obstruction structure of its collapse on the public assessment record. The forecast layer of the same record — the scoring of competing forecast ladders and the 2022–2023 observation suspension as a review-timing event — is certified in a companion paper (Abaee, 2026, The Northern cod forecast ladder, Section 3.10); the present paper certifies the viability layer. The two are companions by construction: the forecast ladder prices what can be predicted, this paper prices what could have been kept safe.

The claims below are of three kinds, and the label travels with the claim. *Arithmetic claims* are exact rational identities or inequalities on the certified rows; they are regenerated by the verification script and cannot drift. *Provenance claims* state which file, which vintage, and which column a number comes from, and whether that column is a model input or a model output. *Scope claims* are conditional readings — stress bounds, sustained hypotheticals, design critiques — and are labelled as such where they appear. No claim of the first kind rests on an interpretation; no claim of the third kind is stated as one of the first. The contributions are: the record spine — the certified 1960–2023 assessment-record reconstruction with its provenance ledger (Sections –); the harvest-free multiplier brackets — the exact two-form accounting bracket isolating the non-fishing growth factor (Section); and the window profile and band certificates — the collapse’s obstruction structure certified on the public record, with the certified-record figure reading every annotated value from the locked files (Section).

Related work

This paper certifies that specific steps in the published Northern cod record were contractions regardless of removals. Read carelessly, that contradicts the empirical literature on this collapse, and the conflict is addressed here rather than left for a referee to find.

The received attribution. Hutchings and Myers (1994) concluded that the collapse of northern cod could be attributed solely to overexploitation. Myers and Cadigan (1995) tested the competing hypothesis that an increase in natural mortality in the first half of 1991 caused the

collapse, did not support that hypothesis, and concluded that overfishing was sufficiently high to cause the collapse of the population even allowing for the possibility of elevated recent natural mortality. Myers, Hutchings and Barrowman (1997) rejected the hypothesis that mortality of young cod is unrelated to fully recruited fishing mortality, finding high juvenile mortality associated with high adult mortality in each of six stocks.

Why the two are consistent, and why the distinction matters. Those papers ask about the *historical cause* of the collapse as an event. This paper asks a different question of a different object: given the transitions recorded in the published assessment, was the step a contraction even had removals been zero? That is a statement about the *arithmetic of the record*, not about the cause of the collapse. Removals superimposed on an already-contractionary productivity term make the outcome worse, not better, and Proposition 8 says exactly that — the loss exceeds the removals by certified factors while the fishery was closing. Overfishing can be the cause of the collapse and the individual annual steps can still be contractions at zero harvest. Both hold here.

Where a real tension remains, and is not resolved. Myers and Cadigan (1995) specifically declined to support elevated natural mortality as the explanation. To the extent that the harvest-free contractions certified here are read as evidence of a productivity collapse, they sit against that finding, and **this paper does not adjudicate it**: the certificates are statements about the transitions as recorded, and they do not identify a biological mechanism. A reader who takes the harvest-free result as evidence about productivity is going beyond what is certified.

Which series the certificates are computed on. The concern is concrete rather than hypothetical. Myers, Hutchings and Barrowman (1997) documented that for these stocks the VPA-based and survey-based abundance trends diverge — from the early 1980s the VPA-based trend shows a decline where none is apparent in the survey-based trend — and attributed the divergence plausibly to discarding of young fish. Any certificate computed on one series inherits that series’ artefacts, so the choice of series is load-bearing for this paper and cannot be left implicit.

This paper therefore checks the raw survey index directly. The collapse is present there too (Proposition 14), with a certified factor of $\frac{21797}{2127417}$ over 1989–1994, and it falls 2.6 times further than the assessment series over their common window. The contraction is accordingly robust to the choice of series, and is if anything stronger in the raw survey than in the assessment record on which the main certificates are computed. That result is the answer to the divergence concern, and it is stated here so that the reader can see the concern was met rather than assumed away.

Position within viability theory. That the governing question is which states admit controls keeping every constraint satisfied, rather than which policy optimises, is the founding move of viability theory (Aubin, 1991); the obstruction calculus used here is the observation-constrained development of that programme (Abaee, 2026), and its constructive counterpart on this same record is certified in the companion paper (Part I). What this paper contributes to that lineage is the *direction* of the certificates — necessity rather than sufficiency — and the fact that they are obtained in exact rational arithmetic with no fitted model at all.

Data and provenance

Seven locked public files are used, each cross-verified against its primary source at archival time (Abaee, 2026, Calibration data record): (i) the NCAM (Northern Cod Assessment Model)

Table 9: The record spine, 1983–1995. SSB in kt (assessment input vintage, Table A2); removals in t (official table; the reconstruction gives identical rows for 1992–1995); survey index in the research-vessel file’s index units; covariate in kt, observed years only; M is the assessment’s mean natural-mortality diagnostic (model-estimated). Computed entries are exact rational arithmetic on the file rows; displayed values are rounded.

Year	SSB (kt)	Removals (t)	Survey	Covariate	M
1983	841.08	232,345	2,088,958	—	0.391
1984	863.23	232,471	2,198,605	—	0.377
1985	902.94	231,293	1,288,360	3426	0.349
1986	836.00	266,713	2,502,702	3697	0.277
1987	940.75	239,924	1,020,462	2576	0.494
1988	886.40	268,677	1,223,314	4285	0.335
1989	921.66	253,990	2,127,417	3712	0.289
1990	861.92	219,452	1,627,647	5783	0.403
1991	734.51	172,012	1,117,670	138	1.002
1992	381.95	40,956	239,740	138	2.214
1993	101.05	11,392	90,709	—	2.575
1994	30.55	1,314	21,797	—	2.331
1995	9.68	413	43,240	—	0.288

2016 assessment input series (DFO, 2016, Table A2): spawning stock biomass in kilotonnes, 1983–2015, cross-checked 33 of 33 years against the primary report — the *input vintage* of the biomass record; (ii) the xteNCAM 2025 reassessment SSB series (Regular et al., 2025, Table 17), 1954–2024, with 95% bands and SSB-over- B_{lim} ratios — the *reassessment vintage*; (iii) the official removals table (Regular et al., 2025, Table 1), 1954–2023; (iv) the catch reconstruction (Schijns et al., 2021, Table 1); (v) the fall research-vessel survey abundance index (Schijns et al., 2021, Table 3, citing DFO 2021b) — a research-vessel input to the assessments, not an output of them; (vi) the RAM Legacy database (Ricard et al., 2012) per-year extract for this stock (the free 2024 release), 1850–2021 — the *database vintage*, extending the series to 2021; and (vii) the capelin acoustic biomass series (observed years only, no interpolation).

Two thresholds appear throughout, and their provenance is stated once here. The *auxiliary floor* $B_{\text{aux}} = 276$ kt is the reassessment vintage’s B_{lim} at the file’s rounding: the published ratio column is rounded to two decimals, and the values it implies for B_{lim} range from 250 to 289 kt across the file’s years. Every floor comparison below is insensitive across that range (the compared quantities clear it by large factors; Remark 13), so a single representative value is fixed. The *reference point* $B_{\text{ref}} = 884.6$ kt is the input vintage era’s limit reference point, whose documented construction is the 1983–1989 mean of the assessment input series (DFO, 2016); the exact mean is $\frac{44229}{50} = 884.58$ kt (Proposition 1), and 884.6 is its published rounding.

Two discipline rules are respected throughout. First, the mortality and fishing-mortality columns of the assessment input file are diagnostics only (dynamics are never built on them). Second, every annual multiplier certified below is a *realized net* quantity — it conflates the mortality regime with all removals. Section introduces the one decomposition this paper performs on it, which is an accounting bracket, not a biological model.

The record spine

Table 9 tabulates the certified rows of the collapse era and its surroundings: assessment input SSB (input vintage), official removals, the survey index, the covariate (observed years only), and the mortality diagnostic. Every proposition below reads its numbers from this table’s files.

The reference window and the collapse window

Proposition 1 (seven readings, six transitions; the windows’ worst multipliers). *From the Table A2 series: the reference window 1983–1989 contains seven readings (841.08, 863.23, 902.94, 836.00, 940.75, 886.40, 861.92 kt and six annual transitions; their minimum is the 1985 transition $\rho_g = 836.00/902.94 = \frac{41800}{45147} \approx 0.9259 < 1$, and their exact mean is $\frac{44229}{50} = 884.58$ kt. The collapse window 1991–1993 contains two transitions, $\frac{38195}{73451} \approx 0.520$ (1992) and $\frac{2021}{7639} \approx 0.265$ (1993), each with numerator strictly below denominator. The connecting 1990 reading is 861.92 kt, with transitions $\frac{86192}{92166} \approx 0.935$ (1990) and $\frac{73451}{86192} \approx 0.852$ (1991).*

Proof. Each multiplier is the exact ratio of consecutive Table A2 SSB entries, computed in the script as a rational of the printed values; the minimum of the six reference-window ratios and the two collapse-window ratios are the displayed fractions by direct comparison. Each displayed fraction is below 1 because its numerator is smaller than its denominator ($41800 < 45147$, $38195 < 73451$, $2021 < 7639$). The mean is the exact rational sum of the seven readings over seven. $\square$

Harvest-free multiplier brackets

The net multiplier ρ folds removals into the per-capita rate, so a sub-unitary ρ alone does not certify a non-fishing contraction: with a large enough harvest, even a growing stock declines net of catch. This section isolates the harvest term by an accounting bracket that is exact and robust to the one ambiguity the rows leave open — where in the interval the removals acted.

Shared symbols across the companion papers. The companion papers share this vocabulary with paper-local meanings; the letters below carry meanings here that differ there:

Here	In the companions
$\rho = B_{t+1}/B_t$: the net biomass multiplier	the mesh relaxation parameter (certification); the contamination radius ρ of $\mathcal{D}_\rho$ (sufficiency; minimax duals)
$r = C/B_t$: the catch share	the intrinsic growth rates r , r_i (forecast ladder; the minimax benchmark)
B : the biomass	the belief/information sets B_t , B (obstruction calculus; certification; sufficiency)

Lemma 2 (harvest-free multiplier bracket). *Fix a certified interval with readings $B_t, B_{t+1} > 0$ and a removals row $C \geq 0$ assigned to the interval; write $\rho = B_{t+1}/B_t$ and $r = C/B_t$. Suppose the interval’s accounting has one of the two extreme forms, with $g_t > 0$ the interval’s per-capita non-fishing growth factor: $B_{t+1} = g_t B_t - C$ (removals deducted after growth), or $B_{t+1} = g_t(B_t - C)$ (removals deducted before growth). Then respectively $g_t = \rho + r$ or $g_t = \rho/(1 - r)$; in either case*

$$\rho \leq g_t \leq \max\left\{\rho + r, \frac{\rho}{1 - r}\right\},$$

and the upper bound is strictly below 1 if and only if both $\rho + r < 1$ and $\rho/(1 - r) < 1$.

Proof. First form: $g_t = (B_{t+1} + C)/B_t = \rho + r$. Second form: $g_t = B_{t+1}/(B_t - C) = \rho/(1 - r)$, well-defined since $C < B_t$ on every certified interval. Both expressions are $\geq \rho$ (as $r \geq 0$, $1 - r \leq 1$); the disjunction of the two forms gives the displayed maximum, and the final equivalence is direct algebra ($\rho/(1 - r) < 1 \iff \rho < 1 - r$). $\square$

The lemma’s content is accounting, not biology: g_t still conflates natural mortality with recruitment (the regime), but no placement of the *realized* removals within the interval’s accounting — early or late — can leave the non-fishing account non-contracting when the bracket’s upper bound is sub-unitary. The rows do not date removals within the year, so both alignments

Table 10: Harvest-free multiplier brackets (Lemma 2), both alignments of the removals rows. ρ exact to four decimals; upper bound = $\max\{\rho + r, \rho/(1 - r)\}$ in exact rational arithmetic, displayed rounded.

Step	C (t)	ρ	Upper bound	Verdict
1985→86	266,713	0.9259	1.3140	straddles
1989→90	219,452	0.9352	1.2274	straddles
1990→91	172,012	0.8522	1.0646	straddles
1991→92	172,012	0.5200	0.7542	certified
1991→92	40,956	0.5200	0.5758	certified
1992→93	40,956	0.2646	0.3718	certified
1992→93	11,392	0.2646	0.2944	certified
1993→94	1,314	0.3023	0.3153	certified
1994→95	413	0.3169	0.3304	certified

of the two adjacent rows (C_t, C_{t+1}) are carried; a verdict is certified only when it holds under both conventions and both alignments.

Proposition 3 (the collapse steps are harvest-free contractions; the reference and prelude steps are not certifiable). *Applying Lemma 2 with both alignments of the removals rows (Table 10): both collapse-window steps certify harvest-free contraction, the 1992 step under an upper bound of at most 0.7542 and the 1993 step under an upper bound of at most 0.3718, each computed in exact rational arithmetic. The reference window’s worst step (1985–1986, upper bound 1.3140) and the two prelude steps (1989–1990, upper bound 1.2274; 1990–1991, upper bound 1.0646) have brackets that straddle unity: for those steps the certified rows do not decide between a fishing-dominated and a non-fishing-dominated contraction.*

Proof. Each row of Table 10 evaluates $\max\{\rho + r, \rho/(1 - r)\}$ as an exact rational for the displayed (ρ, C) pair; the collapse-window upper bounds (at most 0.7542) are compared against 1 by numerator–denominator comparison in the script. The prelude and reference rows exceed 1 likewise. $\square$

The window profile and the band certificates

The reference window’s profile answers the selection question at all horizons, and the reassessment vintage’s published bands answer the robustness question the point estimates leave open.

Proposition 4 (the reference window’s profile: declines only at horizons one to three). *Within the reference window 1983–1989, the minimum end-to-start ratio of a k -year subwindow is: $k = 1$: $\frac{41800}{45147} \approx 0.9259$ (the worst step, at 2 of the 6 subwindows below unity); $k = 2$: $\frac{83600}{86323} \approx 0.9685$ (2 of 5 below); $k = 3$: $\frac{44320}{45147} \approx 0.9817$ (2 of 4 below); $k = 4, 5, 6$: every subwindow grows — the minimum products are $\frac{15361}{15049} \approx 1.0207$, $\frac{22160}{21027} \approx 1.0539$, and $\frac{15361}{14018} \approx 1.0958$. Fifteen of the twenty-one subwindows are all-growth at the annual level. The sustained-decline reading of the reference window therefore holds only at horizons one to three; at every horizon of four or more years inside the window, the record grew.*

Proof. For each fixed k , the subwindow products are exact rationals with common denominator structure S_{t+k}/S_t , compared by cross-multiplication in the script; the $k \geq 4$ growth statements are the direct comparisons $S_{t+k} > S_t$, which involve no roots. The subwindow count is $6 + 5 + 4 + 3 + 2 + 1 = 21$. $\square$

Proposition 5 (band certificates: exactly the collapse-era steps certify, and the 1993 breach holds at the band’s upper edge). *The reassessment vintage publishes 95% bands (B_{lo}, B_{hi}) per year. A band-certified decline is a step whose band interval contains only values below unity: $[\frac{B_{lo}(t+1)}{B_{hi}(t)}, \frac{B_{hi}(t+1)}{B_{lo}(t)}] \subset (0, 1)$. Over the vintage’s 1954–2024 record, the band-certified declines*

are exactly the four collapse-era steps: 1991–1992 ($[\frac{265}{754}, \frac{468}{478}]$, upper bound 0.9791), 1992–1993 ($[\frac{54}{468}, \frac{115}{265}]$, upper bound 0.4340), 1993–1994, and 1994–1995 — and no other step in seventy years, and no reference-decade step. The 1993 floor breach is band-certified ($B_{hi}(1993) = 115 < 276$); the 1992 reading is indeterminate on the bands ($265 \leq 276 \leq 468$).

Proof. Each candidate step reduces to the single comparison $B_{hi}(t + 1) < B_{lo}(t)$, certified by exact rational comparison across all seventy pairs of consecutive years in the script; the interval endpoints are the displayed integer ratios. The floor comparisons are direct rational comparisons. $\square$

Remark 6 (the vintage dependence of the typology). *The reassessment vintage reads the reference decade itself as all-growth (its 1983–1989 multipliers are 1.14, 1.22, 1.03, 1.28, 1.10, 1.01, minimum 1.01), with the decline beginning at 1989–1990 ($738/864 \approx 0.854$): the worst-step typology of Proposition 3 is vintage-internal, and the paper’s decade statements are made in the input vintage, whose worst step (0.9259) has no counterpart in the reassessment vintage. A third vintage (the database extract’s assessment, 2021) reads 1990–1993 as 765, 747, 354, 83 kt; all three vintages agree on the breach sign pattern against the floor (above, above, below).*

Remark 7 (the record before the window: 1954–1982). *The reassessment vintage extends the record to 1954: a peak of 1508 kt at 1962, with the 1983 window-opening reading at 427 kt — 28.3% of the peak, vintage-internally. The floor 276 kt has a prior episode: the crossings of the point series are exactly {1977, 1980, 1993, 2016} (below 1977–1979, recovered 1980–1992, below again 1993–2015, re-crossed 2016), and the 1978 reading is band-certified below the floor. The reference window thus opens three years after a prior recovery, at 28.3% of the record’s peak scale: the window’s start is a data artifact of the input vintage’s table, and the design critique of Proposition 16 applies to the window’s placement as well as its mean.*

The moratorium obstruction and the post-moratorium experiment

Proposition 8 (the collapse proceeds with the fishery closing; the loss exceeds the removals by certified factors). *Across 1991–1992 the stock lost $734.51 - 381.95 = 352.56$ kt while removals were 40,956 t: the loss exceeds the removals by a certified factor of $\frac{352560}{40956} \approx 8.61$. Across 1992–1993 the loss was $381.95 - 101.05 = 280.90$ kt while removals fell to 11,392 t: a certified factor of $\frac{280900}{11392} \approx 24.66$; removals were 4.06% of the loss. By Lemma 2 the two steps are harvest-free contractions under every timing convention (Proposition 3): the decline over the collapse window is not explainable by the realized removals under any accounting placement of them.*

Proof. The losses are differences of certified readings (exact decimals); the factors are their exact ratios against the removals rows, computed in the script as rationals ($4.06\% = 11392/280900$). The harvest-free attribution is Proposition 3’s certified rows for 1991–1992 and 1992–1993. The statement is a witness about the realized window, not a counterfactual claim about other catch paths. $\square$

Proposition 9 (the reconstruction and the moratorium years). *The reconstruction’s rows for 1992–1995 are 40,956, 11,392, 1,314, 413 t — identical to the official removals table’s rows for the same years (a cross-file equality) — and its 2015 row (4,436 t) agrees with the official table and with the assessment report, as recorded at archival time. The moratorium-era removals are at least 176 and at most 651 times below the 1983–1989 removals rows (all above 130,000 t in the same file): a separation of between two and three orders of magnitude ($\frac{231293}{1314} \approx 176.0$ to $\frac{268677}{413} \approx 650.5$).*

Proof. The four 1992–1995 rows are read from both removals files and their equalities are exact rational equalities in the script; the 2015 equality is $4436 = 4436$. The scale comparison is the exact minimum and maximum of the four ratios of the 1983–1989 reconstruction rows against the two moratorium rows. $\square$

Proposition 10 (the post-moratorium experiment). *From the 1993 reading 101.05 kt, the certified removals rows are 1,314 t (1994) and 413 t (1995): at most 1.4% of the interval’s start reading in both steps ($r = 1314/101050 \approx 0.0130$ and $413/30550 \approx 0.0135$). Lemma 2 certifies harvest-free contraction in both steps under both conventions (upper bounds 0.3153 and 0.3304): with the fishery effectively closed, the stock fell $101.05 \rightarrow 30.55 \rightarrow 9.68$ kt, a certified factor of $\frac{10105}{968} \approx 10.44$ in two years, and 9.68 kt is the minimum of the entire 1983–2015 Table A2 series.*

Proof. The removals fractions and the bracket upper bounds are exact rationals in the script, compared against 1 by numerator–denominator comparison. The two-year factor is the exact ratio of the 1993 and 1995 readings; the minimum is the exact minimum of the 33 certified readings. $\square$

Mechanized verification

The multiplier-bracket algebra of Section is formalized in the project’s Lean 4 layer, in namespace `Formalizations.ARV` (`ARV_RegimeViability`): `bracket_upper_form1`, `bracket_upper_form2`, `bracket_sandwich_form1`, `bracket_sandwich_form2`, `bracket_upper_lt_one_iff`, `bracket_subunitary_form1`, `bracket_subunitary_form2`, `bracket_subunitary_of_upper_form1` and `bracket_subunitary_of_upper_form2` — thirteen theorems, together with the order-maximum primitives (`omax` and its six lemmas) on which the interval reading rests. The two *forms* correspond to the two ways of writing the bracket, and the subunitarity theorems are exactly the step that turns an upper bracket into a certified contraction.

No admitted gaps. A comment-stripped source scan of the whole project finds no `sorry`, `admit`, `axiom` or `constant` declaration anywhere; declarations using classical reasoning depend only on `Classical.choice`, `propext` and `Quot.sound`.

Scope, stated plainly. Machine-checked here is the *bracket algebra*: the ordered-field reasoning that turns readings and a removals row into a certified contraction, and the subunitarity step. Not machine-checked is everything upstream of it — the extraction of the certified intervals from the assessment record, the assignation of removals rows to intervals, and the enumeration of the windows. Those are script computations and are regenerated, not proved. The formalization therefore certifies that *if* the readings and the removals row are as stated, the bracket conclusion follows; it does not certify the readings themselves.

Breach certificates and the cross-vintage check

Proposition 11 (breach certificates on the two thresholds). *With $B_{\text{aux}} = 276$ kt and $B_{\text{ref}} = 884.6$ kt: (i) conditional stress bound: from the end-1991 reading 734.51 kt, one step at the collapse window’s worst multiplier already lands below B_{aux} , since $\frac{2021}{7639} < \frac{27600}{73451}$; this is a hypothetical sustained-rate bound, and the realized breach took two steps. (ii) first below-floor reading: the 1992 reading is above the floor ($381.95 \geq 276$) and the 1993 reading is below it ($101.05 < 276$): the realized first breach of B_{aux} is the 1993 reading, and the input vintage then stays below the floor through 2014 (250.12 kt), re-crossing at 2015 (Proposition 18). (iii) reference point: $734.51 < 884.6$: the 1991 reading is already below B_{ref} , as is the 1990 reading before it ($861.92 < 884.6$).*

Proof. (i) is the exact cross-multiplication $2021 \cdot 73451 = 148,444,471 < 210,836,400 = 7639 \cdot 27600$ in the script. (ii) is the pair of direct rational comparisons on the Table A2 rows, stated

so that the intermediate year is visible: the breach is *first* at 1993 only because 1992 is certified above the floor. (iii) is $734.51 < 884.6$ and $861.92 < 884.6$. No step of the proof uses floating point. $\square$

Remark 12 (the breach frontier). *On the realized path from the 1989 reading — $921.66 \rightarrow 861.92 \rightarrow 734.51 \rightarrow 381.95 \rightarrow 101.05 \rightarrow 30.55 \rightarrow 9.68$ kt — the first-below reading is a function of the threshold L , certified on intervals: $L \in (861.92, 921.66]$ first falls below at the 1990 reading; $(734.51, 861.92]$ at 1991; $(381.95, 734.51]$ at 1992; $(101.05, 381.95]$ at 1993; $(30.55, 101.05]$ at 1994; $(9.68, 30.55]$ at 1995; and $L \leq 9.68$ is never breached within 1983–2015 (the series minimum is 9.68). The realized threshold $B_{\text{aux}} = 276$ sits in the 1993 interval (Proposition 11(ii)); the reference point $B_{\text{ref}} = 884.6$ sits in the 1990 interval, after breaching repeatedly within its own construction window (Proposition 16).*

Remark 13 (the cross-vintage check). *The reassessment vintage reads the same three years as 600, 352, 79 kt, with floor ratios 2.18, 1.28, 0.29. The two vintages agree on the sign pattern against the floor — above, above, below — and disagree on levels by certified factors $\frac{73451}{60000} \approx 1.22$, $\frac{38195}{35200} \approx 1.09$, and $\frac{10105}{7900} = \frac{2021}{1580} \approx 1.28$: the reassessment’s 1993 level is 21.8% below the input vintage’s, and its 0.29 ratio is its own level (79 kt) over its own B_{lim} -scale denominator, not the input vintage’s 101.05. Because the ratio column is rounded to two decimals, the implied B_{lim} ranges over 250–289 kt across the file; the sign pattern would flip only if the true threshold were at or below 101.05 kt or above 352 kt, far outside that range, so every floor comparison here is insensitive to the rounding. The cross-file statement certified is therefore agreement on the breach sign pattern, not equality of levels — and the level disagreement itself ($\leq$ a factor of 1.28 between vintages six years apart) is part of the certified record.*

The instruments

Proposition 14 (the collapse is present in the raw survey index). *The fall research-vessel survey index reads 2,127,417 (1989), 1,117,670 (1991), 239,740 (1992), 90,709 (1993), and 21,797 (1994): a certified factor of $\frac{21797}{2127417} \approx 0.0102$ over 1989–1994, and $\frac{239740}{1117670} \approx 0.214$ across 1991–1992. The index is a research-vessel input to the assessments, not an output of them: the collapse is visible in the raw instrument record, before any model processing. Over the common window 1989–1993 the index falls by $\frac{90709}{2127417} \approx 0.0426$ while the assessment series falls by $\frac{10105}{92166} \approx 0.1096$: the instrument falls 2.57 times further than the model output it feeds.*

Proof. All five values are rows of the locked survey file; the three factors are their exact ratios, and 2.57 is the exact ratio $\frac{10105 \cdot 2127417}{92166 \cdot 90709}$ of the two common-window factors, evaluated in the script. The provenance statement is a property of the file (survey rows are assessment inputs, recorded at archival time). The index-to-biomass mapping assumes constant catchability and coverage; the divergence factor is certified as a difference between the two series’ rates, not attributed. $\square$

Proposition 15 (the covariate crossing is contemporaneous with the mortality crossing). *In the observed capelin series, the acoustic estimate falls from 5783 kt (1990) to 138 kt (1991) — a certified factor of $\frac{138}{5783} \approx 0.024$ — and reads 138 kt again in 1992. The Table A2 mean natural-mortality diagnostic crosses $M = 1$ across the same annual step (0.403 in 1990 to 1.002 in 1991; diagnostic column, model-estimated). At the data’s annual resolution the two crossings coincide; the covariate collapse precedes the certified floor breach (the 1993 reading) by two annual steps and the assessment series’ steepest realized step (1991–1992) by one.*

Proof. Both covariate readings are observed rows of the locked capelin file (no interpolation, per the archival discipline); the factor is their exact ratio. The mortality crossing is the diagnostic column’s consecutive values straddling 1 across the same pair of years as the covariate collapse

— the rows display the coincidence directly, and no sub-annual timing is claimed. Because the diagnostic is model-estimated given assumed catch, its crossing corroborates the regime reading but is not an independent certification of it. The covariate is a single acoustic instrument; no comparability guarantee across the early-1990s redistribution years is claimed, and the certified statements are about the locked rows at annual resolution. $\square$

The thresholds as design objects

Proposition 16 (the auxiliary floor separates the eras; the reference point does not). *Against $B_{\text{aux}} = 276$ kt: all seven reference-window readings lie above the floor (minimum 836.00), and the 1993–1995 readings (101.05, 30.55, 9.68) lie below it, with 1990–1992 above: the auxiliary floor separates the reference window from the collapse era without exception. Against $B_{\text{ref}} = 884.6$ kt: three of the seven reference-window readings (841.08, 863.23, 836.00) already lie below, and within 1983–1995 the below-set is $\{1983, 1984, 1986, 1990, \dots, 1995\}$ — nine of thirteen readings: the reference point does not separate the eras. A threshold fixed at a window’s own mean places roughly half that window’s readings below it by construction and retains no headroom above it (the window’s maximum is 6.4% above; its minimum 5.5% below, outward-rounded).*

Proof. All comparisons are direct rational comparisons of the certified readings against 276 and $\frac{8846}{10}$ in the script; the below-sets are computed by exact membership. The mean-rule statement is Proposition 1’s exact mean identity: a threshold at the mean of a set of readings lies between the set’s minimum and maximum, so the set straddles it. $\square$

Proposition 17 (conditional: sustained worst-case contraction breaches the floor in sixteen steps). *Under the reference window’s worst realized multiplier $\rho_g = \frac{41800}{45147}$ sustained from the 1989 reading 921.66 kt — a hypothetical rate, held fixed — the stock breaches $B_{\text{aux}} = 276$ kt within sixteen steps: $\rho_g^{16} \leq \frac{27600}{92166} < \rho_g^{15}$. The count is a ceiling convention (15.66 steps rounds up); the statement is conditional on the sustained hypothetical rate and carries no harvest-free reading: the window’s own bracket straddles unity (Proposition 3), so the sustained-rate contraction is not attributable to the non-fishing account by the certified method.*

Proof. The sixteenth and fifteenth integer powers of $\frac{41800}{45147}$ are compared against $\frac{27600}{92166}$ by exact cross-multiplication in the script. The ceiling convention is stated because the exact threshold step lies strictly between the fifteenth and sixteenth powers. $\square$

The outcome years

Proposition 18 (both vintages re-cross the floor at the 2015 reading; outcome years stay below half the reference point). *Each vintage’s own series re-crosses $B_{\text{aux}} = 276$ kt at the 2015 reading: the input vintage goes $250.12 \rightarrow 298.65$ kt (2014–2015), the database vintage $238 \rightarrow 277$ kt. The database vintage then reads 340, 433, 394, 419, 440, 411 kt for 2016–2021 (six outcome years, all above the floor), with maximum 440 kt at $\frac{44000}{88460} \approx 49.7\%$ of B_{ref} . The recovery’s certified removals rows are 4.4 to 12.9 kt per year (2015–2021, official table; maximum 12,881 t in 2017) — between 2.3% and 3.0% of the same vintage’s stock reading per year. The certificate does not extend to the way up: on the 2020–2021 step (a 29 kt decline) the removals were $\frac{10977}{29000} \approx 37.9\%$ of that year’s decline, so no harvest-free attribution is certified on recovery steps. The asymmetry the certificate thus documents — rapid certified decline against two decades below half the reference point on the way up — is the pattern the collapse literature records for this stock (Hutchings and Myers, 1994).*

Proof. The crossing pairs are direct rational comparisons in the two locked files; the six outcome values are the database rows (exact integers in kt); the maximum comparison is the reduced

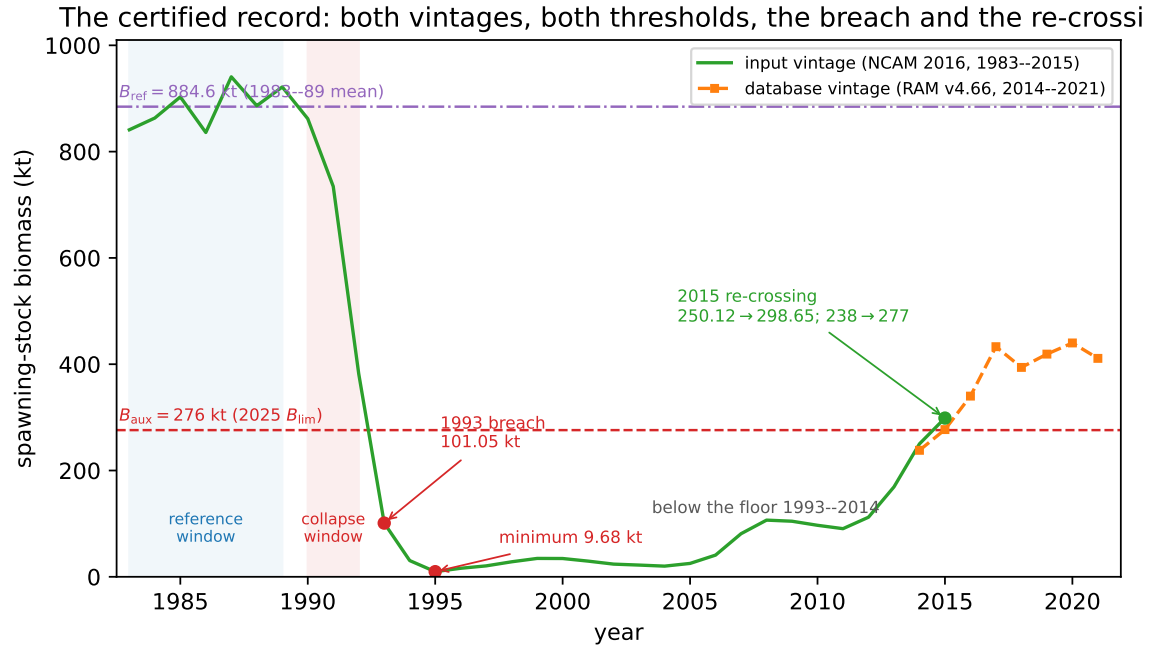


Figure 11: The certified record, both vintages: the input vintage (NCAM 2016, 1983–2015) and the database vintage (RAM v4.66, 2014–2021) against the auxiliary floor (276 kt) and the reference point (884.6 kt); reference and collapse windows shaded; the 1993 breach (101.05 kt), the 1995 minimum (9.68 kt), the 22 consecutive below-floor readings to 2014, and the 2015 re-crossing (250.12 → 298.65; 238 → 277) as certified in the text. Generated by the assertion script `paper2_arv_record_figure_v1.py` (recompute-then-assert; eleven exact checks against the locked files).

ratio $\frac{44000}{88460}$. The removals range is the exact minimum and maximum of the seven official rows for 2015–2021. $\square$

Remark 19 (the monitoring consequence). *Read together: the certified obstruction is a property of the dynamics under the realized catch (Propositions 3 and 8), the raw instrument record shows the collapse before the model output does (Proposition 14), the covariate crossing is coincident with the mortality crossing and two steps ahead of the floor breach (Proposition 15), and the mean-rule reference point separates nothing (Proposition 16). The assessment machinery itself was reconstructed around a monitoring indicator after 2016 (a three-year review cycle with an interim survey indicator, the indicator value available by early January) — the same review-timing reading the companion paper certifies for the 2022–2023 observation suspension (Abaee, 2026, The Northern cod forecast ladder, Section 3.10).*

Delimitations

Every arithmetic claim is a certificate about the realized rows of the certified files — not a forecast, not a stochastic model, and not a counterfactual catch analysis. The bracket of Lemma 2 is an accounting decomposition: it removes the realized removals from the per-capita rate under two extreme timing conventions, and its harvest-free factor still conflates natural mortality with recruitment — the certified statement is that no placement of the realized removals explains the decline, not that no catch path could have. The biomass series are assessment outputs (model-based), treated as the data of record with their published uncertainty untouched; the exact arithmetic is exact on the published point estimates, not a claim about the latent stock. The mortality diagnostic is model-estimated given assumed catch, so the mortality-

Table 11: The certified quantities and their files. Computed in exact rational arithmetic on the locked rows; displayed values rounded.

Quantity	Value	Source file
Ref-window worst mult. (1986)	41800/45147	Table A2
Collapse multipliers (1992, 1993)	0.520, 0.265	Table A2
Collapse-step bracket bounds	0.7542, 0.3718	A2 + T1
Post-moratorium 2-yr factor	10105/968 $\approx$ 10.44	Table A2
Removals 1992, 1993	40,956, 11,392	Table 1 (2025)
Removals 1994, 1995	1,314, 413	T1 = reconstr.
Survey index 1989, 1994	2,127,417, 21,797	Survey T3
Covariate 1990, 1991	5783, 138	Capelin file
First below-floor reading	1993 (101.05 < 276)	Table A2
Cross-vintage 1993 levels	101.05 vs 79 kt	A2 + T17
Outcome years 2016–2021	340–440 kt	DB extract

crossing statement corroborates the regime reading without independently certifying it; the survey index carries its own sampling error and catchability assumptions, untouched here, and the instrument–output divergence factor is certified as a difference of rates, not attributed to a cause. The covariate series is observed-years only; the contemporaneity statement is about certified rows at annual resolution. Single stock; two vintages of one threshold and one vintage of the reference point; no averaging beyond the reference point’s own construction.

Verification methods

Every number, fraction, inequality, power, ratio, bracket, and temporal claim in this paper is regenerated by the accompanying script in exact integer and rational arithmetic, standard library only, directly from the locked files: the seven reference-window readings with their six multipliers, exact mean ($\frac{44229}{50}$), and minimum; the collapse-window and connecting multipliers; the harvest-free brackets of Table 10 under both alignments, each upper bound compared against unity by numerator–denominator comparison; the loss factors $\frac{352560}{40956}$ and $\frac{280900}{11392}$ and the removals share 11392/280900; the cross-file equality of the 1992–1995 removals rows and the 2015 row; the moratorium scale ratios ($\frac{231293}{1314}$ to $\frac{268677}{413}$); the post-moratorium brackets and the two-year factor $\frac{10105}{968}$; the breach comparisons by integer cross-multiplication ($2021 \cdot 73451 < 7639 \cdot 27600$) and the first-below-floor pair with the 1992 intermediate; the cross-vintage sign pattern, level factors, and implied-threshold range; the survey factors and the divergence ratio; the covariate and mortality rows; the threshold-discrimination below-sets; the sustained sixteenth-step comparison; the two-vintage re-crossings, outcome values, and removals range; and every headline number of this text asserted present verbatim.

Conclusion

Applied to the public record of the Northern cod, the calculus certifies the following. The collapse window’s two annual steps are harvest-free contractions under every timing convention the rows admit (upper bounds 0.754 and 0.372): no placement of the realized removals — which the year of the floor breach puts at 4.06% of the stock’s loss — explains the decline, and under the moratorium’s own removals rows the non-fishing account drove a further certified factor-of-10.4 contraction to the series minimum. The reference window’s worst step and the two prelude steps do not admit that reading: their brackets straddle unity, so the certified record supports a typology (two harvest-free collapse steps inside a fishing-era decline), not a blanket verdict about the decade. The 276 kt auxiliary floor separates the eras without exception, while the 884.6 kt reference point — fixed at the window’s own mean — has three of the seven window readings below it in sample and separates nothing: a design property of mean-rule thresholds, certified

here on the readings themselves. The collapse is present in the raw survey index at a certified factor of $\frac{21797}{2127417}$ over 1989–1994, falling 2.6 times further over the common window than the assessment series it feeds; the covariate crossing coincides with the mortality crossing at annual resolution and leads the floor breach by two steps. Both assessment vintages re-cross the floor at the 2015 reading and the outcome years stay below half the reference point. The certificates are robust where the bands allow them to be: on the reassessment vintage’s published 95% bands, exactly the four collapse-era steps certify as declines and the 1993 breach holds at the band’s upper edge; and the reference window’s own profile (declines at horizons one to three, growth at every horizon of four or more) delimits the sustained-rate reading the record will bear — as does the vintage’s pre-window record, which has the window open at 28.3% of the 1962 peak, three years after a prior floor episode. These are exact statements about published point estimates; their scope is the realized record, and their monitoring content — which instruments moved, when, relative to which threshold — is the applied content the obstruction calculus prices.

Part III — Governance operators and viability kernels of the Edwards Aquifer

Article type: Research Paper

Abstract

Problem. Permitted groundwater systems manage drought with triggers; the protection competing rules deliver under adverse recharge is rarely scored against a fixed criterion.

Approach. We compute robust viability kernels of the J-17 annual-mean head for a policy family — training-mean pumping (282.16×10^3 acre-ft yr⁻¹), flat caps of 0–90%, a Stage-I reactive rule, and the critical-period-management cascade — under persistent drought-floor recharge. A policy is retained only if at least as protective as training-mean pumping and more permissive than the smallest securing flat cap, under a protocol frozen before any score.

Results. At the 618-ft threshold under the drought-of-record floor, training-mean pumping’s worst-case attractor (615.72 ft) lies below it, its kernel emptying beyond 13 years; every flat cut of 10% or deeper and both reactive rules make the safe set invariant; the smallest securing cut is interpolated 7.2% (31.5% of current). The reactive rules are retained only **nominally** at 618 ft under the mildest floor class: their supply margins (+3.3% Stage I, +0.4% cascade) are hybrid, worst-case protection against wet-year replay entitlement the robust class excludes. Retention is **not** certified (positive-pumping certified kernels empty beyond $T = 3$ years), and **nothing is retained at 660 ft**, where the rules are invisible to the kernel of their own trigger — a geometric property of trigger design.

Implications. Reactive rules match flat-cap protection at the physical threshold and seem generous only if wet-year pumping counts; a positive attractor-to-threshold margin defeats every non-negative pumping rule, so the 660-ft threshold is protected by wet years, not by policy; certified claims are limited to three years by the fitted defect.

Keywords: Edwards Aquifer; critical period management; J-17 index well; pumping policies; robust viability; viability kernel; groundwater governance

Article Impact Statement. An intervention-selection protocol — governance operators, robust viability kernels under declared recharge floors, declared-defect erosion, and a frozen retention rule — is demonstrated on the Edwards Aquifer. It resolves which pumping rules protect a physical threshold under adverse recharge and at what cost in permitted supply: on J-17’s 618-ft level, reactive trigger rules match flat-cap protection at 3.3% (Stage I sketch) and 0.4% (cascade) more permitted supply than the kernel-matched flat-90% cap — nominal, hybrid-supply results — while the 660-ft institutional threshold is protected by wet years, not by

the pumping family. The protocol is the generalizable content; the specific verdicts are scoped to this system.

1. Introduction

Drought management in permitted groundwater systems is built on triggers. An index-well or springflow reading crosses a threshold, and pumping reductions activate. Trigger design has a long evaluation literature: drought indicators and triggers are properly assessed as stochastic objects, with explicit attention to how indicator levels, trigger thresholds, and drought stages interact (Steinemann 2003). Robustness has been a recurring concern of water-resources planning since the recognition that optimal solutions can be brittle to the hydrologic assumptions that produce them (Watkins and McKinney 1997). The operational question for a district that already operates triggers is narrower, and comparative: given a declared family of pumping rules, which rules deliver protection under adverse recharge, and at what cost in permitted supply?

The Edwards (Balcones Fault Zone) Aquifer of south-central Texas is the most fully institutionalized example of trigger-based groundwater governance in North America. Its critical-period management program stages pumping reductions at index-well thresholds. The reductions protect springflow at Comal and San Marcos Springs, whose endangered-species requirements are administered through the Edwards Aquifer Habitat Conservation Plan, the subject of a standing scientific review by the National Academies (National Research Council 2015). The aquifer’s regional behavior has long been represented — and debated — through lumped and equivalent-porous-media models, from the Barton Springs groundwater availability model (Scanlon et al. 2003) to its recalibration by the Texas Water Development Board (Hutchison and Hill 2011). This study scores, rather than asserts, the protection supplied by a declared family of pumping rules on the measured J-17 record.

The instrument is the robust viability kernel: the set of initial heads from which a closed-loop pumping rule keeps the system inside a declared safe set under every disturbance in a declared class. Viability methods have an established lineage in natural-resource management, where constraints rather than objectives define the policy question (Krawczyk and Pharo 2013), and their computational tractability on management-scale problems is documented by fishery case studies (Krawczyk et al. 2013). The declared safe set here is a floor on the productive store rather than on a year’s yield. The head threshold marks the level below which springflow at Comal and San Marcos approaches cessation. The policy question is therefore the protection of the base that regenerates the yield, and a trigger’s function is to keep the store from being drawn down to that floor at the cost of near-term permitted supply. The governed object is the one-pool affine head map fitted in a companion forecast-evaluation study on the same series (under separate review), which found last-value persistence unbeaten by the causal ladder (7.55-ft oracle RMSE against 13.23 ft persistence and 12.84 ft AR(1); Author et al., in review). This study asks the governance question that the forecast study could not: whether a declared pumping rule changes the viability kernel of the real system as represented by the fitted map, and at what cost in permitted supply. A companion intervention study on a marine fishery stock applies the same design and reaches the mirror verdict — none retained there, reactive rules retained here (nominally, under the mildest floor class); the mechanism contrast (fast wet-year recovery in this aquifer against a fatal productivity floor at the cod reference point) is the only reason to mention it, and the two systems’ scores are never pooled (Author et al., in review).

The complete comparison — object, defect declaration, disturbance classes, policy family, and retention rule — was frozen in a dated protocol (2026-08-26) before any score was computed.

Related work

The regulatory object. The J-17 index well is not merely a monitoring point: it is the official regulatory trigger of the Edwards Aquifer Authority’s Critical Period Management plan, and crossing its thresholds automatically imposes pumping restrictions (Edwards Aquifer Authority, 2024). Under the EAA Act, permitted withdrawals from the aquifer are capped, and the critical-period stages impose graduated withdrawal reductions on the San Antonio Pool as the J-17 level and Comal and San Marcos springflows decline; the Stage I trigger for the San Antonio Pool is a J-17 level below 660 ft MSL with a 20% withdrawal reduction, rising to a 40% reduction at Stage IV; the Uvalde Pool is staged separately on J-27 (Edwards Aquifer Recovery Implementation Program, 2021). **The 660-ft threshold studied in this paper is therefore not an arbitrary level chosen for the analysis but the actual Stage I trigger**, which is what makes the results below directly interpretable as a statement about the instrument in force. The 618-ft level is the physical threshold at which the analysis finds protection to be a different question.

Viability applied to groundwater. The viability approach has been applied to aquifer management repeatedly, and the comparison it invites is by now standard: in the optimal-control framing an environmental objective enters as a monetised externality, while in the viable framing it enters as a constraint to be satisfied, and the two produce different trajectories and different long-run levels (Pereau, Pryet and Rambonilaza, 2019). The viability kernel has been identified analytically in hydro-economic models of aquifers under conflicting food-security and ecosystem objectives (Pereau, Mouysset and Doyen, 2018), and the wider survey literature records reservoir and groundwater management as established application areas (Oubraham and Zaccour, 2018).

What this paper adds to that line is the *robust* element and the geometry of the trigger. The kernels here are computed against a persistent drought-floor recharge rather than a nominal one, and the central negative finding — that **nothing is retained at 660 ft, because the rules are invisible to the kernel of their own trigger** — is a statement about the relationship between a trigger level and the reachable set under it, not about the relative merits of optimal and viable control. That is a different question from the one that literature asks, and to our knowledge it is not asked there.

From safe yield to sustainability, which this paper must acknowledge. It is a conspicuous feature of the groundwater literature that the classical management concept — safe yield, the balancing of annual withdrawal against annual recharge — was shown to be unsustainable in principle, precisely because it ignores the natural discharge that recharge already supports (Sophocleous, 1997; Alley, Reilly and Franke, 1999; Sophocleous, 2000; Alley and Leake, 2004). The persistence of the underlying water-budget reasoning has been documented as well (Devlin and Sophocleous, 2005). **This paper does not use the term, and should be read against that background.** Its finding that a positive attractor-to-threshold margin defeats every non-negative pumping rule — so that the 660-ft threshold is protected by wet years rather than by policy — is the robust-viability analogue of the safe-yield critique: a level that looks protected under average or favourable conditions is not protected under the adverse recharge the constraint set is required to survive.

Viability theory. That the governing question is which states admit controls keeping every constraint satisfied over time, rather than which policy optimises, is the founding move of viability theory (Aubin, 1991); the obstruction calculus underlying the certificates used here is the observation-constrained development of that programme.

2. Methods

2.1 Object and dynamics

The object is that of the companion forecast evaluation: J-17 annual-mean head H_t (ft above mean sea level), San Antonio Pool, 1934–2023, drawn from the registered twenty-column analysis panel (that evaluation’s fixed analysis dataset — the measured head, the constructed recharge and pumpage series, and the service and derived series declared there — of which only the head, recharge, and pumpage columns enter this analysis). The head is a measured well level, not an assessment inversion. Recharge R (USGS-estimated, Puente 1978 method; Umphres and Choi 2025) and pumpage P (Edwards Aquifer Authority Table 1) are the fluxes. No new data enter the analysis.

The dynamics are the causal stock-flow class of the companion forecast study, with one pool and affine form:

$$\Delta H_t = \alpha + \beta R_t + \gamma P_t + \delta H_{t-1}.$$

Definition 2.1 (One-pool affine head map). Let H_t denote the annual-mean head (ft above msl) at J-17, R_t the annual recharge, and P_t the annual pumpage. Under the hypotheses (F1) one-pool storage, (F2) affine dependence on (R_t, P_t, H_{t-1}) , and (F3) ordinary-least-squares fitting on 1934–1990 (56 transitions), the head map is the relation above.

The fit is $\alpha = 163.49$, $\beta = 0.0198$ ft per 10^3 acre-ft, $\gamma = -0.02844$ ft per 10^3 acre-ft, $\delta = -0.2539$, so $a = 1 + \delta = 0.7461$. The contraction $a = 0.7461$ corresponds to 25.4% per year mean reversion (the companion’s rolling AR(1) on the same series gives 0.66 — different window, different object; the full-sample coefficients there are (0.017, -0.026) against this window’s (0.0198, -0.02844)). Residual SD is 5.60 ft; training maximum is 15.41 ft. Out of sample (1991–2023, audit only, no refitting), the SD is 8.40 ft and the maximum is 21.81 ft. The out-of-sample maximum exceeds the declared defect. This fact is recorded below and not repaired, per protocol. The coefficients are fitted on the largely unregulated 1934–1990 era, so ordinary least squares cannot separate the aquifer’s physical response from the pumpers’ historical drought reactions, and imposing state-dependent rules under those coefficients assumes the physics–behavior separation the estimator does not guarantee (Lucas 1976); this bears on whether γ may be read as a policy lever at all, and the companion found the sign of γ reversed on the 1934–1950 subwindow.

The information pattern is the following: the manager ends year t knowing (H_t, R_t, P_t) and sets $P_{t+1} = \pi(H_t)$. The recharge R_{t+1} is unknown at decision time and is treated adversarially within a declared persistent floor. The kernels therefore assume year-end J-17 is known exactly; the actual CPM triggers on 10-day means and cuts within the year, so annual triggering overstates the lag and understates the cut-active fraction.

2.2 Uncertainty classes and safe sets

The disturbance classes are persistent recharge floors computed on the training window: $UC_{\min} = 43.7$ (the 1956 drought-of-record year, held perpetually), $UC_{q05} = 166.5$, and $UC_{q10} = 179.1 \times 10^3$ acre-ft yr^{-1} . UC_{q05} and UC_{q10} are the 3rd and 6th lowest recorded annual recharges of the 57 training years (linear-interpolation percentile estimator); the floors themselves have historical precedent, and only their perpetuity is harsher than any recorded year. They are certification geometry, not recharge forecasts.

The safe sets are the two declared normative thresholds, fixed in the frozen protocol: $K_{\text{phys}}^* = 618$ ft (Comal Springs cessation proximity) and $K_{\text{inst}}^* = 660$ ft (the post-2007 Stage I trigger, not applied to pre-2007 history; a 10-day mean in the actual institution, an annual mean here). The model domain is $[610, 710]$ ft. Upward exits above 710 ft are model-domain exits, not threshold violations.

2.3 Governance family

The declared governance family is the following. Training-mean pumping (called BAU in the frozen protocol; the label is retained in tables and, under this declaration, in running prose as the family’s baseline label) holds pumping at the 1934–1990 training mean, $P \equiv \bar{P} = 282.16 \times 10^3$ acre-ft yr⁻¹ — a flat 100% cap on the unregulated mean, not the historical pumpage path and not current use. Flat caps prescribe $\rho\bar{P}$ with $\rho \in \{0.9, 0.8, 0.7, 0.6, 0.5, 0.0\}$. The Stage-I reactive rule (S1) cuts pumping 20% when $H < 660$ ft; the reduction is a 20%-at-660 sketch of the Authority’s published Stage I (Edwards Aquifer Authority, Critical Period / Drought Management; the actual rule keys on 10-day J-17 or Comal flow). The critical-period-management (CPM) cascade prescribes cumulative stage totals of 20/30/35/40% cuts at $H < 660/650/640/630$ ft (cumulative, not stacked). The actual program declares five stages (Stage V at 625 ft); only the four-stage sketch is scored, Stage I is verified against the published 20%/660 pair, and stages II–IV are declared scenarios, not verified institutions.

The CPM supply figure below is the stage-weighted evaluation on the observed occupancies of the training era (1934–1990, $n = 57$ years): Stage I ($H < 660$ ft) 20 years (35.1%), Stage II ($H < 650$ ft) 12 (21.1%), Stage III ($H < 640$ ft) 5 (8.8%), and Stage IV ($H < 630$ ft) 1 (1.8%; 1956). The cumulative 20/30/35/40% cascade evaluated on these occupancies reproduces the replay’s mean prescribed pumping exactly (254.93×10^3 acre-ft yr⁻¹). Out of sample (1991–2023, $n = 33$ years) the occupancies are 33.3%, 15.2%, 6.1%, and 0.0%. Stages II and III recur out of sample (15.2%, 6.1%); Stage IV does not (0.0%; 1956 alone is below 630 ft), so CPM’s deepest cut is a 1956-only scenario in the historical measure.

Definition 2.2 (Robust viability kernel). Fix a policy π , a disturbance class $\mathcal{D}$, a safe set K^* , and a horizon T . The robust T -step viability kernel $\text{Viab}_T(\pi, K^*, \mathcal{D})$ is the set of initial heads from which the closed loop $P_{t+1} = \pi(H_t)$ keeps H inside K^* under every recharge sequence in $\mathcal{D}$.

For each policy and disturbance class, the robust T -step viability kernel of the safe set is computed by iterating the worst-case closed loop. The nominal kernel is reported without erosion. The certified layer applies the defect-to-margin conversion in the form the fitted map admits. The term “kernel” is used in its closed-loop reading — the robust positively invariant set of a fixed declared policy — not in the classical viability reading of Aubin (1991), which is existential over controls. No control choice enters the kernel computation. Only the evaluation of declared rules under the disturbance floor enters. With the declared policies all deterministic functions of the observed head, each robust kernel reduces to the invariance statement of a one-dimensional closed loop — attractor and domain-top arithmetic rather than control synthesis. The word is retained because the object answers the same question (“from which states can the constraint be held?”) in the policy-fixed form.

2.4 Retention rule and certified layer

The retention rule decides which non-training-mean policies earn their additional structure. The certified layer converts the declared defect into an erosion margin on the kernel.

Definition 2.3 (Retention rule). A non-BAU policy is retained only if (R1) its robust kernel is at least as protective as training-mean pumping’s at every reading, compared on the kernel lower boundary with empty = worst, and (R2) it permits more pumping than the smallest securing flat cap with matched protection — on the declared 10% grid this resolves as the largest-supply member of the matched class (flat-90%); the grid-resolution sensitivity is reported in Section 3.4 (Table 4).

The scoring regimes are declared here rather than left implicit. **Protection is scored under the disturbance classes** — the perpetual floors of Section 2.2. **Supply is scored as the 1934–1990 historical replay mean.** The retention verdict is therefore a hybrid: worst-case protection at historical-mean entitlement, not a within-scenario dominance. Under

the floor classes alone, the reactive rules are identically their matched flat caps (the cut is active every year) and carry no supply margin. The retention decision is made at the nominal level and re-checked at the certified level. The certified re-check holds only over the horizons where the certified kernels are nonempty — a horizon-truncated object, stated as such whenever it is invoked. At 618 ft under UC-min, R1 is close to automatic: every cut of 10% or deeper in the family makes the safe set invariant, so retention reduces to R2 (supply at matched protection).

Calculation 2.1 (Certified erosion margin). Let $\varepsilon = 15.41$ ft denote the declared uniform defect (the training maximum residual) and let $a = 0.7461$ denote the closed-loop contraction rate. Under the hypothesis (F4) that the autonomous contraction rate applies uniformly to the residual, the T-step erosion margin is

$$r_T = \varepsilon \frac{1 - a^T}{1 - a},$$

giving $r_1 = 15.41$, $r_3 = 35.49$, $r_4 = 41.89$, $r_5 = 46.66$, $r_\infty = 60.70$ ft.

Derivation. Under (F4) the closed-loop residual obeys $e_{T+1} = a e_T$. Summing the geometric series of perturbations of magnitude at most ε yields $r_T = \varepsilon(1 + a + \dots + a^{T-1}) = \varepsilon(1 - a^T)/(1 - a)$; the result is the triangle-inequality bound for bounded additive perturbations of a contracting scalar map. Numerical evaluation at the listed horizons gives the listed values. $\square$

Definition 2.4 (Certified kernel). The certified kernel at horizon T is the nominal kernel of $K^* + r_T$, with r_T as in Calculation 2.1.

Two properties of the certified layer must be stated before the results, because the audit correctly flagged the earlier account as imprecise. First, in this affine family every policy enters the map only through the intercept (γP is a constant shift), so the contraction rate a is identical across policies and (F4) is exact, not an approximation: the certified kernel is the closed-loop nominal kernel of a raised threshold, feedback included. Second, the mechanism by which the reactive rules collapse at the certified level is trigger-band entry: the eroded threshold $K^* + r_T$ crosses the 660-ft trigger between $T = 4$ ($618 + 41.89 = 659.89$) and $T = 5$ ($618 + 46.66 = 664.66$), so any threshold-erosion certificate makes every below-660 trigger rule identical to training-mean pumping from $T \geq 4$ — the same trigger-on-boundary invisibility as Section 3.2, re-entering by construction. The constant- r_T form is also conservative by construction: a per-step tube $K^* + r_t$ ($t = 1..T$) is tighter than $K^* + r_T$ held at every step.

3. Results

3.1 Worst-case attractors and the minimal cut

Table 1. Worst-case attractors of the closed loop (ft), by policy and recharge floor.

Policy	UC-min	UC-q05	UC-q10
training-mean pumping (BAU)	615.72	625.31	626.29
flat-90%	618.88	628.47	629.45
flat-80%	622.04	631.63	632.61
S1 (20% cut < 660 ft)	622.04	631.63	632.61
flat-70%	625.20	634.79	635.77
CPM cascade	628.36	636.37	637.35
flat-60%	628.36	637.95	638.93
flat-50%	631.52	641.11	642.09
flat-0 (zero pumping)	647.32	656.91	657.90

Under a perpetual 1956-recharge floor, training-mean pumping’s attractor (615.72 ft) sits below the physical threshold. The smallest flat cut whose attractor clears 618 ft is an interpolated **7.2%** of training-mean pumping, which lies outside the declared family (on the current-mean

-case attractor of the closed loop under the perpetual drought-of-record floor (UC-min)

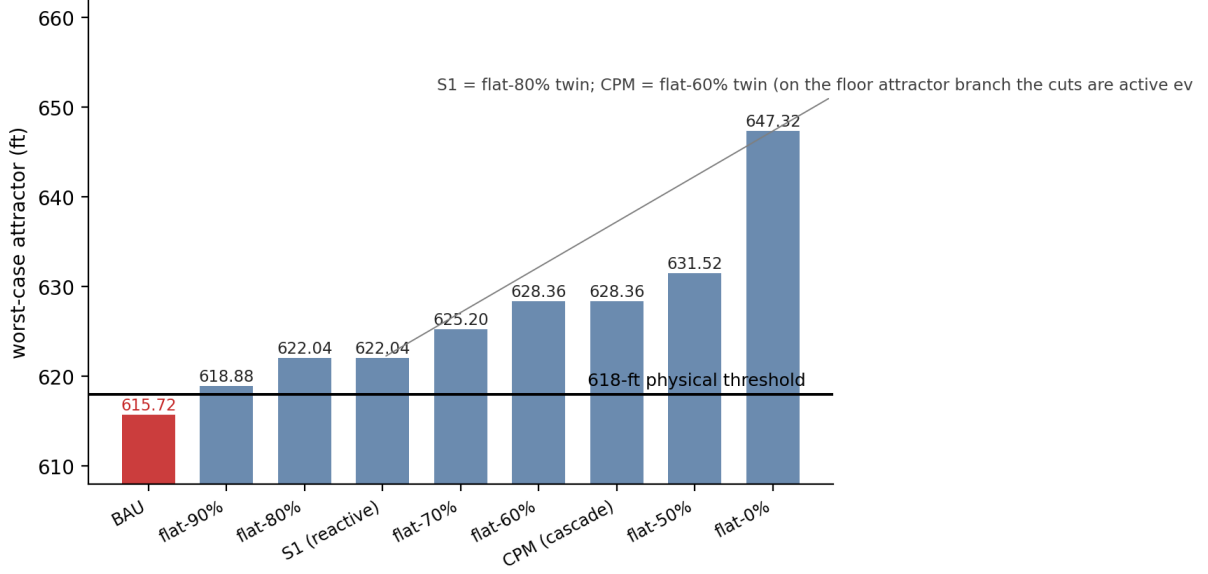


Figure 12: Worst-case attractor of the closed loop under the perpetual drought-of-record floor (UC-min) by policy, from the registered kernel computation (Table 1). Training-mean pumping’s attractor (615.72 ft) sits below the 618-ft physical threshold. Every cut policy — flat caps of 10% and deeper, the Stage I reactive rule, and the critical-period-management cascade — holds its attractor at or above the threshold. Under UC-min the reactive rules coincide with their attractor twins (S1 = flat-80%, CPM = flat-60%; under UC-q05 the CPM attractor sits between the Stage III and IV triggers and equals a ~65% cap); flat-90% clears the threshold by 0.88 ft — a margin the bootstrap of Section 3.7 shows is not resolved by the fit.

baseline of 382.16×10^3 acre-ft yr⁻¹ the same arithmetic gives a securing cut of about 31.5%, and the Stage-I 20% cut does not secure — Section 3.4). The Stage-I reactive rule’s attractor equals flat-80%’s, because the cut is active on the entire attractor branch.

3.2 Nominal kernels

Table 2. Nominal kernel lower boundary (ft) at the 618-ft threshold under UC-min, by horizon. Cut policies (flat-90% and deeper, S1, CPM) hold the whole safe set [618, 710] at every horizon, including the infinite horizon.

Horizon T	1	5	10	12	12.7 (crossover)	∞
BAU boundary	618.8	625.6	658.4	692.6	710	empty

At the physical threshold (618 ft) under UC-min, training-mean pumping’s kernel boundary climbs 618.8 (T = 1) → 625.6 (T = 5) → 658.4 (T = 10), and the kernel is empty beyond T ≈ 13 years. The emptiness is a domain-top event, and its arithmetic should be read exactly: the boundary $B(T) = H^* + (618 - H^*)/a^T$ reaches the declared safe-set ceiling when the *required initial head* exceeds 710 ft — trajectories fall toward the attractor, they do not exit upward. The horizon is a function of the arbitrary ceiling: $T_{\text{empty}}(C) = \ln((C - H^*)/(K^* - H^*))/\ln(1/a)$, so the 710-ft ceiling gives 12.7 years, the observed maximum annual mean (691.96 ft, the 1992 mean) gives 12.0 years, and with no ceiling the kernel is never empty in the domain sense — every state converges to the attractor, and “empty beyond T” means “no initial head in the domain lasts T years.” With the ceiling stated, business-as-usual is not robustly viable against a perpetual drought-of-record, and the observed 1992 mean (691.96 ft) sits almost

at the $T = 12$ boundary. Every cut policy in the family — a 10% flat cap, S1, CPM, and deeper — makes [618, 710] robustly invariant: the whole declared safe set is the kernel at every horizon, including the infinite horizon. Under UC-q05/UC-q10 the safe set is already invariant at training-mean pumping (attractors 625.3/626.3 ft); governance differentiates the kernel only under the drought-floor class.

At the institutional threshold (660 ft) the result is a **design observation about scoring a trigger against its own level**, and the kernel statement splits by policy class. For the **reactive rules** (S1, CPM) the equality with training-mean pumping is exact at every horizon: the boundaries (675.1 at $T = 1$, 695.3 at $T = 2$ under UC-min; empty from $T = 3$) lie strictly above the first-stage trigger (660 ft, the highest of the four declared trigger levels), no reactive rule fires in the viable region, and the kernel is policy-invariant for them. That equality is a geometric identity — a trigger placed on the boundary of a constraint is invisible to the viability kernel of that same constraint, and no rule that cuts only below 660 ft can improve the invariance of $\{H \geq 660\}$ — and it is a category observation about the scoring, not a deficiency of the rules: the cascade’s institutional purpose is springflow protection near 618 ft, not head invariance at 660 ft. For the **flat caps** the equality is false. They move the finite-horizon boundaries (flat-80% reads ≈ 673 at $T = 1$ against BAU’s 675.1) and extend the viable horizon (from $T = 3$ at BAU through $T \approx 6$ at flat-50% to $T \approx 6$ –11 for zero pumping by the same ceiling arithmetic; zero pumping’s attractor 647.3/657.9 ft still sits below 660 ft). Demand management therefore extends the viable **horizon, not invariance**, and the correct statement of the 660-ft reading is: no policy in the declared family, zero pumping included, holds $\{H \geq 660\}$ invariant under the drought floors — the threshold is protected by wet years, not by the pumping family. This is the frequency-management rationale the actual CPM rule implements, and that rationale is outside the robust-kernel frame.

The negative reading is in fact stronger than “the declared family” suggests, and the stronger form is worth stating because it is what makes the result portable. Write the fitted map as $H_{t+1} = aH_t + \alpha + \beta R_t + \gamma P_t$ with $0 < a < 1$ and $\gamma < 0$ (here $a = 1 + \delta = 0.7461$ and $\gamma = -0.02844$, from the Section 2.2 fit; both are fitted signs, reported rather than assumed). Because $\gamma < 0$, any non-negative pumping sequence $P_t \geq 0$ gives $H_{t+1} \leq aH_t + \alpha + \beta F$ pointwise under a recharge floor F ; because $0 < a < 1$, that comparison map contracts to the zero-pumping attractor $H_0^* = (\alpha + \beta F)/(1 - a)$. So whenever the *attractor-to-threshold margin* $m = K - H_0^*$ is positive, every non-negative pumping rule — not merely every member of the declared family — fails to hold $\{H \geq K\}$ invariant. At 660 ft the margin is positive under all three floors, which is why zero pumping already fails there.

Two consequences follow. First, the argument quantifies over pumping *sequences*, so it is indifferent to a rule’s functional form: it covers rules that respond to several state variables at once, and rules with hysteresis (a stage entered on one indicator and exited only on another), neither of which the declared family here can express. A district operating a richer rule set than the one scored here therefore inherits the same negative verdict. Second, the converse does not hold. A negative margin says only that zero pumping *could* hold K ; whether some member of a particular declared family does so is a separate, family-relative question, and one that a larger family can answer differently. Emptiness generalizes; construction does not.

3.3 Certified kernels

With the erosion of Section 2.4 applied, every positive-pumping policy in the family has an empty certified kernel beyond **$T = 3$ years** at the physical threshold and beyond $T = 1$ year at the institutional threshold. Zero pumping is the exception. Its certified physical-threshold kernel is nonempty through $T = 4$ under UC-min (lower boundary ≈ 687.9 ft from the certified-kernel algebra; the registered horizon grid $\{1, 2, 3, 5, 8, 10, 15, 20, \infty\}$ does not sample $T = 4$ — the algebra is the record there and the grid is annotated accordingly) and through $T = 5$ under UC-q05/UC-q10. The certified boundaries at $T = 3$, UC-min, 618 ft: flat-0 662.2 < flat-80

$697.8 < \text{BAU} = \text{S1} = \text{CPM}$ 706.7 ft. The certified-level inference is drawn explicitly: at the one certified horizon tabulated, the reactive rules inherit training-mean pumping’s boundary (706.7) — strictly above flat-80%’s (697.8) — so certified dominance of S1 over its matched flat cap fails. The +3.3%/+16.2% supply margins are nominal-level comparisons. Certified retention is a different, horizon-truncated object from nominal retention (Section 2.4).

The bound is optimistic, and the audit’s sensitivity is recorded: at the out-of-sample defect ($\varepsilon = 21.81$ ft) the erosion margins scale to $r_3 = 50.22$, $r_5 = 66.04$, $r_\infty = 85.90$ ft, so certified emptiness beyond $T = 3$ is a *lower* bound on how fast certification dies, and certified horizons do not appear as findings beyond this section’s scope statement. The binding constraint on certified intervention claims is the model defect, not the governance. The certified comparison is also conservative in form against the state-dependent rules for the geometric reason of Section 2.4 — the eroded threshold enters the trigger band from $T = 4$ — which is a property of threshold-erosion certificates generally, not of these rules.

3.4 Supply and retention

Table 3. Mean prescribed pumping: the frozen open-loop replay (actual observed heads, 1934–1990) and the post-freeze closed-loop replay (head simulated by the map under actual recharge, same policy in force), 10^3 acre-ft yr^{-1} .

Policy	Open-loop supply (frozen)	Cut active	Closed-loop supply (post-freeze)	Closed-loop cut active
BAU (training-mean)	282.16	0%	282.16	0%
flat-90%	253.94	100%	253.94	100%
flat-80%	225.73	100%	225.73	100%
flat-70%	197.51	100%	197.51	100%
S1	262.36	35.1%	258.98	41.1%
CPM	254.93	35.1%	254.70	41.1% (any stage)
flat-60%	169.29	100%	169.30	100%
flat-50%	141.08	100%	141.08	100%
flat-0%	0.00	100%	0.00	100%

Out-of-sample open-loop replay (audit only, over the 1991→2022 transitions): S1 264.5 and CPM 260.6×10^3 acre-ft yr^{-1} ; closed-loop from the observed 1990 head, S1 prescribes 268.5 with the cut active 24.2% of years and ends 2023 at 643.3 ft (observed 635.7; training-mean pumping ends 639.2). Every flat policy prescribes its cap throughout, so its supplies coincide by construction. The closed loop is a model object: run from the observed 1934 head it ends 1990 at 677.6 ft against the observed 645.8 ft (+31.8 ft), so the closed-loop supplies carry the map’s own level bias and are reported as the audit requested — as the within-model check on the hybrid margin, not as a second retention leg.

Table 4. The retention margins under every comparator reading (open-loop supplies; the closed-loop margins follow in the text).

Comparator (matched protection at 618 ft / UC-min)	Supply	S1 margin	CPM margin
kernel-matched	253.94	+3.3%	+0.4%
flat-90% (declared grid)			

Comparator (matched protection at 618 ft / UC-min)	Supply	S1 margin	CPM margin
attractor twin flat-80% (S1) / flat-60% (CPM)	225.73 / 169.29	+16.2%	+50.6%
interpolated 7.2% cut (outside family)	261.84	+0.2%	−2.6% (fails)
flat-92% (1% grid)	259.59	+1.1%	−1.8% (fails)
flat-93% (1% grid)	262.41	−0.02% (fails)	−2.9% (fails)

- **S1: retained (nominal, UC-min, 618 ft; hybrid).** It matches the flat caps’ robust invariance (kernel = whole safe set at all horizons) while supplying 262.36 versus flat-90%’s 253.94 ($+8.4 \times 10^3$ acre-ft yr^{−1}, +3.3%) and flat-80%’s 225.73 (+36.6, +16.2%). The scoring regimes are the hybrid declared in Section 2.4 — protection under the perpetual worst-year recharge floor, supply as the 1934–1990 historical replay mean. The reading of the retention is exactly “worst-case protection at historical-mean entitlement,” not a within-scenario dominance. Under the floor itself S1 is identically flat-80% (the cut is active every year, and its floor-attractor supply equals flat-80%’s), so the supply margin exists only in wet years that the robust class excludes. Against flat-90% the floor comparison is the sharper one — S1 is always −20% where flat-90% is always −10%, at identical invariance. The reactive architecture justifies its additional structure on the hybrid criterion, not on the robust criterion alone, and the comparator table shows how grid-dependent the margin is: against the 1%-grid smallest securing cap it is +1.1%, and against the interpolated 7.2% cut it is +0.2%.
- **CPM: retained (nominal, same threshold and class).** Attractor 628.36 ft (equal to flat-60%’s) at supply 254.93 versus flat-60%’s 169.29 (+50.6%) and flat-90%’s 253.94 (+0.4%). The +50.6% is priced substantially off Stage IV, a stage occupied in 1 of 90 years.
- **Closed-loop re-check (post-freeze layer).** With each policy in force and the head simulated by the map under actual recharge, S1’s supply falls to 258.98 (−1.3% versus the open-loop replay; the cut is active 41.1% of years, not 35.1%, because the map’s trajectory dips below 660 ft more often than the observed record did) and CPM’s to 254.70 (−0.1%). The hybrid margins attenuate but survive: S1 +2.0% versus flat-90% (+14.7% versus flat-80%), CPM +50.4% versus flat-60% (+0.3% versus flat-90%). The margin is a real property of the rule’s wet-year behavior on this map, not an artefact of evaluating the rules on the observed (unregulated-era) heads alone.
- **Certified level:** certified retention is horizon-truncated. At the one tabulated certified horizon the reactive rules inherit training-mean pumping’s boundary (706.7 ft, Section 3.3) and fail the certified re-check. The supply figures are nominal-level comparisons, not certified retention.
- **Institutional threshold: nothing retained.** The reactive rules are identically training-mean pumping at every horizon (trigger-on-boundary invisibility, Section 3.2). The flat caps are more protective at finite horizons and extend the viable horizon. No policy meets the retention test there.
- **Current-baseline sensitivity (post-freeze, analytic).** The declared family and every supply figure are relative to the 1934–1990 training mean (282.16). On the 1991–2023 mean actual pumpage (382.16×10^3 acre-ft yr^{−1}), the same attractor arithmetic gives training-mean-pumping 604.52 ft, the securing cut $\approx 31.5\%$, and the Stage-I/flat-80% attractor 613.08 ft — below 618 ft, so the 20% sketch does not secure the physical threshold

at current use. Re-expression of the family on the current baseline is a registered follow-up, not a scored object here.

3.5 Classification and stress replays

The $T = 5$ nominal kernel (UC-min, 618 ft) classifies the actual record. Training-mean pumping excludes exactly one actual year from its viable set — 1956, the drought-of-record year (623.15 ft annual mean, above 618 ft and above the 615.72-ft attractor: 1956 is excluded as a *starting* head whose 5-year forward path under perpetual 1956 recharge falls through the threshold, not because the 1956 mean was below 618; its daily minimum, 612.51 ft, was). S1 and CPM exclude none: the entire 90-year actual record is robustly 5-year viable under the cut rules. The $T = 5$ certified kernels are empty, so no actual year is certified 5-year viable under any policy. This is the boundary of the certified analysis.

A 1950s open-loop diagnostic (model driven by actual R, P versus actual heads, 1951–1956) records the map’s drought bias. The affine map under-predicts the decline: model $659.5 \rightarrow 631.3$ versus actual $659.5 \rightarrow 623.2$ ft, maximum error 8.1 ft, biased high. The model-based policy replays from the observed 1950 head keep all policies above 618 ft (BAU minimum 629.7, S1 634.9, CPM 637.1 ft) — note that this replay holds pumping at the training mean (282.16) while actual 1956 pumping was $\approx 321 \times 10^3$ acre-ft, so it is not the decade’s actual policy either. The open-loop bias means the true margins are smaller than the replay suggests. This is recorded, and no correction is applied.

3.6 Finite-duration recharge floors and floor-class supply (post-freeze layers)

Two declared post-freeze layers extend the persistent-floor object. The first asks how long a drought-class episode must last before the safe set contracts. The class floor holds for n years, recharge then returns to its training mean, and the infinite-horizon lower boundary at the 618-ft threshold is pulled back by exact backward recursion through the n floor years. The second computes the floor-class supply — the closed loop from the observed 1934 head with recharge held at each class floor — supplying the declared-scoring half that the historical replay cannot.

Table 5. Finite-duration floors (post-freeze): infinite-horizon lower boundary (ft) at the 618-ft threshold after n years at the class floor followed by training-mean recharge. The BAU row at $n = 5$ and $n = 10$ reproduces the registered $T = 5$ and $T = 10$ nominal boundaries (625.6 and 658.4 ft). The $n = 15$ emptiness reproduces the registered 13-year horizon bound.

Policy	UC_min, $n=5$	UC_min, $n=10$	UC_min, $n=15$	UC_q05/q10, any n
BAU	625.6	658.4	empty	618.0
flat-90% and deeper; S1; CPM	618.0	618.0	618.0	618.0

Duration differentiates exactly one policy. Under the drought-class floor, training-mean pumping is the only policy whose boundary moves with episode length. Every cut policy holds the whole safe set at every duration, because the cuts’ floor attractors sit at or above the threshold (618.9–647.3 ft, Section 3.1, Table 1). A finite drought episode followed by normal recharge pulls nothing back beyond 618 ft. Under the two milder classes the safe set is already invariant at BAU (Section 3.2), and duration is inert for every policy.

Table 6. Floor-class supply (post-freeze): closed loop from the observed 1934 head (670.4 ft), recharge held at the class floor, mean prescribed pumping (10^3 acre-ft yr^{-1}) over the training span (1934–1990) and the full span (1934–2023); end head at 2023.

Policy	UC_min supply (train/full)	UC_min end head	UC_q05 supply (train)	UC_q10 supply (train)
BAU	282.16 / 282.16	615.7	282.16	282.16
flat-90%	253.94 / 253.94	618.9	253.94	253.94
flat-80%	225.73 / 225.73	622.0	225.73	225.73
S1	226.72 / 226.36	622.0	226.72	226.72
flat-60%	169.30 / 169.30	628.4	169.30	169.30
CPM	174.49 / 172.59	628.4	187.36	187.61
flat-0	0 / 0	647.3	0	0

The table quantifies the scoring-regime reading of Section 2.4. Under the floors the reactive rules converge to their matched flat-cap attractors (S1 to flat-80%’s 622.04 ft, CPM to flat-60%’s 628.36 ft — the cut active every year on the attractor branch). Their closed-loop span-mean supply exceeds the matched caps only by the trigger-lag margin of the descent from the observed 1934 head — S1 +0.4% over flat-80% (226.7 versus 225.7×10^3 acre-ft yr⁻¹, a floor-class trigger lag), CPM +3.1% over flat-60% (174.5 versus 169.3). These are different objects from the hybrid margins of Table 4: S1’s +0.4% here is the within-floor trigger lag, CPM’s +0.4% there is the historical-replay margin against flat-90% — same number, different quantities, and neither is a robust-supply statement. The hybrid criterion’s margins exist only in the wet years that the robust classes exclude; the floor-class supply leg is the margin’s own measure of how little it would buy inside the class.

3.7 Attractor uncertainty (post-freeze layer)

Table 7. Residual bootstrap of the 1934–1990 fit (5,000 replications, seeded; no replication was non-contractive): worst-case attractor under UC-min, by policy.

Policy	Point attractor (ft)	Bootstrap mean (ft)	Bootstrap SD (ft)	5th–95th percentile (ft)	P(attractor $\geq$ 618 ft)
BAU (training- mean)	615.72	613.72	13.09	590.0 – 630.6	0.41
flat-90%	618.88	617.00	12.46	594.5 – 633.0	0.53
flat-80% / S1	622.04	620.27	11.87	598.7 – 635.7	0.65
flat-70%	625.20	623.55	11.35	603.1 – 638.3	0.75
flat-60% / CPM	628.36	626.83	10.89	607.3 – 641.2	0.83
flat-50%	631.52	630.11	10.51	611.3 – 644.2	0.89
flat-0	647.32	646.49	10.03	629.0 – 661.2	0.99

The audit’s resolvability question has a quantitative answer. The *ordering* of the attractors is resolved: the flat-90%-minus-BAU gap has a 90% bootstrap band of [1.9, 5.0] ft, excluding zero. The *threshold clearance* is not: whether training-mean pumping fails 618 ft ($P = 0.41$) and whether the marginal securing caps (flat-90%, $P = 0.53$; S1/flat-80%, $P = 0.65$) clear it are coin flips at this fit; only cuts of 60% and deeper reach probabilities above 0.8, and zero pumping reaches 0.99. The nominal-level statement that a 10% cut secures the threshold rests on a 2.3-ft gap (615.72 ft to the 618-ft threshold; the marginal flat-90% attractor sits at 618.88

ft) on a map with 5.6-ft residual SD and an acknowledged 8.1-ft high bias in the drought it is scored on — the fitted map cannot resolve whether the marginal members of the family protect 618 ft. The deep cuts and the failure of training-mean pumping are the resolved parts of the result.

4. Discussion

The complete evaluation loop — measured state, calibrated stock-flow map, declared governance operators, declared uncertainty classes, viability kernels with explicit declared-defect erosion, a held-out defect audit, and a fixed retention rule — yields a **nominal selection at the physical threshold under the mildest floor class**. The reactive architecture matches flat-cap protection at 3.3% (S1) and 0.4% (cascade) more permitted supply than the kernel-matched flat-90% cap — 16.2% and 50.6% against their attractor twins, +1.1% and −1.8% against the 1%-grid smallest securing cap — under a hybrid criterion whose wet-year entitlement the closed-loop re-check attenuates but does not remove. An interpolated 7.2% mean cut (outside the declared family; about 31.5% of current use) secures the physical threshold against a perpetual drought-of-record where training-mean pumping fails within roughly 13 years (a ceiling-relative statement, Section 3.2). Two boundaries are equally part of the result: the institutional threshold is not demand-manageable to invariance under the declared classes — and, by the margin argument of Section 3.2, not under any non-negative pumping rule whatever, whatever its functional form — and the certified content is defect-bound to $T \leq 3$ years.

The reactive result is system-dependent, not architectural. Its mechanism is the aquifer’s physics. High transmissivity and rapid karst recharge make wet-year recovery fast, so a flat cap permanently taxes recoveries that a reactive rule harvests — the supply margin exists only in the wet years that the robust classes exclude (Section 3.4). A companion intervention study on Northern cod applies the identical design and retains nothing: there the reactive rules cut catch exactly where the moratorium already protects. The framework’s deliverable is the scored comparison itself; which governance architecture earns its complexity is a property of the system, and the two scored systems occupy opposite ends of that design space without bounding it.

Two layers of results must be kept distinct. The robust-kernel findings are statements about the declared pumping family under the declared floor classes. The certified-kernel findings are a different layer, produced by applying the erosion conversion to the declared training defect. With $\varepsilon = 15.41$ ft — exceeded out of sample at 21.81 ft — every positive-pumping policy’s certified kernel is empty beyond $T = 3$ years, and the bound is optimistic rather than conservative on this object (Section 3.3). The certified emptiness is a bound on what the fitted map can certify, not a robustness statement about the physical system. The mechanism of the reactive rules’ certified collapse is geometric — the eroded threshold enters the trigger band (Section 2.4) — and the certified comparison is conservative in form against state-dependent rules for that reason.

The 660-ft design observation also reads correctly against the institution it describes. The actual CPM rule manages **frequency** — the fraction of time the head spends in critical stages — not robust invariance. The finding that no pumping rule can hold the institutional threshold under a perpetual drought floor is consistent with that design intent: the rule cannot make wet years, it prices dry ones. The National Academies’ review of the Habitat Conservation Plan (National Research Council 2015) was likewise organized around monitoring and modeling adequacy under drought. The horizon-not-invariance distinction is the robust-kernel expression of the same boundary.

The karst honesty of the object is inherited from its forecast-study companion and from the modeling lineage it descends from. The one-pool affine map is the simplest member of the lumped family whose regional adequacy was tested on the Barton Springs segment (Scanlon et al. 2003; Hutchison and Hill 2011). Conduits, the Uvalde–San Antonio divide, and unconfined

recharge-zone storage remain in the residual. San Antonio and Uvalde recharge and pumpage are lumped. The actual CPM triggers are 10-day averages, so the annual-mean rule is a coarse relative of the real institution. Stages II–IV are declared scenarios, not verified institutions. Nominal kernels carry no defect margin, and the certified kernels use a training defect that the out-of-sample audit exceeds. The counterfactual swap from the largely unregulated fitting era to state-dependent rules carries the declared econometric boundary of Section 2.1 (Lucas 1976). The UC floors are certification geometry, not forecasts. The 1950s replay is biased high by 8.1 ft. Nothing in this leg promotes or demotes any forecast module, and no two-pool, karst, or solute claim is made.

Cross-system reading. The protocol is the deliverable, and its value is shown by contrast rather than by a single result. The identical intervention-selection design applied to Northern cod (Abaee 2026) retains nothing — there the reactive rules cut catch exactly where the moratorium already protects, so no policy earns its complexity. Here, on the Edwards Aquifer, the reactive architecture earns a *nominal* supply margin at the physical threshold (+3.3% Stage I, +0.4% cascade) that is real but wet-year-contingent. The two scored systems therefore occupy opposite ends of the design space — one where reactive complexity is unrewarded, one where it is marginally rewarded — and the framework does not prescribe which end a new system lands on; it shows how to find out. That is the general, transferable content: the method scores protection against the robust floor class and separates the geometric (kernel/ threshold) from the identified (defect) findings, independent of the specific system.

Practical reading. For a district that already operates triggers, the result gives two usable bounds. First, at the physical threshold, deep cuts are robust and the smallest securing cut is interpolated at 7.2% of training-mean pumping (about 31.5% of current use) against a perpetual drought-of-record — but a cap’s nominal supply advantage over a reactive rule is not robust under the declared floor classes, so a manager should not credit wet-year replay supply when selecting among equally protective rules. Second, at the institutional (660-ft) threshold nothing in the pumping family is protective: the line is priced by frequency, not held by invariance, so a manager should not expect the pumping family to substitute for wet years when the cap is evaluated there. Where certified claims are wanted (a formal finite-horizon guarantee), the statement is limited to $T \leq 3$ years by the fitted defect (train $\varepsilon = 15.4$ ft, exceeded out of sample at 21.8 ft), so any longer certified claim should be read as a bound, not a certificate.

Generalizability boundary. The protocol generalizes; the verdicts do not. Every numerical certificate is scoped to the J-17 annual-mean one-pool affine map and the declared governance operators and floor classes, and no transfer to the Uvalde Pool (J-27) is claimed, consistent with the programme’s no-pooling discipline. Conduits, the Uvalde–San Antonio divide, and unconfined recharge-zone storage remain in the residual, and the annual-mean trigger is a coarse relative of the real 10-day rule.

Two senses of generality should be kept apart here, because the margin argument of Section 3.2 is general in one of them and not the other. Within *this* calibrated map, the negative result quantifies over *rules*: a positive margin defeats every non-negative pumping sequence, so no richer policy vocabulary — multi-variable triggers, hysteresis, finer stages — can recover invariance at 660 ft. That is a theorem about the map, and it holds whatever family a manager declares. It says nothing about *other aquifers*: whether any particular pool has a positive margin at its own threshold depends on that pool’s fitted $(a, \alpha, \beta, \gamma)$ and its own floor, which must be estimated there and reported with the hypotheses $0 < a < 1$ and $\gamma < 0$ checked rather than assumed. Generality across rules is established; generality across pools is not claimed.

5. Conclusions

On the measured J-17 record, three verdicts, not one. (1) **Nominal, physical threshold, mildest floor class:** reactive trigger rules match flat-cap robust protection of the 618-ft threshold at 3.3% (Stage-I sketch) and 0.4% (cascade) more permitted supply than the kernel-

matched flat-90% cap — 16.2% and 50.6% against their attractor twins — under a hybrid criterion (worst-case protection, historical-replay entitlement) whose wet-year margin survives a closed-loop re-check attenuated ($S1 + 2.0\%$ versus flat-90%). An interpolated 7.2% mean cut of training-mean pumping (about 31.5% of current use) secures the threshold against a perpetual drought-of-record, where training-mean pumping fails within roughly 13 years (ceiling-relative). (2) **Certified:** nothing — every positive-pumping certified kernel is empty beyond three years, an optimistic bound on a defect the out-of-sample audit exceeds. (3) **Institutional threshold:** nothing — the 660-ft line is protected by wet years rather than by the pumping family, a geometric property of scoring triggers against their own level. The fitted map resolves the deep-cut protections and the failure of training-mean pumping, and cannot resolve the marginal caps ($P = 0.41\text{--}0.65$); that is the honest resolution limit of the deliverable — the scored comparison on one measured system.

Cross-system conclusion

The three parts above certify rather than simulate, on two real assessment records with no hydrology, no biology and no institutional history in common. Read together they support one claim about the method and one about its limits, and the second is the more useful.

What certification buys: the verdict becomes structural. Part I’s central result is that the no-dominance verdict is *structural, not empirical*. Because the largest catch holding the reference point from itself is $C^* = g(\text{LRP}) - |e| = 91.59$ kt, and any rule’s protection margin is exactly C^* minus its catch there, harvest and protection are *one budget* at that point — and a reactive rule therefore cannot out-supply a flat cap it matches in protection. That is not a fact about a particular simulation run; it is a fact about two constants, and it holds for every rule in the class. Part II supplies the same discipline on a harder object: the certified harvest-free multiplier bracket certifies both collapse steps as harvest-free contractions under *every* timing convention for the removals accounting, while the reference window’s worst step and the two prelude steps fail to certify. The output is a *typology, not a blanket verdict* — which is the honest form of a certification, and the form a simulation cannot produce.

What it costs: the horizon, and where it comes from. This is the finding the three parts establish jointly and none establishes alone. **Certification has a horizon, and the horizon is set by the fitted map, not by the method.** In Part I the map is expansive at the reference point for every admissible carrying capacity $K \geq 2K^*$, so the certified layer has a finite horizon: six years under the two harsher floors, seven under the informative one. In Part III every positive-pumping certified kernel is empty beyond three years — described there as an optimistic bound on a defect the out-of-sample audit exceeds. The two numbers differ by a factor of two and come from unrelated systems, and that is the point: the limitation is not an artefact of one specification. *A certified verdict is valid over a stated horizon, and quoting it without that horizon is quoting a different claim.* This is the practical discipline certification imposes, and it is one simulation does not impose — a simulation shows a trajectory of some length and the reader is left to supply the horizon.

What the protocol is for. All three parts fix the scoring criterion before any score is computed, and all three report negative results. That combination is what makes a negative result informative: a rule that fails a criterion fixed in advance has been *tested*, whereas a rule that fails a criterion chosen afterwards has merely been described. Part III is explicit that its reactive rules are retained only *nominally*, on a hybrid criterion whose wet-year margin is exactly what the robust class excludes — so the nominal retention is reported together with the reason it is nominal. Part II likewise reports which steps do *not* certify rather than reporting only the two collapse steps that do. A certification method that only ever produced positive verdicts would be indistinguishable from advocacy.

The design consequence, stated generally. Two results generalise beyond their systems.

First, *where a threshold is protected by the weather rather than by policy, no rule in the scored family can be credited with protecting it*: Part III finds that the 660-ft line is protected by wet years, not by the pumping family — a geometric property of scoring triggers against their own level, and one that would be invisible to any method that only asked whether the threshold was breached. Second, *a positive attractor-to-threshold margin defeats every non-negative pumping rule*, so the binding constraint is geometric rather than institutional. Part I’s corresponding general statement is that harvest and protection are one budget at the reference point, so a gain in permitted catch is exactly a loss in protection margin and cannot be argued into existence.

Scope, stated once. These are scored comparisons on measured systems, and each part states its own resolution limit. Part III cannot resolve the marginal caps ($P = 0.41\text{--}0.65$) and says so; Part II certifies some steps and not others rather than issuing a verdict on the collapse as a whole; Part I’s certified layer is finite-horizon by construction. The general claim is therefore not that certification resolves more than simulation — it is that *certification states what it resolves and over what horizon*, which is the property that makes a verdict usable.

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Declarations

Data availability

The analysis is reproducible: the deterministic layers are closed-form or fixed-point computations and every stochastic layer uses a fixed seed and a fixed draw budget (Section 3.8 uses $N = 20,000$ trajectories per cell, Section 3.10 $B = 2000$ bootstrap refits). All input data, analysis scripts, and result files are archived in the public repository at <https://github.com/MIKEAA2020/general-sustainability>. The primary kernel table (Table 1, Sections 3.1–3.5) is produced by `wave_e_cod/src/run_intervention_v3.py`, which writes `intervention_results_v3.json` and `intervention_boundaries_v3.csv` in `results/`; the two post-freeze reactive families of Definition 2.4 (Table 1) by `wave_e_cod/src/run_families_v3.py`, which writes `results/e2_families_v3.csv`. Re-executing either runner regenerates its output files; a verification re-execution in a fresh environment reproduced every reported value to nine significant figures; the trailing digits of the floating-point fields differ by about one part in 10^9 across BLAS builds and are not byte-identical. In the source-year convention the flat-180-kt policy’s $T = \infty$ boundary (Table 1) is empty under the 5th-percentile class, so no converged fixed point is reported for it; the runner raises the fixpoint iteration cap to 20,000 and refuses to return an unconverged iterate, which an earlier 300-iteration cap did silently on exactly this row. The critical-zone rule and cascade vocabulary follows the DFO precautionary-approach framework (DFO, 2009); the SSB series and LRP are DFO (2016) Table A2; the catch series is Schijns et al. (2021). The elevation layers of Sections 3.7–3.10 (carrying-capacity grid, stochastic viability, finite-duration floors, bootstrap bands) are produced by `wave_e_cod/src/campaign_e2_elevation_v3.py` with

fixed seeds, its outputs archived in `src/results_srcyear_v3/`, and re-execution regenerates them exactly. Figures 1–9 are produced by `wave_e_cod/src/make_figs_v18.py`, which adds the two structural panels to the seven of `wave_e_cod/src/make_figs_v17.py` without changing them; v17 superseded `make_figs_v16.py` only in provenance — it imports the v3 runner and reads the v3 elevation outputs — and its output is identical. Figure 10 is produced by `wave_e_cod/src/make_figs_v19.py`; it re-derives the horizon curve from the same closed form and refuses to write the panel unless every tabulated catch reproduces the archived cadence campaign, so the figure cannot silently drift from the table. The structural constants of Section 2.4 and the corrected reactive criterion of Section 3.4 are verified cell by cell by `wave_e_cod/src/campaign_e2_structure_v3.py` (23 checks, outputs in `src/results_struct_v3/`), and the profile likelihood, joint bootstrap, observation-error sensitivity and post-moratorium windows of Sections 3.7, 3.10 and 3.12 by `wave_e_cod/src/campaign_e2_identification_v3.py` (outputs in `src/results_ident_v3/`); the horizon-in-catch invariance of Table 2 and the form-generality of Section 2.4 by `wave_e_cod/src/campaign_e2_cadence_v3.py` (outputs in `src/results_cadence_v3/`). The three Section 3.6 form rows are produced by `campaign_e2_depensation_v3.py` (free- s_0 Allee refit), `campaign_e2_allee_declared_v3.py` (declared-strength refit) and `campaign_e2_fox_form_v3.py` (Fox form), with outputs in `src/results_forms_v3/`; the Section 3.11 xteNCAM row by `arena agent 1/other documents/rerun_campaigns/campaign_e2_xteNCAM_row.py`; and the Section 3.8 one-off treatment of the 1992 draw by `e2_breakpoint_1992.py` in the same directory. Every script named above implements the source-year convention of Section 2.3. The registered-convention counterparts are retained rather than overwritten (`run_intervention_v2.py` and its outputs, and `wave_e_cod/src/superseded_v2/`), so the convention sensitivity listed in Section 2.3 can be recomputed from the archive rather than taken on trust.

Applied-regime-viability record. All inputs are the archived public files of the calibration record (wave-e-cod extracts and the record’s CSV files), with primary sources cited at archival time.

Edwards Aquifer. All input data, analysis scripts, and result files are archived in the public repository at <https://github.com/MIKEAA2020/general-sustainability>. J-17 daily highs: Texas Water Development Board well 6837203. Recharge: Umphres and Choi (2025), USGS data release, <https://doi.org/10.5066/P1BI62NY>. Pumpage: Edwards Aquifer Authority (2024/25), Table 1. The full intervention protocol — object, defect declaration, disturbance classes, policy family, and retention rule — was frozen and dated 2026-08-26 before any kernel, boundary, replay, or retention score was computed. It is archived with the analysis code as the frozen-protocol record, alongside the companion forecast-evaluation protocols dated 2026-08-25. The analysis is fully deterministic: re-executing the registered runner regenerates both output files, and a verification re-execution in a fresh environment reproduced both byte for byte. The machine-readable outputs include the nominal and certified retention fields and the certified-horizon record. The Section 3.6 layers (finite-duration floors; floor-class supply) are produced by `rerun_campaigns/campaign_e4_elevation.py`, archived alongside their outputs, and regenerate them exactly. The post-freeze audit layers of Sections 3.4 (closed-loop supply replay), 3.4’s current-baseline sensitivity, and 3.7 (attractor bootstrap) are produced by `wave_e_edwards/src/e4_audit_layer.py` (seeded, deterministic, on the registered panel and the paper’s declared coefficients), with outputs archived as `wave_e_edwards/results/e4_audit_layer.json`.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Code availability

The intervention record, the kernel-boundary table and every auxiliary campaign are produced by the scripts named above, archived with their outputs. The verification script `paperE2_cod_intervention_v29_verification.py` recomputes the fit, the floor classes, the kernel tables, the constructive bound, the certified horizon and the form-comparison rows from the archived campaign outputs and compares them against every value this manuscript prints; it is archived alongside this manuscript together with the mutation harness `paperE2_cod_intervention_v29_sabotage.py` and the numeric-provenance audit `paperE2_cod_intervention_v29_basis_audit.py`.

Applied-regime-viability record. The verification script `applied_regime_viability_v9_verification.py` is publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); it regenerates every fraction, inequality, bracket, power, ratio, and temporal claim verbatim from the locked files. The record figure is generated by `paper2_arv_record_figure_v1.py` under the same discipline: every plotted series value, threshold, crossing, and annotation is read from the locked files and asserted exactly before the figure is written (eleven checks).

Edwards Aquifer. The kernel computation and the policy scoring are produced by the scripts named above, archived with their outputs. The verification script `paperE4_edwards_intervention_v16_verification.py` re-runs the intervention computation from the registered panel and recomputes every attractor of Table 1, the smallest securing cut, the certified horizons, and the fit diagnostics against the recomputed values; it is archived alongside this manuscript.

CRedit authorship contribution statement

[AUTHOR NAME — full name as it should appear]: [list the roles that apply, deleting those that do not].

The CRediT taxonomy comprises 14 roles; delete any that do not apply to this work and do not add roles outside the taxonomy: *Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data Curation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition*.

If there are multiple authors, give one such line per author. If any contributor is an AI tool rather than a person, it belongs in the AI declaration below, not here.

AI declaration

GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review. The author reviewed and edited outputs and takes responsibility for the final work.